The Rise of Discounters and its Impact on Concentration, Market Power and Welfare

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Abstract

We quantify changes in concentration, market power, and welfare in the UK grocery retail sector from 2002 to 2021. We document that discounter expansion coincided with declining retail and manufacturer concentration across narrowly defined product categories. We develop an equilibrium model of consumer choice, retailer pricing, and manufacturer–retailer bargaining that recovers how market power is divided along the supply chain. Applied to breakfast cereals, we find that discounter expansion, through store entry, efficiency gains, and changes in product offerings, reduced concentration and prices, increased consumer and total surplus, and generated larger gains for households near newly opened stores.

Keywords: discounters, concentration, market power, distributional effects

JEL classification: D12, L11, L13, L81

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1 Introduction

The grocery retail sector is of substantial economic importance, accounting for a significant share of household expenditure. Competition authorities in many countries have expressed concerns over rising market concentration, growing retailer market power, and barriers to entry, particularly those stemming from zoning and planning regulations. These concerns have shaped decisions to block further consolidation in the sector, such as the rejection of the 2024 Kroger–Albertsons merger in the US and the 2019 ASDA–Sainsbury's merger in the UK.[1]

In recent years, a retail format often referred to as *discounters* or *limited assortment stores* has grown rapidly in both store count and revenue share. This trend has occurred in parallel in the UK, the US, and across much of the EU. Discounters follow a no-frills business model centered on a limited product range composed primarily of private-label goods, i.e., products exclusive to a single retailer and sold without manufacturer branding. In contrast to traditional retailers, discounters make relatively limited use of branded goods, i.e., products that carry manufacturer branding and are sold across multiple retailers. Firms in this format, such as Aldi and Lidl, operate smaller stores than traditional supermarket chains, enabling them to largely circumvent government planning restrictions that constrain rivals using large-store formats. Since discounters combine a distinct retail format with heavy reliance on private-label goods, their expansion has potentially important implications not only for retail concentration and prices, but also for manufacturer concentration, vertical relations, and the division of surplus between consumers, retailers, and manufacturers.

In this paper, we quantify changes in concentration, market power, and welfare in the UK grocery retail sector from 2002 to 2021 and provide evidence on the impact of the rise of discounters. We combine longitudinal microdata, spatial and temporal variation in discounter store expansion, shaped in part by changes in planning and land-access conditions, and a model of consumer choice, retailer pricing, and manufacturer–retailer bargaining that recovers how market power is divided along the supply chain. We document that the expansion of discounter-format retailers coincided with declining retail and manufacturer concentration across a wide range of narrowly defined product categories. Applying the model to breakfast cereals, we find that discounter expansion, through store entry, efficiency gains, and changes in product offerings, reduced concentration and prices, increased consumer and

[1] See, respectively, Federal Trade Commission (Feb. 26, 2024) complaint and Competition and Markets Authority (2019).

total surplus, and generated larger gains for households near newly opened stores. Branded manufacturers lost profits as sales shifted to discounters' private labels.

In the multi-category descriptive analysis, we exploit microdata that track purchases of disaggregate products (UPCs) brought into the home by over 100,000 households from 2002 to 2021, covering all major grocery retailers, including discounters and traditional retailers. These products are grouped into more than 100 narrowly defined categories, each designed to represent a distinct product market. We document substantial changes in market structure over this period, with broadly similar patterns across categories. At the retail level, concentration, measured using a Herfindahl–Hirschman Index (HHI) of retail market shares within each category, initially rose and then declined. The average category-level HHI increased from 1,551 in 2002 to 1,851 in 2011, before falling back to 1,560 by 2021. At the manufacturer level, concentration remained stable from 2002 to 2011, then declined as the market share of branded manufacturers fell. The reduction in concentration at both levels coincided with the rapid expansion of the discounters. This expansion was facilitated by a planning regime that was relatively favorable to the discounters, and further supported by the Controlled Land Order (2010), which prohibited incumbent traditional retailers from restricting nearby discounter entry through restrictive land-control practices.

We then use a model of retail market power and retailer–manufacturer relations to quantify the evolution of market power and the distribution of economic surplus between consumers, retailers, and manufacturers. To enable a disaggregated analysis of the supply chain that accommodates the rise of discounters and their private-label suppliers, we focus on a single category: breakfast cereals. This category exhibits concentration patterns that are broadly representative of those in other categories.

The model captures the differences between branded and private-label suppliers in their vertical relations with retailers. Retail prices are determined by Bertrand–Nash competition among retailers, who optimize against product- and retailer-specific marginal costs that depend on wholesale prices. Private-label suppliers have limited bargaining power, so we model their wholesale prices as equal to marginal cost. Branded-goods suppliers negotiate wholesale prices, which we model using a Nash-in-Nash bargaining framework.[2] Bargaining outcomes depend on each party's gains from trade relative to breakdown; firms with larger portfolios outside the negotiation have stronger bargaining positions. Conditional on product–retailer demands, the bargaining-power and marginal-cost parameters can

[2]Empirical applications of this framework include Draganska et al. (2010), Crawford and Yurukoglu (2012), Ho and Lee (2017), Crawford et al. (2018), and Noton and Elberg (2018).

be estimated using a linear instrumental-variable strategy. The model also accounts for cross-category demand and pricing effects (Thomassen et al., 2017).

We model product–retailer demand using a discrete-choice framework in which consumers choose among the breakfast cereals available at each retailer. A key determinant of retailer choice is travel distance to the nearest store, which we incorporate using store-location data spanning 2002 to 2021. Our model allows for heterogeneous preferences over product and retailer attributes, including price sensitivity, the taste for breakfast cereal relative to the outside option, preferences across cereal brands and types (e.g., wheat, rice), and persistent tastes for retailers. We identify the parameters governing this heterogeneity using the panel structure of our microdata, constructing moments based on the persistence of household choices over time and variation in household-specific choice sets.

Our estimates indicate that the distribution of additive markups (i.e., price–cost margins) across breakfast cereal products exhibits a modest inverted U-shape from 2002 to 2021, peaking in 2008 and mirroring the changes in retailer concentration described above. This pattern is primarily driven by traditional retailers, whose markups initially rose but then declined as competitive pressure from discounters intensified. In contrast, markups on products sold by discounters increased over the period, narrowing much of the initial gap with traditional retailers. We show that this reflects strengthening portfolio effects among discounters: the diversion ratio from an average discounter product to other products sold by the same discounter rose by approximately 50% between 2002 and 2021. This increase in within-firm substitutability, combined with the expansion of discounter store networks, enabled discounters to raise markups while continuing to exert downward pressure on traditional-retailer prices and expand market share, ultimately increasing their share of total industry profits by more than 8 percentage points.

To quantify the impact of the rise of discounters, we simulate what the market would have looked like had Aldi and Lidl not expanded beyond their 2002 position. One counterfactual holds their store network fixed at 2002 levels, isolating the role of store entry and local access. Another additionally eliminates post-2002 efficiency gains, product-quality growth, and portfolio changes, isolating changes to the in-store business model. Comparing observed and counterfactual outcomes allows us to quantify the importance of these markups for market performance.

We show that the rise of discounters reduced retailer- and manufacturer-level HHIs by 242 and 231 points, respectively, equivalent to more than a 10% decline. This expansion also reduced average prices by 4.1%. The decline reflects both discounter efficiency gains, through lower marginal costs partially passed through

to prices, and reduced markups on products sold by traditional retailers. Thus, discounters contributed not only to greater product variety but also to lower prices through their own cost advantages and by intensifying competition. Overall, total market surplus increased by 3.0% of total revenue, while consumer surplus rose by 5.1% of total revenue. Discounter profits grew, but these gains were more than offset by losses incurred by traditional retailers and manufacturers.

We also implement a policy-motivated counterfactual that captures the acceleration in discounter expansion following the Controlled Land Order (2010) by restricting store growth to its previous trend. This exercise suggests that the increase in store openings following the Controlled Land Order accounts for roughly half of the welfare gains from the store-entry counterfactual, highlighting the importance of entry and land-access conditions. This is consistent with previous work showing that the removal of entry barriers can generate substantial welfare gains (e.g., Bourreau et al., 2021).

Our model generates counterfactual predictions at the consumer level, enabling us to quantify the distributional effects of the discounters' rise. We find substantial heterogeneity in consumer welfare gains: in 2021, the interquartile range of gains, as a share of total cereal spending, spans 2% to 6%. These gains are broadly distributed across the income distribution, but exhibit systematic spatial variation. Households that saw the opening of a nearby discounter where previously none existed experienced the largest gains. Nevertheless, even households without a new nearby store benefited substantially, reflecting the competitive pressure that discounters exerted on traditional retailers.

Related literature We contribute to a literature measuring the evolution of market power (see Miller, 2025 for a recent survey). One strand estimates markups using production-function approaches across industries (e.g., De Loecker et al., 2020), while a complementary approach uses demand models in specific sectors to infer markups and their distribution. Work in the latter literature typically either abstracts from retailers (e.g., Grieco et al., 2024) or relies on simplifying assumptions that limit the role of retailers (e.g., Döpper et al., 2025 and Atalay et al., 2023, which assume constant and zero retail markups, respectively). De Loecker and Scott (2026) compare demand- and production-based markup estimates in the beer industry and find support for a highly competitive retail sector, while emphasizing the importance of accounting for downstream costs. We instead treat retailers as strategic intermediaries with market power both downstream and in their negotiations with manufacturers. Consumers choose among product–retailer options, and

the estimated demand system implies economically important differentiation driven by travel distance to stores and persistent retailer preferences.

Our results also relate to a growing literature documenting trends in market concentration. Several recent studies show that conclusions about national concentration trends are sensitive to market definition (e.g., Rossi-Hansberg et al., 2020; Affeldt et al., 2021; Benkard et al., 2021; Smith and Ocampo, 2025; see also Peltzman, 2014; Shapiro, 2018). We provide new evidence for the UK by tracking concentration across a wide set of narrowly defined fast-moving consumer goods markets at both the retail and manufacturer levels of the supply chain.

We also contribute to the literature on new retail formats. Prior work links the entry of non-traditional retailers to productivity growth and consumer gains (e.g., Foster et al., 2006; Hausman and Leibtag, 2007; Atkin et al., 2018), while work on Walmart's expansion emphasizes effects on rival retailers and the role of economies of density (see Basker, 2007 for a survey; Jia, 2008; Holmes, 2011; Arcidiacono et al., 2020). Cleeren et al. (2010) document rising competitive pressure from discounter entry, and a more recent literature studies the effects of dollar store entry on food consumption and welfare (Cao et al., 2024; Caoui et al., 2026; Schneier, 2025b). We quantify the effects of discounter entry on market power, surplus, and its distribution across consumers, retailers, and manufacturers.

Finally, the paper relates to work on vertical relations in retailing (e.g., Villas-Boas, 2007; Bonnet and Dubois, 2010; Bonnet et al., 2025) and on private-label products and retail competition (e.g., Meza and Sudhir, 2010; Dubois and Jodar-Rosell, 2010; Griffith et al., 2018). We distinguish private-label and branded products in their vertical relations with retailers. For branded products, we model retailer–manufacturer negotiations using a Nash-in-Nash framework, building on Draganska et al. (2010), Crawford and Yurukoglu (2012) and Ho and Lee (2017). We also build on work modeling market power in breakfast cereals (e.g., Nevo, 2000, 2001; Backus et al., 2021; Barahona et al., 2023) by allowing both retailers and manufacturers to exert pricing power.

The rest of the paper is structured as follows. Section 2 introduces the microdata and documents the evolution of concentration across grocery markets. Section 3 presents the equilibrium model of the breakfast cereal market, and Section 4 discusses identification and estimation. Section 5 presents estimates and documents the evolution of markups. Section 6 quantifies the effect of the discounters' rise on market performance. A final section concludes.

2 Data, Market, and Trends in Concentration

Consumer data We use longitudinal microdata from Worldpanel by Numerator's Take Home Purchase Panel (Worldpanel by Numerator, 2021), which covers households residing in Great Britain over 2002–2021. The sample includes approximately 15,000 households in 2002, rising to 30,000 from 2011 onward. Households typically remain in the panel for several years and record all fast-moving consumer-goods purchases, including food, beverages, toiletries, pet food, and cleaning products. They record purchases by scanning UPCs using a handheld scanner or mobile app, and by submitting receipts electronically or by post.[3] For each transaction, we observe quantity, expenditure, retailer and UPC characteristics, including product category and manufacturer. For panel members, we observe annual household income, household size, the age of household members, and residential postal district, one of approximately 2,500 districts in Great Britain.

Store location and input price data We use a dataset compiled from multiple sources that records retailer store locations over 2002–2021. For 2014–2021, we use data from Geolytix Retail Points (Geolytix, 2023); for 2002–2006, we use data from the Institute of Grocery Distribution (IGD, 2006); and for intervening years, we use Glenigan data on new supermarket construction projects (Glenigan, 2018). We combine store locations with household locations to construct household-specific, time-varying distances to the nearest store of each retailer (see Appendix A). We also use official input-price data for cereal grains and sugar (Defra, 2023; European Commission, 2023; FAO, 2023) (see Appendix B).

Retail formats We refer collectively to Asda, Morrisons, Safeway, Sainsbury's, and Tesco as *traditional retailers*. Safeway operated separately until its acquisition by Morrisons in 2005. These retailers have long had a significant presence in UK grocery retailing and typically operate large stores carrying a wide range of products. We refer to Aldi and Lidl as *discounters*. They follow an everyday-low-pricing strategy, carry a limited product range, and rely heavily on private-label products (also known as own-label products): on average, private-label products account for 90% of Aldi's sales and 85% of Lidl's, compared with about 50% for traditional

[3]For non-barcoded items, such as loose fruit and vegetables, households scan a code in a book provided by Numerator.

retailers. Like the traditional retailers, discounters operate stores throughout the UK, and both groups follow *national pricing* policies.[4]

Private labels For private-label goods, the retailer controls quality, marketing, pricing, and quantity decisions (see CMA, 2023, para. 3.8), so that in practice the vertical structure comes close to vertical integration. This allows retailers to switch between private-label manufacturers without consumers noticing, weakening manufacturers' bargaining positions. The Competition and Markets Authority (CMA)—the UK's antitrust regulator, which succeeded the Competition Commission (CC)—investigated ten representative food categories and found that retailers generally secure competitive prices from private-label suppliers.

> Own-label food and drink manufacturers compete with each other to win and retain contracts from retailers. ... competition to win and retain supply contracts appears to be strong, switching does occur, and retailers generally appear to obtain competitive prices, assisted by the transparency of the costs of their own-label suppliers. (CMA 2023, para. 10)

This cost transparency is often facilitated by "Open Book Accounting", under which suppliers share their costs with retailers (para. 4.39). The CMA also reports that "very few of the manufacturers we spoke to make own-label as well as branded products" (CMA, 2023, para. 4.28).

Even where a branded manufacturer supplies private-label products, it faces competition from specialist private-label firms, and the CMA finds that "for most products, own-label margins are low and have generally fallen in the most recent financial year" (CMA, 2023, para. 11). Net margins average around 2% for private-label manufacturers, compared with about 14% for branded producers (CMA, 2023, Figure 1).[5]

Planning policy and controlled land use Two features of the planning system favored discounters. First, the system was highly restrictive for retailers seeking to open large stores, defined as having sales areas above 1400 square meters (CC, 2000,

[4] "Most retailers set their prices uniformly, or mostly uniformly, across their store network [...]. Various other facets of the retail offer, such as promotions, may also be applied uniformly, or mostly uniformly, across a retailer's store network" (CC 2008, paragraph 4.98, pp. 498–501).

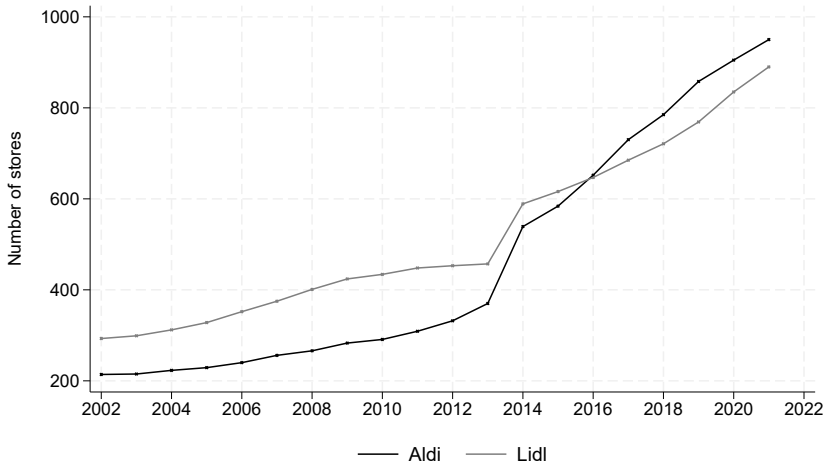
[5] Pasirayi and Richards (2023) provide evidence from emerging markets that branded-goods manufacturers can exert market power when they supply private-label goods, but distinguish these settings from Western economies like the UK, where private-label products have a large market share and a developed manufacturing base.

2008). Traditional retailers primarily use this format, whereas discounters typically operate mid-sized stores, with sales areas between 280 and 1400 square meters, which face fewer planning restrictions. Second, before 2010, traditional retailers limited discounter expansion through land controls, such as restrictive covenants and exclusivity clauses. The Controlled Land Order (2010) prohibited these practices, releasing many mid-sized sites well suited to discounters. The Order exempted discounters, allowing them to continue using such controls against rivals and thereby strengthening their incentives to expand store coverage.[6] See Appendix C.

Market trends A major change in the UK grocery market over the first two decades of the 21st century was the growing popularity of discounters relative to traditional retailers, with important consequences for both downstream and upstream concentration.

A key element of this growth was the expansion of store coverage. Figure 2.1 shows the sharp rise in Aldi and Lidl stores, from 507 stores in 2002 (214 Aldi and 293 Lidl) to 757 in 2011, and 1840 in 2021. The acceleration after 2010 coincides with the Controlled Land Order. More than 90% of discounter stores opened between 2002 and 2021 were in postal districts already served by an incumbent traditional retailer. Over the same period, traditional-retailer store counts grew by only 40%,[7] reflecting continued planning restrictions.

Figure 2.1: *Discounter store coverage growth over time*



Notes: Authors' calculations using a store coverage dataset based on Geolytix Retail Point, Institute for Grocery Distribution, and Glenigan data.

[6]Schneier (2025a) shows that land-use restrictions of the type banned by the Order are also widespread in the US and have substantial effects on entry, reducing entry by firms subject to them and promoting entry by firms able to use them against others.

[7]This figure excludes stores some of these firms operate in the "convenience store" format, which have low floorspace (less than 280 square meters).

The expansion in store coverage was accompanied by a substantial rise in Aldi and Lidl's average market share across the 226 fast-moving consumer goods categories, from about 3% in 2002 to about 14% in 2021 (Figure 2.2, panels (a) and (e)).[8] This growth in market share had important implications for retail concentration. Panel (b) plots the Herfindahl–Hirschman Indexes (HHIs), calculated using retailer revenue shares within each product category. Retail concentration rose between 2002 and 2006, coinciding with Morrisons' acquisition of Safeway and several smaller mergers, and remained relatively stable until 2011. It then declined substantially as discounters expanded. Panel (e) shows that average retail HHIs rose from 1551 in 2002 to 1851 in 2011, before falling to 1560 by 2021.

The rise of Aldi and Lidl also affected manufacturer concentration, calculated using manufacturer revenue shares within each category. For private-label goods, we treat the manufacturer as a distinct firm for each retailer, because, as discussed above, the private-label supply chains are effectively retailer-controlled and the identities of private-label manufacturers are often unobserved. Panels (c) and (d) of Figure 2.2 summarize the manufacturer-level patterns. Since private-label products account for a high share of discounter sales, discounter private-label shares follow a similar pattern to discounter retail shares, rising sharply after 2011. The mean share increased from around 4% in 2011 to over 12% in 2021. This contributes to a decline in manufacturer concentration, with the mean HHI falling from 1676 to 1437 between 2011 and 2021.

Breakfast cereals

In our analysis of market power and surplus, we focus on the market for multi-portion breakfast cereals. As shown in Figure 2.2, where breakfast cereal trends are highlighted in red, this category exhibits changes in retailer and manufacturer concentration that are broadly representative of other product categories.

We exclude single-portion cereal products and cereal bars from the market definition. We also exclude products and manufacturers with very small market shares. Specifically, we require a manufacturer to account for at least 1% of breakfast cereal spending in any year over 2002–2021, a brand to account for at least 0.1% of spending in any year, and a brand–pack-size combination either to account for at least 0.1% of spending in any year or to be the most popular available pack size for that brand in any year. These conditions leave us with approximately 90% of total breakfast cereal spending.

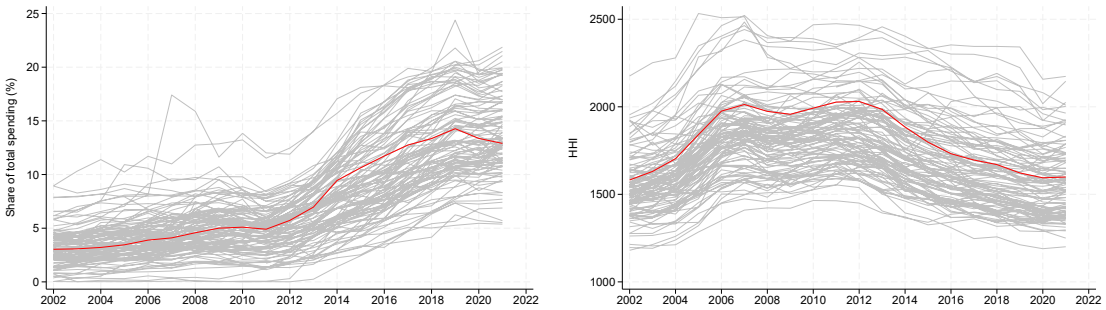
[8] These categories are based on a classification developed by Numerator and designed to reflect distinct consumer markets, with slight adjustments for consistency over time. We report them in Appendix S.

We define products at the barcode level, capturing differences in brand and pack size. For instance, "Kellogg's Cornflakes 750g" is a specific product within the Kellogg's Cornflakes brand.

Figure 2.2: *Evolution of market concentration*

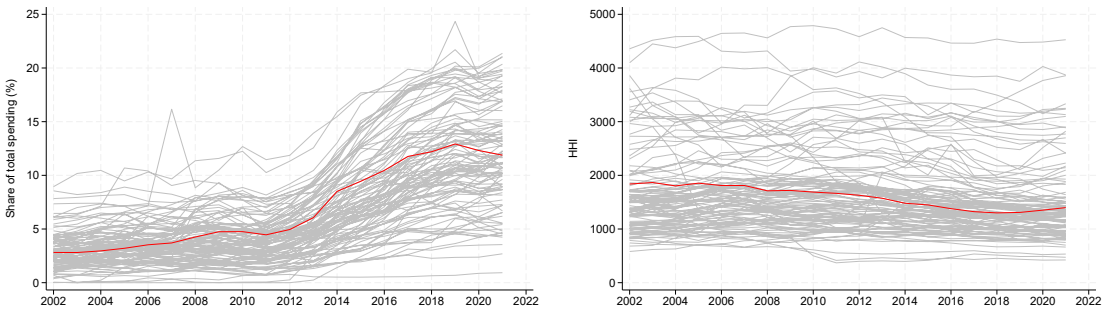
(a) Discounter retail market shares

(b) Retail HHIs



(c) Discounter private-label market shares

(d) Manufacturer HHIs



Breakfast Cereals

(e) Across category mean shares and HHIs

	Discounter share:		HHI	
	Retail	Manufacturer	Retail	Manufacturer
2002	3.23	3.04	1551	1636
2011	4.81	4.03	1851	1676
2021	14.05	12.44	1560	1437

Notes: Authors' calculations using Numerator's Take Home Purchase Panel, 2002–2021. Panel (a) shows the share of spending made in discounters, and panel (c) shows the share of spending on discounter private-label products. Panels (b) and (d) show the evolution of retail and manufacturer HHIs. The red line corresponds to breakfast cereal; the gray lines represent all other product categories with average spending shares greater than 0.25% over 2002–2021 (they collectively account for 88% of fast-moving consumer goods spending). Panel (e) reports the mean discounter shares and HHIs across categories in 2002, 2011, and 2021, weighted by each category's mean revenue-share over 2002–21.

Market structure Table 2.1 summarizes the structure of the breakfast cereal market. Panel A focuses on retailers, and panel B on manufacturers of branded

products. Retailers stock both branded products and their own private-label products, which are exclusive to a single retailer.

Column (1) reports the average number of branded products sold by each retailer and produced by each manufacturer. Column (2) shows the average number of private-label products sold by each retailer, and column (3) shows the average number of vertical links per firm—for instance, in a typical year, Asda stocks products from all six branded manufacturers, and Kellogg's sells its products through six retailers. The remaining columns report average prices and firm-level spending shares.

Table 2.1: *Breakfast cereal retailers and manufacturers*

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Number of:			Price (£/kg)		Spending share (%)		
	branded products	private-label products	vertical links	branded products	private-label products	Mean	2002	2021
Panel A: Retailers								
Traditional retailers								
Asda	97	37	5	5.01	2.87	16.57	16.34	14.57
Morrisons	83	22	5	5.16	3.14	9.93	5.80	9.94
Safeway	70	9	5	5.03	3.17	6.05	8.55	-
Sainsbury's	92	31	5	5.20	3.09	15.60	16.29	14.02
Tesco	114	46	6	5.11	3.05	32.88	29.04	29.53
Discounters								
Aldi	3	28	1	3.86	2.80	4.68	1.83	8.06
Lidl	4	23	1	3.95	2.50	3.09	0.98	5.30
Small retailers								
Other	86	-	5	4.81	-	16.04	21.16	18.59
Panel B: Branded manufacturers								
Dorset	5	-	4	4.34	-	0.56	0.16	-
Halo	4	-	5	4.78	-	1.58	3.14	0.20
Kelloggs	60	-	6	5.47	-	34.52	40.57	32.11
Nestle	39	-	6	4.96	-	17.60	21.01	13.65
Jordans	10	-	5	4.28	-	2.85	2.51	3.56
Whitworths	28	-	6	4.53	-	14.36	13.17	15.65

Notes: Authors' calculations using Numerator's Take Home Purchase Panel, 2002–2021. Numbers describe the sample we use to estimate our model and cover 90% of total breakfast cereal spending (see text for details). Columns (1)–(6) are means across years firms were in operation. All firms were in operation over 2002–2021 except Safeway, which ceased operation as an independent firm in 2005, and Dorset, which was acquired by Jordans in 2014. Prices are deflated by the all-items CPI (ONS, 2023b) and expressed in 2021 £.

On average, 76% of annual breakfast cereal spending takes place at traditional retailers. These retailers typically stock products produced by all six branded man-

ufacturers and offer broad private-label ranges. The discounters' share of breakfast cereal retailing rose from under 3% in 2002 to 13% in 2021, mirroring their broader expansion. Over this period, Aldi and Lidl increased their links with branded manufacturers, from zero to two for Aldi and from one to three for Lidl. We aggregate smaller national retailers with relatively small market shares into a composite "Other" retailer.[9]

Each branded cereal product is produced by one of six manufacturers. The largest is Kellogg's, followed by Nestle, Whitworths, and three smaller manufacturers, Dorset, Halo, and Jordans. All manufacturers sell through several retailers, with the mean annual number ranging from four for Dorset to six for Kellogg's, Nestle, and Whitworths. The largest two manufacturers have seen a decline in their market share from 2002 to 2021, in part due to the rise in private-label sales.

It is typically not observable which manufacturers supply a given supermarket's private-label breakfast cereal products, although some public information is available. Some branded manufacturers also supply private-label goods to UK supermarkets, while others do not, and several firms produce private-label cereals without producing branded products (see Appendix D).

Local effects of discounter entry Using the store–location and household data, we present event studies around local discounter entry. We construct a postal-district–year panel and define entry as the first year in which Aldi or Lidl opens within 5 km of the district centroid. For each postal district and year, we compute a local cereal price index using cereal products available within 5 km and national market-share weights, the share of breakfast cereal purchases made at discounters, and the average price paid for breakfast cereal.

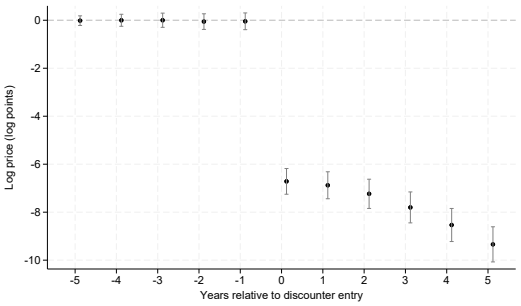
Figure 2.3 plots event-time coefficients estimated using the imputation estimator of Borusyak et al. (2024), with postal-district and calendar-year fixed effects. Discounter entry is associated with an immediate decline of around 7 log points in the local cereal price index and an increase of around 4.5 percentage points increase in the discounter share of cereal purchases. Both effects grow over subsequent years, in part reflecting additional discounter openings within the local area. The average price paid for cereal also falls by around 1–2 log points following entry. There is little evidence of differential pre-trends apart from some movement in the year immediately preceding recorded entry, which likely reflects the annual frequency at

[9]These include retailers focused on convenience store formats (Co-op, Kwik Save, Somerfield), high-quality products (Waitrose and Marks & Spencer), and internet-only shopping (Ocado). We exclude minor outlets, including independent stores, which collectively account for 6% of breakfast cereal spending.

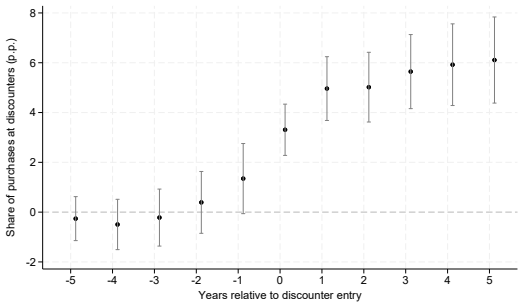
which store entry is measured.[10] These patterns show that discounter entry expands access to lower-priced options, households shift purchases toward them,[11] and the prices they pay decline.[12]

Figure 2.3: *Local effects of discounter entry*

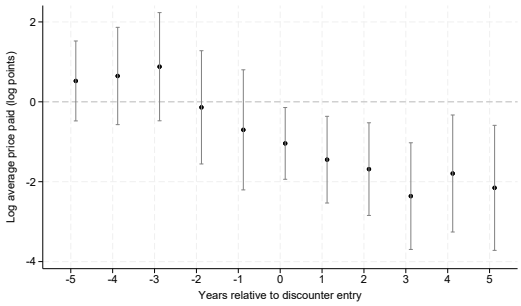
(a) Price of locally available cereals



(b) Discounter cereal purchase shares



(c) Average price paid for breakfast cereal



Notes: The figure plots event-study coefficients around the first recorded year in which Aldi or Lidl operates within 5 km of a postal-district centroid; event time 0 denotes this year. Coefficients are estimated using the imputation estimator of Borusyak et al. (2024), with postal-district and calendar-year fixed effects. The estimation sample comprises 652 treated postal districts and 231 never-treated postal districts, where never-treated districts have no Aldi or Lidl within 5 km at any point during the sample period. The local cereal price index in panel (a) is a weighted-average price of breakfast cereal products available within 5 km, using national market shares as weights. Panel (b) shows the share of breakfast cereal purchases made at discounters, and panel (c) shows the average price paid for breakfast cereal. Log-price outcomes are multiplied by 100, so coefficients in panels (a) and (c) are reported in log points. Vertical lines are 95% confidence intervals.

[10] Specifically, the flat pre-entry coefficients for the local price index in panel (a) are mechanical, because discounter products enter the local choice set only from the recorded entry year onward. For the purchase outcomes in panels (b) and (c), entry is measured using annual snapshots of store locations, whereas household purchases are observed throughout the year. The recorded entry year therefore need not coincide exactly with the date on which a store begins operating, which can generate some movement immediately before recorded entry. Consistent with this interpretation, when we assign treatment one year earlier, the joint pre-trend test for these outcomes no longer rejects the null of no pre-trends.

[11] Access to discounters increased substantially across income groups and in both urban and non-urban areas, while all groups increasingly shopped at discounters; see Appendix E.2.

[12] Arcidiacono et al. (2020) study incumbent supermarket price responses to Walmart entry. In our setting, the major traditional retailers follow national pricing policies, so local entry provides little scope for a local incumbent-price response. Consistent with this, an event study restricting the local cereal price index to non-discounter products shows essentially parallel treated and counterfactual price paths following entry; see Appendix E.1.

In summary, the growing market penetration of the discounters coincided with substantial declines in both retail and manufacturer concentration in the breakfast cereal market, a pattern typical of other fast-moving consumer goods categories. At the local level, discounter entry expands access to lower-priced cereals, increases the share of purchases made at discounters, and lowers prices paid. To evaluate how the rise of discounters affected market power and economic surplus, we turn to a model of demand and supply.

3 A Model of the Breakfast Cereal Market

3.1 Overview

We develop a model of the breakfast cereal market that allows us to recover retail and manufacturer markups and simulate counterfactual outcomes absent the rise of discounters. On the demand side, households choose both a product and a retailer. The supply side captures horizontal competition among retailers and manufacturers, as well as vertical bargaining between retailers and branded-good manufacturers. The timing is as follows: in each year–quarter market, manufacturers and retailers negotiate wholesale markups and retailers set retail prices. Household then choose either a product–retailer pair or the outside good.

3.2 Demand

Consumer options are indexed by j, where $j = [k(j), r(j)]$ denotes a product–retailer combination. Let i index a household-week choice occasion, with $h = h(i)$ denoting the household and $t = t(i)$ the year–quarter market. On each occasion, the household chooses among the available breakfast cereal options, $\mathcal{J}_t$, and the outside option, $j = 0$, corresponding to shopping without purchasing breakfast cereal.

The utility from option $j \in \mathcal{J}_t$ is

$$U_{ij} = \underbrace{\boldsymbol{\theta}_2 \mathbf{x}_{jt} + \Delta\xi_{jt}}_{\delta_{jt}} + \underbrace{\left[-\alpha_i \tilde{p}_{jt} + \sigma^{\phi}\nu_i^{\phi} + \textstyle\sum_{l\in\mathcal{L}} \sigma^l \nu_h^l x_j^l + \boldsymbol{\lambda}'\mathbf{d}_{hj} - (\tau_0 + \tau_1 u_h)\log \text{dist}_{ir}\right]}_{\mu_{ij}(\boldsymbol{\theta}_1)} + \epsilon_{ij} \quad (3.1)$$

where $\tilde{p}_{jt}$ is per-pack price,[13] $\mathbf{x}_{jt}$ includes observed option characteristics and market fixed effects, $[x_j^l]_{l\in\mathcal{L}}$ are time-invariant option characteristics, and dist_{ir} is the

[13]The pack price $\tilde{p}_{jt} \equiv p_{jt} \times (\text{kg})_j$, where p_{jt} is the price per kilogram and $(\text{kg})_j$ is the weight of j in kilograms.

distance to the nearest store of retailer $r = r(j)$. The term $\Delta\xi_{jt}$ is an option–market-level unobservable, $\mathbf{d}_{hj}$ collects interactions between selected option characteristics and household demographics, and u_h indicates whether the household resides in an urban area.[14] The price coefficient is

$$\alpha_i = \exp(\bar{\alpha} + \alpha^y y_{ht} + \sigma^\alpha \nu_h^\alpha).$$

The taste shocks $\boldsymbol{\nu}_i = (\nu_h^\alpha, \nu_i^\phi, [\nu_h^l]_{l\in\mathcal{L}})$ have independent standard normal components, $\boldsymbol{\sigma} = (\sigma^\alpha, \sigma^\phi, [\sigma^l]_{l\in\mathcal{L}})$ are the corresponding scaling terms, and y_{ht} is annual equivalized household income.[15] The parameters of the model are $\boldsymbol{\theta} = (\boldsymbol{\theta}_1, \boldsymbol{\theta}_2)$, where $\boldsymbol{\theta}_1 = (\bar{\alpha}, \alpha^y, \boldsymbol{\sigma}, \boldsymbol{\lambda}, \tau_0, \tau_1)$ denotes parameters in the household-week specific term μ_{ij} and $\boldsymbol{\theta}_2$ denotes parameters in mean utility δ_{jt}.

For all inside goods, utility includes a household-level random coefficient on the inside-good constant, ν_h^1, capturing persistent tastes, and a household-week shock, ν_i^ϕ, capturing short-run variation in whether the household wishes to buy breakfast cereal.

We interpret the outside good as grocery shopping without purchasing breakfast cereal. The outside good is a composite option encompassing the same set of retailers as the inside options. We normalize the mean utility of the outside option so that $U_{i0} = \epsilon_{i0}$, implying that δ_{jt} is interpreted as the mean utility difference between option j and the outside good.

The mean utility δ_{jt} also captures non-cereal factors that influence the choice of retailer r for cereal, such as utility from other services or product categories co-purchased with cereal. In Section 6, we outline the assumptions under which it is possible to separately identify changes over time in cereal-specific and non-cereal components of mean utility for each retailer.

We assume the idiosyncratic term ϵ_{ij} follows a Type I Extreme Value distribution. The probability that household $h(i)$ chooses option $j \in \mathcal{J}_t$ on choice occasion i is

$$s_{ij} = s_j(\boldsymbol{\delta}_t, \boldsymbol{\mu}_i(\boldsymbol{\theta}_1)) = \frac{\exp(\delta_{jt} + \mu_{ij}(\boldsymbol{\theta}_1))}{1 + \sum_{j'\in\mathcal{J}_t} \exp(\delta_{j't} + \mu_{ij'}(\boldsymbol{\theta}_1))}.$$

Integrating over the distribution $F_t(\boldsymbol{\mu}|\boldsymbol{\theta}_1)$ gives the market share of option j in market t: $S_{jt} = S_j(\boldsymbol{\delta}_t, \boldsymbol{\theta}_1) = \int_\mu s_j(\boldsymbol{\delta}_t, \boldsymbol{\mu}) \mathrm{d}F_t(\boldsymbol{\mu}|\boldsymbol{\theta}_1)$. We approximate the integral by simulation, using the household data as described in Section 4.1 (see Appendix I.1). To obtain demand q_{jt} in units of weight, as in Section 3.3, we multiply the market

[14] We construct the urban indicator from the Rural–Urban Classification in the ONS Postcode Directory (ONS, 2021). See Appendix A.

[15] Household income is reported in bands; we assign band midpoints and deflate by the all-items CPI (ONS, 2023b). Where a household's income is missing in a year, we impute it from an adjacent year using growth in mean real household disposable income (ONS, 2023a)

share of j by its weight: $q_{jt} = M \times (\text{kg})_j \times S_{jt}$, where M is the market size in terms of number of consumers.[16]

3.3 Supply

In this subsection, we condition on a given market and suppress the market subscript t. Define the retailer and manufacturer additive markups as $\Gamma_j^R = p_j - w_j - c_j^R$ and $\Gamma_j^F = w_j - c_j^F$, respectively, where w_j is the wholesale price and c_j^R and c_j^F are retailer and manufacturer marginal costs. We collect these markups in the vectors $\boldsymbol{\Gamma}^R = (\Gamma_j^R)_{j\in\mathcal{J}}$ and $\boldsymbol{\Gamma}^F = (\Gamma_j^F)_{j\in\mathcal{J}}$. Let $q_j(\boldsymbol{p}, \mathcal{J})$ denote market demand, measured in kilograms, for option j, given the vector of retail prices $\boldsymbol{p}$ and choice set $\mathcal{J}$. Let the manufacturer of product j be denoted $f(j)$. At retail prices $\boldsymbol{p}$ and manufacturer markups $\boldsymbol{\Gamma}^F$, the profits of retailer r and manufacturer f are

$$\pi_r(\boldsymbol{p}, \boldsymbol{\Gamma}^F) = \textstyle\sum_{j\in\mathcal{J}_r}(p_j - \Gamma_j^F - c_j)q_j(\boldsymbol{p}, \mathcal{J}), \text{ and} \tag{3.2}$$

$$\pi_f(\boldsymbol{p}, \boldsymbol{\Gamma}^F) = \textstyle\sum_{j\in\mathcal{J}_f}\Gamma_j^F q_j(\boldsymbol{p}, \mathcal{J}), \tag{3.3}$$

respectively, where $\mathcal{J}_r$ denotes the set of options sold by retailer r, $\mathcal{J}_f$ is the set of options supplied by manufacturer f (i.e., product–retailer pairs for products it produces), and $c_j = c_j^R + c_j^F$ is total marginal cost.

Retailer r sets prices to maximize profits in equation (3.2) treating manufacturer markups and rival retailers' prices as given. The first-order conditions for retail prices are

$$\boldsymbol{\Gamma}^R(\boldsymbol{p}) = \boldsymbol{\Delta}(\boldsymbol{p})^{-1}\boldsymbol{q}(\boldsymbol{p}, \mathcal{J}), \tag{3.4}$$

where $\boldsymbol{\Delta}$ is a $|\mathcal{J}|\times|\mathcal{J}|$ matrix of (negative) demand own- and cross-price derivatives, with cross-price derivatives set to zero for pairs of options sold by different retailers.

We assume static Bertrand–Nash pricing by retailers. The UK's competition authorities conducted several in-depth investigations into the supermarket industry over our sample period, including two market investigations (CC, 2000, 2008) and two merger inquiries (CC, 2003; CMA, 2019). None of these investigations concluded that the grocery retailers were engaged in market-wide coordination, providing support for our Bertrand–Nash pricing assumption.[17]

[16] We define the potential market as household-weeks with a grocery shopping trip. In each year–quarter, market size equals the number of households multiplied by ten, which is the average number of weeks per quarter in which a household undertakes a grocery shopping trip in our sample. Since M enters demand and its derivatives multiplicatively, and marginal costs are constant in quantity, it cancels from the markup equations and can be normalized to one for this purpose.

[17] For example, CC (2008, para. 8.40) conclude that "we did not find that grocery retailers were engaged in tacit coordination." More recently, CMA (2019, para. 9.36) state that "overall, we

Manufacturer markups are determined through bilateral negotiations between the retailer–manufacturer pairs $n = (r, f) \in \mathcal{N}$, where $\mathcal{N}$ is the set of trading pairs, treating retail prices as given.[18] The disagreement point for pair n is that retailer r no longer stocks manufacturer f's products, implying quantity gains from trade for each $j \in \mathcal{J}$ given by $\Delta_n q_j(\boldsymbol{p}, \mathcal{J}) = q_j(\boldsymbol{p}, \mathcal{J}) - q_j(\boldsymbol{p}, \mathcal{J} \setminus \mathcal{J}_n)$ where $\mathcal{J}_n$ is the set of options covered in negotiation n.[19] Holding retail prices $\boldsymbol{p}$ fixed, changes in wholesale markups redistribute the gains from trade without changing their total. A single negotiating instrument is therefore sufficient to determine how this surplus is divided. We assume each retailer–manufacturer pair negotiates over the per-unit manufacturer markup Γ^F_n, which is common across all $j \in \mathcal{J}_n$. Thus, for each option j, $\Gamma^F_j = \Gamma^F_{n(j)}$.[20]

The Nash Bargaining problem for pair n is

$$\max_{\Gamma^F_n \geq 0} \left[\sum_{j' \in \mathcal{J}_r} (p_{j'} - \Gamma^F_{n(j')} - c_{j'}) \Delta_n q_{j'}(\boldsymbol{p}, \mathcal{J}) \right]^{(1-b_n)} \left[\sum_{n' \in \mathcal{N}_f} \sum_{j' \in \mathcal{J}_{n'}} \Gamma^F_{j'} \Delta_n q_{j'}(\boldsymbol{p}, \mathcal{J}) \right]^{b_n} \quad (3.5)$$

where $\mathcal{N}_f$ denotes the set of bilateral negotiations involving manufacturer f and b_n and $(1 - b_n)$ represent the bargaining power of the manufacturer and retailer respectively. The solution to equation (3.5) is

$$\rho_n \textstyle\sum_{j' \in \mathcal{J}_r} (p_{j'} - \Gamma^F_{n(j')} - c_{j'}) \Delta_n q_{j'}(\boldsymbol{p}, \mathcal{J}) = \sum_{n' \in \mathcal{N}_f} \Gamma^F_{n'} \sum_{j' \in \mathcal{J}_{n'}} \Delta_n q_{j'}(\boldsymbol{p}, \mathcal{J}), \quad (3.6)$$

where $\rho_n = b_n/(1 - b_n)$ is the manufacturer's relative bargaining power. This condition links the two parties' gains from trade according to their relative bargaining power. It implies that a manufacturer's markups are increasing in its *leverage*: the extent to which the gain from trade is high for the retailer relative to the manufacturer. For private-label products, we set $b_n = 0$, which implies $\Gamma^F_n = 0$, so retailers

considered the evidence to be more consistent with competition and we therefore find that there is no pre-existing coordination in the markets for in-store groceries." During our sample period, we are only aware of investigations involving collusion in two grocery categories: dairy (OFT case number CE/3094-03, 2011) and cigarettes (OFT case number CE/2596-03, 2010).

[18] The simultaneous approach to retail prices and manufacturer markups is used in Draganska et al. (2010), Ho and Lee (2017), Crawford et al. (2018) and Noton and Elberg (2018). An alternative assumption, with greater computational cost, is that wholesale and retail prices are determined sequentially (see Bonnet et al., 2025); to make the sequential model computationally feasible it is necessary to focus on a small number of options. See Lee et al. (2021a) for a discussion.

[19] An alternative assumption for the disagreement point is that the retailer no longer stocks product j, while continuing to stock all other products sold by the manufacturer. Since disagreement points are out-of-equilibrium behavior there is no direct evidence for this assumption. However, we note that CC and CMA investigations detail cases where retailers threaten to delist a range of a manufacturer's products wider than just a single product. See also Bonnet et al. (2025), who find their results are robust to alternative disagreement point specifications.

[20] As these are per-unit, the manufacturer's per-pack markups for $j \in \mathcal{J}_n$ are proportional to pack quantity, $(\text{kg})_j$.

set prices optimizing against marginal cost, c_j. We use the Nash-in-Nash solution concept: in equilibrium the vector of negotiation-level manufacturer markups, $\boldsymbol{\Gamma}^F_{\mathcal{N}} = (\Gamma^F_n)_{n\in\mathcal{N}}$, solves the system of Nash bargaining problems defined in equation (3.5) for all $n \in \mathcal{N}$.[21] To express the bargaining problem compactly, let $\boldsymbol{A}(\boldsymbol{p})$ be the $|\mathcal{N}| \times |\mathcal{J}|$ matrix where row n gives quantity gains from trade for all options sold by retailer $r(n)$ in the negotiation (with zeroes for other options), i.e., the element in row n and column j is

$$A_{nj}(\boldsymbol{p}) = 1[j \in \mathcal{J}_{r(n)}] \times \Delta_n q_j(\boldsymbol{p}, \mathcal{J}).$$

Let $\boldsymbol{B}(\boldsymbol{p})$ denote the $|\mathcal{N}| \times |\mathcal{N}|$ matrix giving the manufacturer's quantity gain for products in the column negotiation if the row negotiation n is agreed, i.e., the element in row n and column n' is

$$B_{nn'}(\boldsymbol{p}) = 1[f(n) = f(n')] \times \textstyle\sum_{j'\in\mathcal{J}_{n'}} \Delta_n q_{j'}(\boldsymbol{p}, \mathcal{J}).$$

The system of $|\mathcal{N}|$ Nash bargaining solutions in (3.6) is

$$\boldsymbol{\Gamma}^F_{\mathcal{N}} = \boldsymbol{B}(\boldsymbol{p})^{-1}\boldsymbol{\rho}\boldsymbol{A}(\boldsymbol{p})(\boldsymbol{p} - \boldsymbol{\Gamma}^F - \boldsymbol{c}), \tag{3.7}$$

where $\boldsymbol{\rho}$ is a diagonal matrix of ρ_n terms. Equilibrium retail prices and manufacturer markups are obtained when (i) the $|\mathcal{J}|$ retail pricing first-order conditions in (3.4) and (ii) the $|\mathcal{N}|$ bargaining solutions in (3.7) are jointly satisfied.

We assume that private-label manufacturers have zero bargaining power. This is motivated by the fact that, for a private-label product, a retailer can switch producers without the customer noticing, unlike for a branded good. If there are multiple private-label producers with identical marginal costs, the Nash Bargaining negotiation with a private-label supplier can be viewed as constrained by an "outside option" (in the sense of Binmore et al. (1989)) of sourcing the product from an alternative supplier at marginal cost. This delivers the same outcome as setting $b = 0$ for private-label manufacturers. This logic applies when a branded manufacturer supplies private-label goods: its private-label lines are in a different bargaining

[21] We use the standard Nash-in-Nash solution in which each bargaining problem is based on gains from trade relative to the disagreement point of not trading. An alternative specification would include an "outside option" in the sense of Binmore et al. (1989), where the retailer can threaten to replace a supplier with an alternative supplier that is not one of its trading partners in equilibrium; see, for example, Ho and Lee (2019). For traditional retailers, this alternative specification is unlikely to be important because they typically stock products from the full set of major branded manufacturers. For discounters, outside replacement options may be more relevant, since they stock fewer branded products and could in principle replace one branded supplier with another. We do not model such replacement options because only a small set of discounter negotiations involve branded manufacturers, and branded products sold through discounters account for a small share of the overall breakfast cereal market.

position from their branded goods and therefore do not increase its bargaining leverage vis-à-vis the retailer and in negotiations over branded goods. In practice, we also do not observe when manufacturers produce each retailer's private-label cereals.

Retailers may take cross-category effects (i.e., substitution to non-cereals) into account when pricing breakfast cereals (Thomassen et al., 2017). Our model accommodates this (see Appendix F). Under this interpretation, breakfast cereal retail markups equal the difference between price and the retailer's marginal cost, net of the marginal benefit—through increased profits in other categories—from generating an additional unit of breakfast cereal demand. We return to this point when discussing our counterfactual analysis in Section 6. The simultaneous timing assumption in our model facilitates the incorporation of cross-category effects. This is because it does not require us to keep track of the impact of changes in wholesale prices (or the effect of bargaining breakdowns) on prices in other categories.

Some branded manufacturers in the breakfast cereal category sell products in other product categories to supermarkets. We assume that these manufacturers do not link negotiations across categories.

Alternative supply models When $b_n > 0$, the bargaining model implies double marginalization and negotiations are not bilaterally efficient. There is empirical evidence of double marginalization—for example, Luco and Marshall (2020) find support for it in the US branded soft drink market, and Noton and Elberg (2018) document wholesale prices above marginal costs in the Chilean branded retail coffee market—but vertical contracts can also be designed to avoid it. We therefore also present results under two bilaterally efficient benchmarks: retailer pricing and manufacturer pricing.

Under *retailer pricing*, manufacturer markups are zero ($\Gamma_j^F = 0$ for all j), and total markups are determined by the retailers' pricing condition (3.4). This case is nested in our model when $b_n = 0$ for all n. It is also consistent with a two-stage model of negotiated two-part tariffs in which manufacturers and retailers simultaneously negotiate wholesale prices w_j and transfers, after which retailers set retail prices. Wholesale prices are set equal to the manufacturer marginal cost to ensure bilateral efficiency, while transfers determine the division of surplus. See Rey and Verge (2019) and Appendix G. Thus, although our baseline assumes simultaneous determination of wholesale and retail prices, it is consistent with a sequential model of two-part tariffs in the special case in which $b_n = 0$. For $b_n > 0$, however, simulta-

neous timing does not readily accommodate two-part tariffs, as noted by Panhans (2024).

Under *manufacturer pricing*, manufacturers set retail prices directly. Each manufacturer simultaneously sets the retail price of its options to maximize total variable profits, optimizing against marginal costs, c_j. Manufacturers may share some of the resulting surplus with retailers through fixed transfers, which we do not model. This benchmark is not nested within our bargaining model: $b_n = 1$ does not correspond to manufacturer pricing.[22]

4 Identification and Estimation

We estimate demand in the first stage, without imposing any supply-side restrictions. In the second stage, we use the estimated demand system to recover the supply-side parameters. We discuss our identification strategy for each stage in turn.

4.1 Demand Parameters

We estimate demand parameters using a method-of-moments estimator that combines market-level moments with household-level micro moments. The estimator matches market shares and exploits the panel structure of the data to identify preference heterogeneity. This delivers the substitution patterns and elasticities required for the supply-side pricing and bargaining model.

Market-level moments We assume that product attributes and assortment decisions are determined prior to the realization of the option–market demand shock, $\Delta\xi_{jt}$, but allow prices to be correlated with this shock.[23] Following Berry (1994) and Berry et al. (1995), given a parameter vector $\boldsymbol{\theta} = (\boldsymbol{\theta}_1, \boldsymbol{\theta}_2)$, we solve for the mean utilities $\boldsymbol{\delta}(\boldsymbol{\theta}_1)$ by equating observed market shares, S_{jt}, with model-predicted shares: $S_{jt} = S_j(\boldsymbol{\delta}_t(\boldsymbol{\theta}_1), \boldsymbol{\theta}_1)$ for all (j, t). We then recover the structural errors as $\Delta\xi_{jt}(\boldsymbol{\theta}) = \delta_{jt}(\boldsymbol{\theta}_1) - \boldsymbol{x}_{jt}\boldsymbol{\theta}_2$. We assume that the unobserved utility component satisfies $\mathbb{E}(\Delta\xi_{jt}(\boldsymbol{\theta})|\mathbf{z}_{jt}) = 0$. This yields the market-level moments

[22]Note that, when $b_n = 1$ for all n, manufacturers make take-it-or-leave-it offers, but no equilibrium exists because retailers have an incentive to deviate from zero retail markups whenever demand is positive (see Lee et al., 2021b).

[23]Observed breakfast cereal attributes are fixed over time, so the product-attribute assumption is automatically satisfied. We assume assortment decisions are made prior to pricing decisions. For recent work that explicitly addresses demand estimation with endogenous product assortment, see Aguirregabiria et al. (2023).

$\mathbf{g}_A(\boldsymbol{\theta}) = |N_A|^{-1} \sum_{jt} \mathbf{z}'_{jt} \Delta\xi_{jt}(\boldsymbol{\theta})$, where $|N_A|$ is the number of option-market observations. The instrument vector $\mathbf{z}_{jt}$ includes observed characteristics $\boldsymbol{x}_{jt}$ (indicators for year–quarters, retailers, and products, used as their own instruments), eight cost-shifter instruments,[24] and BLP-style instruments constructed from observable characteristics of rival options to capture the intensity of competition (see Berry et al. (1995), Gandhi and Houde (2019), and Appendix H for details). Distance does not enter $\mathbf{z}_{jt}$ and therefore plays no role in identifying $\boldsymbol{\theta}_2$. Instead, its coefficient is identified from household-level micro moments.

Micro moments The micro moments are particularly informative about taste heterogeneity, governed by $\boldsymbol{\theta}_1$. For moment m and household h, let Y_h^m denote the observed moment and $y_h^m(\boldsymbol{\theta})$ its model-predicted counterpart. The population moment condition is

$$\mathbb{E}(Y_h^m) - \mathbb{E}(y_h^m(\boldsymbol{\theta})) = 0, \quad \forall m$$

where expectations are over households.

We compute the observed micro moments using the full consumer dataset. Evaluating predicted moments on the full sample is computationally costly, so following O'Connell et al. (2025), we use the full dataset for the observed moments and a subsample for predicted moments. Let N_H denote the full set of households. We draw a random subset N_H^* of 2000 households and, for each household, three household-weeks from different markets, yielding 6000 household-weeks. For each sampled household-week, we observe demographics and location and simulate the unobserved taste shocks entering $\boldsymbol{\mu}_i$.

The sample analog of the mth micro moment is

$$\mathrm{g}_M^m(\boldsymbol{\theta}) = \frac{1}{|N_H|} \sum_{h \in N_H} Y_h^m - \frac{1}{|N_H^*|} \sum_{h \in N_H^*} y_h^m(\boldsymbol{\theta}). \tag{4.1}$$

We use three sets of micro moments; Appendix H gives details.

The first set exploits persistence in household choices across choice occasions to identify the spread parameters of the random coefficients. These moments measure whether a household persistently chooses options with high or low values of the relevant characteristics. We construct one moment for each dimension with persistent taste heterogeneity: price, the inside-good indicator, retailer, brand, and three moments for cereal base.[25] These panel moments use repeated household choices to

[24] These are the eight input prices in $\mathbf{x}_{\mathbf{jt}}^{\mathbf{s}}$ listed in Table 5.1(c), from maize price × corn base, through to sugar price.

[25] For retailers and brands, we constrain the spread parameters σ^l to be common within each group and aggregate across retailers and brands, respectively, to construct one moment for each.

identify preference heterogeneity and play a role similar in spirit to second-choice data when such data are unavailable (Thomassen and Zhang, 2026).[26]

The second set matches the relationship between choices and household characteristics. These moments include the interaction between option price and household income; distance between the household and the retailer, separately for households in major urban areas and other households; the interaction between the inside-good indicator and urban residence; and the interaction between a chocolate-cereal indicator and the presence of children in the household.

The coefficient on distance is identified from variation in households' distances to retailers and their retailer and product choices. The key identifying assumption is that, conditional on observed household characteristics, market and retailer controls, and the realized store network, households living closer to a retailer do not systematically have stronger unobserved preference for shopping at that retailer. As Cao et al. (2024) emphasize, this assumption can fail if retailers choose locations in response to spatial variation in consumer tastes. Three features of our setting help mitigate this concern. First, the model conditions on observed household characteristics, including household income, household composition, and urban residence, that affect product and shopping choices. This reduces the scope for sorting on observable household characteristics to be absorbed by the distance coefficient. Second, repeated household choices allow us to model persistent unobserved heterogeneity in retailer preferences, so persistent differences in retailer affinity need not be attributed to distance sensitivity. Third, as discussed in Section 2, the location and expansion of UK grocery stores were materially shaped by planning restrictions and land-use controls, including the 2010 Controlled Land Order, which relaxed constraints on sites particularly well suited to discounters.

The third set of micro moments is informative about the household-week-specific shock to the utility of the inside good, governed by σ^{ϕ}. These moments relate the probability of purchasing any breakfast cereal to household-specific measures of the local choice set. We use four such measures: the number of available options within 2 kilometers; an indicator for whether this number is positive; the distance to the 50th nearest option; and the distance to the 200th nearest option. As with the distance moments, we treat these choice-set measures as predetermined with respect to idiosyncratic taste shocks, conditional on the market, retailer, and the realized store network.

For cereal bases, we allow separate spread parameters for three groups and construct one moment for each group.

[26] The demand model treats the household as a single decision-making unit and therefore abstracts from preference heterogeneity across household members. The random coefficients are interpreted as persistent household-level taste heterogeneity.

Estimation We estimate utility parameters by solving $\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \mathbf{g}(\boldsymbol{\theta})'\mathbf{W}\mathbf{g}(\boldsymbol{\theta})$ where $\mathbf{g}(\boldsymbol{\theta}) = [\mathbf{g}_A(\boldsymbol{\theta}), \mathbf{g}_M(\boldsymbol{\theta})]$ stacks the market-level and micro moments, with $\mathbf{g}_M(\boldsymbol{\theta}) = [\mathbf{g}_M^m(\boldsymbol{\theta})]_{\forall m}$. The weighting matrix $\mathbf{W}$ is block diagonal, $\boldsymbol{W} = \text{diag}(\boldsymbol{W}_A, \boldsymbol{W}_M)$. For the market-level moments, $\boldsymbol{W}_A$ is the standard 2SLS weighting matrix. For the micro moments, $\boldsymbol{W}_M$ is diagonal, with each entry equal to the inverse square of the observed component of the corresponding moment. This normalization expresses the micro moment discrepancies as unit-free percentage deviation between predicted and observed values (see Low and Meghir, 2017). See Appendix I for further details of the estimator and Appendix J for details on how we compute standard errors.

4.2 Supply-Side Parameters

The supply-side parameters consist of cost and bargaining parameters. To identify these, we combine the model's equilibrium pricing conditions (3.4) and (3.7) to obtain

$$\boldsymbol{\Gamma}^F_{\mathcal{N},t} = \boldsymbol{\rho}\boldsymbol{\ell}_t \quad \text{where} \quad \boldsymbol{\ell}_t \equiv \boldsymbol{B}_t^{-1}\boldsymbol{A}_t\boldsymbol{\Gamma}^R_t(\boldsymbol{p}_t). \tag{4.2}$$

The leverage term $\boldsymbol{\ell}_t$ is known given the estimated demand system and observed prices. It depends on retailer markups, $\boldsymbol{\Gamma}^R_t$, which we recover using equation (3.4). The corresponding option-level manufacturer markups, $\boldsymbol{\Gamma}^F_t$, are obtained by assigning each negotiated markup $\Gamma^F_{n,t}$ to the options covered by negotiation n.

We specify total marginal cost for option j in market t as

$$c_{jt} = \boldsymbol{\gamma}\boldsymbol{x}^s_{jt} + \omega_{jt}, \tag{4.3}$$

where $\boldsymbol{\gamma}$ are cost parameters, $\boldsymbol{x}^s_{jt}$ is a vector of observed cost shifters, and ω_{jt} is an unobserved cost shock. The cost-shifter vector $\boldsymbol{x}^s_{jt}$ includes eight cereal input prices, quarter effects, year effects interacted with retailer dummies, year effects interacted with indicators for whether the product is private label or branded, and product effects. These controls absorb broad cost changes at the retailer–year level and common time variation by product ownership type, so that ω_{jt} captures the residual option–market cost shock.

By definition $\Gamma^R_{jt} + \Gamma^F_{n(j)t} = p_{jt} - c_{jt}$, so we can write

$$[p_{jt} - \Gamma^R_{jt}] = \rho_{n(j)}\ell_{n(j)t} + \boldsymbol{\gamma}\boldsymbol{x}^s_{jt} + \omega_{jt}, \tag{4.4}$$

where $\ell_{n(j)t}$ is the element of $\boldsymbol{\ell}_t$ corresponding to bargaining pair $n(j)$. We set $\rho_{n(j)} = 0$ for private-label products, consistent with pricing under a vertically integrated structure, and impose a common coefficient $\rho_{n(j)} = \rho$ across all branded retailer–manufacturer bargaining pairs.

We identify $(\rho, \boldsymbol{\gamma})$ from the condition that the residual cost shock is mean independent of the supply-side instruments $\mathbf{z}^s_{jt}$, conditional on the controls included in $\boldsymbol{x}^s_{jt}$, i.e., $\mathbb{E}[\omega_{jt}|\mathbf{z}^s_{jt}, \boldsymbol{x}^s_{jt}] = 0$. The leverage term depends on endogenous retailer markups, so excluded instruments are required. We construct these instruments from measures of the parties' product portfolios. For a bargaining pair (r, f), let the negotiated set be the set of product–retailer options sold by retailer r and supplied by manufacturer f, i.e., the options whose terms are determined in that bilateral negotiation. The instruments measure the size of this set relative to each party's overall portfolio. Specifically, we use: the number of options offered by the retailer and by the manufacturer; the number of options in the negotiated set; the share of the retailer's portfolio accounted for by the negotiated set; the share of the manufacturer's portfolio accounted for by this set; and the share of the manufacturer's options outside the negotiated set that belong to brands represented within the negotiated set. Because $\boldsymbol{x}^s_{jt}$ includes retailer–year effects and year effects interacted with whether the product is private label or branded, identification of the leverage coefficient relies on variation in these portfolio variables after netting out common retailer–year cost components and common time variation by product ownership type. The maintained exclusion restriction is that the portfolio instruments are orthogonal to the remaining option–market cost shock.[27]

For private-label goods, ρ would in principle be identifiable if supplier identities, and hence the corresponding leverage variable, were observed. In our data, however, we do not observe which manufacturer produces each private-label product and therefore cannot construct the leverage term. We instead set the private-label bargaining weight to zero, an assumption justified both by the theoretical arguments in Section 3.3 and the evidence from the UK's competition authority discussed in Section 2.

[27] When we allow for cross-category pricing effects (see Appendix F), equation (4.4) identifies an *effective* marginal cost, equal to technological marginal cost net of the marginal benefit—through increased profits in other categories—of generating an additional unit of breakfast cereal demand. This cross-category benefit reflects markup-weighted diversion to other categories. Retailer–year effects absorb any retailer-year component of this term, while the private-label/branded year effects absorb common time variation by product ownership type. The maintained exclusion restriction is therefore that, conditional on these controls, the portfolio instruments are orthogonal to the remaining option-level variation in this cross-category adjustment. This restriction is plausible because cross-category pricing incentives depend primarily on retailer-level markups and consumers' propensity to switch retailers when cereal purchases are displaced from the retailer, whereas the portfolio instruments primarily capture variation in the structure of individual retailer–manufacturer negotiations.

5 Model Estimates

5.1 Parameters

Table 5.1(a) reports estimates of the demand parameters. Households prefer geographically closer retailers, and the disutility of distance is 25% higher, on average, for those residing in major urban areas. Households with children have stronger preferences for breakfast cereals containing chocolate. We allow for unobserved preference heterogeneity in price sensitivity and tastes for breakfast cereal (i.e., inside options), brand, retailer, and cereal base. The estimated spread parameters are large and statistically significant across these dimensions. We do not report mean taste parameters, as these are absorbed by the product and retailer effects. We also allow for a household-time-varying shock to preferences for breakfast cereals, which has a substantial spread parameter (σ^{ϕ}).[28]

We estimate these parameters in part by targeting a set of micro moments. Table 5.1(b) reports the observed moments alongside their model-predicted counterparts, which closely matches across the persistence, demographic, and inside-option moments. The model also closely reproduces the relationship between purchase probabilities and travel distance to the nearest store selling an option, indicating that it captures the spatial pattern of household choices (see Appendix K.2).

Table 5.1(c) reports supply estimates from equation (4.4). Column (1) presents OLS results and column (2) our preferred IV estimates, which instrument leverage using retailer and manufacturer portfolio variables. The OLS leverage coefficient is small and negative, whereas the IV estimate is positive and statistically significant. This pattern is consistent with downward OLS bias from estimation error in retailer markups: estimated markups enter the dependent variable negatively but the leverage term positively through the retailer's gains from trade, inducing a mechanical negative correlation between measured leverage and the regression residual. In our main specification, the estimated bargaining parameter, $\hat{b} = \hat{\rho}/(1+\hat{\rho})$, is 0.58. The cost-shifter coefficients are consistent with input prices feeding into marginal costs: interactions with related cereal bases, such as maize for corn-based products, wheat for wheat- and bran-based products, and oats for granola, are positive and statistically significant, while some are less precisely estimated or have unexpected signs, perhaps reflecting imperfect mapping between inputs and products.

[28] As a diagnostic for time variation in price sensitivity, we also estimated a specification allowing the baseline price-sensitivity parameter to shift before and after 2010; the shift is economically small, and the implied elasticities are essentially unchanged.

Table 5.1: *Parameter estimates and model fit*

(a) Demand parameters

	Estimate	Standard error
Price		
Baseline ($\bar{\alpha}$)	1.69	0.10
× income (α^y)	-0.71	0.21
Spread (σ^α)	0.29	0.03
Inside option		
i spread (σ^ϕ)	11.93	1.99
h spread (σ^1)	6.67	1.45
Option attributes		
Brand (σ^2)	2.64	0.08
Retailer (σ^3)	3.39	0.10
Wheat (σ^4)	0.53	0.13
Rice, corn (σ^5)	1.64	0.14
Oats, bran (σ^6)	1.95	0.13
Option–household interactions		
Inside × urban (λ_1)	0.48	0.64
Chocolate × children (λ_2)	1.68	0.19
Log distance		
× non-urban (τ_0)	1.82	0.15
× urban (τ_1)	2.28	0.27
Product effects	Yes	
Retailer effects	Yes	
Market effects	Yes	

(b) Demand model fit: Micro moments

	Data	Model
Panel moments		
Price	0.169	0.167
Inside good	0.036	0.036
Brand	0.168	0.162
Retailer	0.370	0.350
Wheat	0.046	0.046
Rice, corn	0.047	0.047
Oats, bran	0.055	0.055
Demographic moments		
Price × income	0.759	0.762
Inside × urban	0.165	0.163
Chocolate × children	0.093	0.093
Log distance × non-urban	0.470	0.504
Log distance × urban	0.213	0.228
Inside option nest moments		
# opt. in 2km	53.7	56.3
1[# opt. in 2km> 0]	0.243	0.249
km to 50th nearest opt.	0.696	0.652
km to 200th nearest opt.	1.32	1.20

(c) Supply parameters

	(OLS)		(IV)	
	Estimate	Standard error	Estimate	Standard error
Bargaining parameter (b)	-0.088	0.051	0.580	0.044
Leverage (ρ)	-0.081	0.043	1.384	0.250
Maize price x corn base	0.089	0.048	0.081	0.014
Wheat price x wheat base	0.111	0.044	0.085	0.016
Rice price x rice base	0.021	0.071	0.002	0.024
Oats price x oats base	-0.031	0.031	-0.043	0.016
Wheat price x bran base	0.082	0.061	0.077	0.022
Maize price x multi base	0.026	0.105	-0.030	0.050
Oats price x granola base	0.089	0.032	0.090	0.015
Sugar price	0.025	0.021	0.002	0.010
Product effects	Yes		Yes	
Retailer–year effects	Yes		Yes	
Private-label–year effects	Yes		Yes	
Quarter effects	Yes		Yes	

Notes: Panel (a) reports demand parameter estimates. The cereal-base spread parameters group rice with corn, and oats with bran, multigrain, and granola. Panel (b) reports the observed and model-predicted values of the micro moments used in demand estimation. Panel (c) reports supply parameter estimates; the bargaining parameter applies to branded products, while private-label bargaining power is set to zero. Standard errors in panel (a) are clustered at the household level, and those in panel (c) are heteroskedasticity-robust. Weak-instrument diagnostics for demand estimation are reported in Appendix K.1. For the supply equation, the Lewis–Mertens robust weak-IV diagnostic (local-to-zero, absolute-bias version) is 62.8, exceeding the 5% critical value for a 5% absolute-bias threshold (24.9).

5.2 Markups and their Vertical Division

We estimate a sales-weighted average own-price elasticity of -7.02 for breakfast cereal options (i.e., product–retailer options), compared with -0.62 for the breakfast cereal category level. To facilitate comparison with the literature, we also construct brand-level elasticities by aggregating across all retailer-product options belonging to a brand. The average brand-level elasticity is -5.16.

Our option-level elasticities are larger in magnitude than the brand-level elasticities reported for the US in Nevo (2001) and Backus et al. (2021). Aggregating to the brand level narrows this gap, indicating that part of the difference reflects our more disaggregate product–retailer definition of the choice object. Our brand-level elasticities nevertheless remain somewhat larger in magnitude than those reported in these US studies, potentially reflecting differences in country, period, retailer structure, and product coverage. By contrast, our category-level elasticity is closer to that reported in Backus et al. (2021), although somewhat larger in magnitude.

Table 5.2 reports average option-level and brand-level elasticities for branded and private-label options, separately for traditional retailers and discounters, in 2002, 2011, and 2021.[29] Using equations (3.4), (4.2), and (4.4), we recover retailer and manufacturer additive markups, $\mathbf{\Gamma}_t^R$ and $\mathbf{\Gamma}_t^F$, and thus total marginal costs, defined as $\boldsymbol{c}_t \equiv \boldsymbol{p}_t - \left(\mathbf{\Gamma}_t^R + \mathbf{\Gamma}_t^F\right)$. The table summarizes these variables alongside Lerner indexes and the share of total markups accruing to retailers.

Several notable patterns emerge. First, in all product–retailer groups, own-price elasticities decline in magnitude over time. The average option-level elasticity across all options moves from about -8.8 in 2002 to -7.4 in 2021, with similar patterns for branded and private-label products and across traditional retailers and discounters. Brand-level elasticities also decline in magnitude over time.

Second, marginal costs decline, especially after 2011. For discounter private-label options, for example, average marginal costs fall from around £1.9/kg in 2002 to £1.1/kg in 2021. This pattern is consistent with economies of density associated with discounter store expansion (see Figure 2.1 and Holmes 2011), as well as improvements in supply-chain efficiency over time. Branded products typically have higher marginal costs than private-label products. This pattern is broadly

[29] Option-level elasticities depend not only on the price sensitivity of the households choosing a given option, but also on the option's price level and the substitution patterns it faces across products and retailers. In the estimated model, households with higher price sensitivity are more likely to choose private-label and discounter options (see Appendix L).

Table 5.2: *Average elasticities, costs and markups*

		Traditional retailers		Discounters		All options
		Branded	Private-label	Branded	Private-label	
2002	Option own-price elasticity	-9.24	-7.75	-11.45	-5.60	-8.79
	Brand own-price elasticity	-6.36	-6.38	-5.43	-5.29	-6.66
	Marginal cost c (£/kg)	2.35	2.14	2.72	1.91	2.25
	Total markup $\Gamma^R + \Gamma^F$ (£/kg)	2.63	1.14	1.37	0.83	2.28
	Lerner index ($\frac{\Gamma^R+\Gamma^F}{p}$)	0.55	0.36	0.33	0.31	0.51
	Retailer share (%)	51%	100%	42%	100%	63%
2011	Option own-price elasticity	-9.21	-7.05	-9.22	-5.65	-8.51
	Brand own-price elasticity	-6.38	-5.96	-5.10	-4.84	-6.22
	Marginal cost c (£/kg)	2.44	1.74	1.95	1.76	2.25
	Total markup $\Gamma^R + \Gamma^F$ (£/kg)	2.84	1.33	1.63	1.01	2.40
	Lerner index ($\frac{\Gamma^R+\Gamma^F}{p}$)	0.56	0.44	0.46	0.38	0.52
	Retailer share (%)	55%	100%	49%	100%	65%
2021	Option own-price elasticity	-8.56	-5.40	-7.66	-4.54	-7.43
	Brand own-price elasticity	-6.02	-4.53	-5.68	-3.68	-4.93
	Marginal cost c (£/kg)	2.12	1.25	1.78	1.12	1.80
	Total markup $\Gamma^R + \Gamma^F$ (£/kg)	2.57	1.22	2.07	1.04	2.13
	Lerner index ($\frac{\Gamma^R+\Gamma^F}{p}$)	0.56	0.52	0.54	0.52	0.55
	Retailer share (%)	52%	100%	48%	100%	67%

Notes: The table reports average own-price elasticities (option and brand level), total marginal costs, total additive markups, Lerner indexes, and retailer shares of total markups in 2002, 2011 and 2021. For private-label products, the retailer share is 100%. Marginal costs and markups are expressed in 2021 £ per kg.

consistent with evidence from CC (2000), which suggests that private-label cereals have lower production or sourcing costs.[30]

Third, average Lerner indexes rise from 2002 to 2021, driven primarily by increases for private-label products and branded products sold by discounters.

These patterns—declining marginal costs, less elastic demand, and rising Lerner indexes—are broadly consistent with recent evidence from US consumer product markets based on scalable demand estimation and manufacturer pricing (Döpper et al., 2025; Atalay et al., 2023). Our focus on a single category allows us to trace these mechanisms in greater detail and, in particular, to show how vertical relationships and retailer conduct shape the evolution of markups.

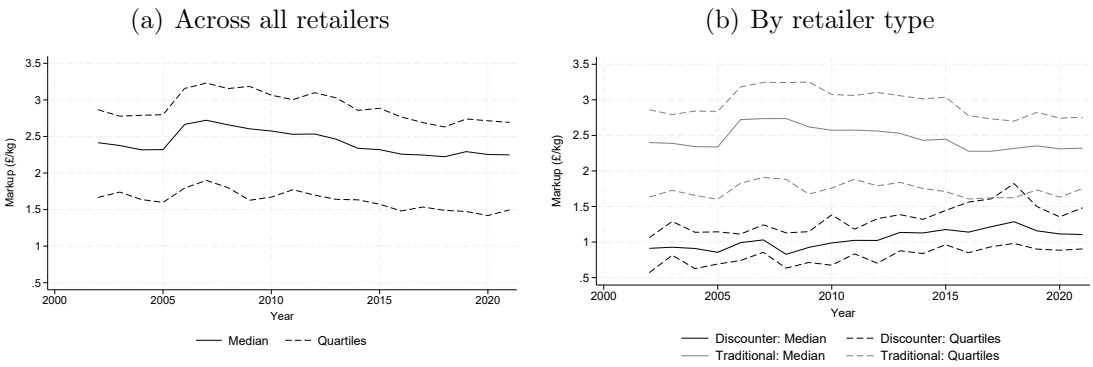
Finally, for branded products, retailers account for approximately half of total vertical markups, with manufacturers accounting for the remainder. This suggests that retailers obtain a substantial share of markups, consistent with significant market power in both pricing and negotiations with manufacturers. Our findings are broadly consistent with direct evidence in Alvarez-Blaser et al. (2025), who use

[30]CC (2000, para. 7.206) report that "most [retail] companies agreed that higher margins could be obtained from own-label products because of lower costs."

cost, wholesale, and retail price data for a large global manufacturer of non-durable household products and show that, in the UK, retailers capture a large share of vertical markups, with average retail markups exceeding manufacturer markups.

In Figure 5.1, we summarize the evolution of the markup distribution (measured by total vertical additive markups) from 2002 to 2021. Panel (a) shows the interquartile range and median of markups over time. Markups follow a shallow inverted U-shape, peaking in 2008. The subsequent decline coincides with the main phase of discounter expansion. Panel (b) separates options by whether they are sold by traditional retailers or discounters. The distribution for the traditional retailers closely mirrors that in panel (a). In contrast, markups for discounter products display a clear upward trend beginning around 2010. In that year, the 25th, 50th and 75th percentiles of the discounter markup distribution (per kilogram) were £0.67, £0.99 and £1.38, respectively—similar to their levels in 2002. By 2021, these had risen to £0.90, £1.11 and £1.48, indicating a substantial increase in discounter markups over the period. Thus, discounter expansion is associated with both lower average local prices (see Section 2) and rising markups on the products they sell, alongside declining markups at rival retailers. This pattern underscores the role of retail conduct in shaping the evolution of market power.

Figure 5.1: *Markups over time*



Notes: Graphs show the 25th, 50th and 75th percentiles of total vertical additive markups across all options (panel (a)) and separately across options sold by discounters and traditional retailers (panel (b)). Markups are expressed in 2021 £ per kg.

Each option's additive markup can be decomposed as:

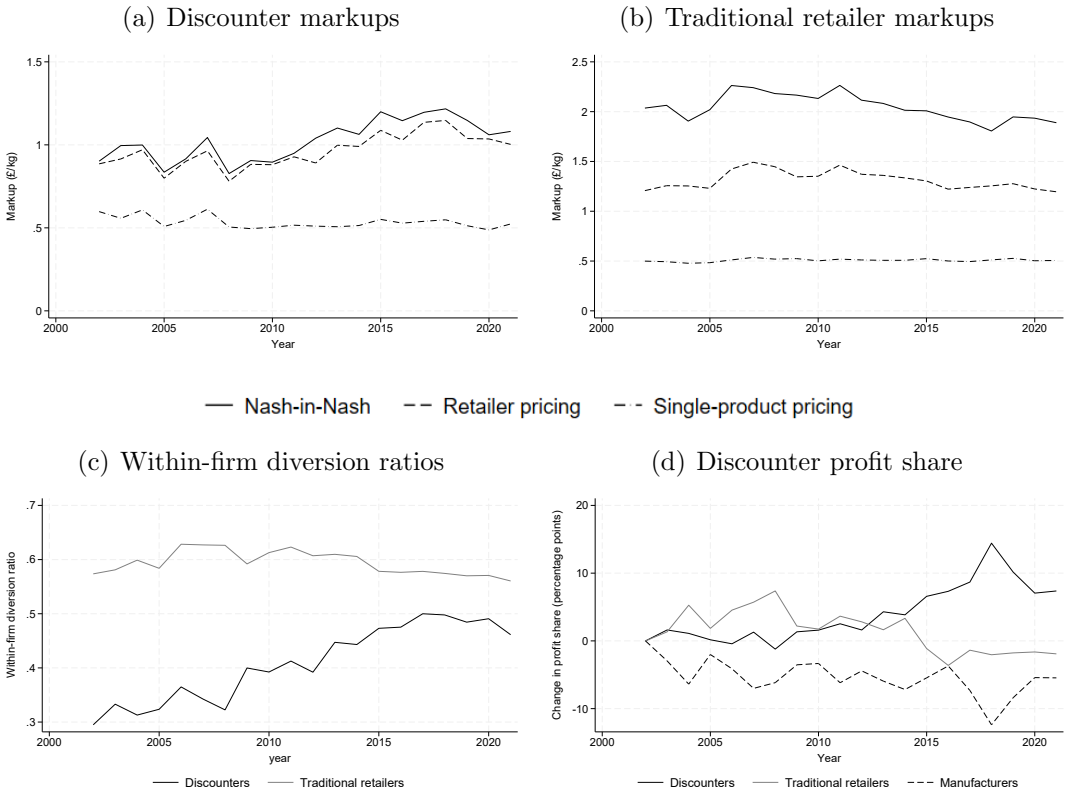
$$p - c = \tfrac{1}{\varepsilon} + \left(\Gamma^R - \tfrac{1}{\varepsilon}\right) + \Gamma^F,$$

where ε is the absolute value of the option's own-price semi-elasticity. The first component, $\frac{1}{\varepsilon}$, is the markup that would arise under single-product pricing. The term $\Gamma^R - \frac{1}{\varepsilon}$ is zero under single-product retailer pricing and captures the additional downstream markup generated by retailers internalizing diversion to other products

in their portfolios. Finally, Γ^F represents the markup accruing to manufacturers through bargaining with retailers.

Figure 5.2 presents the decomposition for the sales-weighted average additive markup for discounters (panel (a)) and traditional retailers (panel (b)). Total markups exhibit different patterns across retailer types. Discounter markups increase from 2010, while markups for traditional retailers first rise and then fall, with the decline coinciding with the expansion of discounters. These changes are driven almost entirely by retailer portfolio effects (i.e., the difference between retailer and single-product pricing). Manufacturer market power plays a minor role for discounters, as discounters primarily stock private-label products. In contrast, manufacturers account for a larger share of markups for traditional retailers, reflecting the importance of branded products in their assortments.

Figure 5.2: *Markup decomposition, within-firm diversion ratios and profits*



Notes: The top two panels show sales-weighted averages of additive markups under alternative supply models for options sold by discounters (panel (a)) and traditional retailers (panel (b)). Panel (c) shows revenue-weighted average within-firm diversion-ratios for options sold by discounters and traditional retailers. Panel (d) shows the profit share accruing to discounters, traditional retailers and manufacturers of branded products, expressed as differences from their 2002 shares. Markups are expressed in 2021 £ per kg.

Panel (c) further illustrates the role of within-retailer portfolio effects using only the demand estimates. It reports within-firm diversion ratios over time, separately for options sold by discounters and traditional retailers. For each option, we com-

pute the share of lost demand following a price increase that is diverted to other options sold by the same retailer, and we plot the revenue-weighted average across options. The diversion ratio for traditional retailers initially rises and then gradually declines, reflecting increasing competitive pressure from discounters as consumers become more likely to substitute to discounters. The corresponding diversion ratio for discounters rises strongly over time, indicating that, as their assortments and store networks expand, a larger share of demand lost by one product is recaptured by other products sold by the same retailer. Together with panels (a) and (b), this pattern shows that substantial within-firm diversion translates into significant downstream pricing power.

Panel (d) shows that these trends translate into discounters capturing an increasing share of total breakfast cereal profits at the expense of traditional retailers and branded manufacturers. Prior to 2008, discounters accounted for 4.0% of total profits, with the remainder split between non-discounter retailers (58.8%) and manufacturers of branded products (37.2%). By 2017–2021, the discounter share had risen by 8.9 percentage points to 12.9%, with other retailers' share falling by 4.9 percentage points and manufacturers' share declining by 4.0 percentage points. This shift in the division of vertical markups underlines the key role played by retailers in driving the evolution of market power over our sample period.

Our approach is related to De Loecker and Scott (2026), who emphasize the importance of accounting for downstream costs and combine demand- and production-based markup estimates in the beer industry to assess retail conduct. Their exercise summarizes the degree of substitutability between retailers with a single parameter, ranging from perfect substitutability, which delivers competitive retailing, to no substitutability, which delivers monopoly. This allows them to test for a highly competitive retail sector. Our model does not have a single parameter summarizing the degree of substitutability between retailers and therefore does not provide an equivalent test of perfect retail competition. In contrast to De Loecker and Scott (2026), we directly observe household panel choices and store locations, allowing us to estimate substitution across retailers using consumer–store distances. The estimated demand system implies imperfect substitutability between retailers, driven by spatial frictions and persistent retailer-specific preferences. Under our Bertrand–Nash pricing model, these substitution patterns imply economically meaningful downstream retail markups. The role of spatial frictions is consistent with prior store-choice estimates for the UK and US (CC, 2008; Ellickson et al., 2020).

6 Counterfactual Analysis

The goal of the counterfactual analysis is to measure the impact of discounter expansion between 2002 and 2021 on equilibrium outcomes. We do this by reverting different dimensions of the discounters' store networks and in-store primitives to their 2002 configuration, simulating market outcomes, and comparing them to the corresponding observed equilibrium. This approach isolates the impact of different dimensions of the discounters' expansion on market performance while avoiding extrapolation beyond the observed range of the data.

6.1 Counterfactual Specification

We consider three counterfactual scenarios that isolate different dimensions of the rise of the discounters. In each case, we solve for the resulting equilibrium wholesale and retail prices and compute outcomes for each year–quarter market in 2021.[31] We outline the three counterfactuals below, with further details provided in Appendix N.

Store-entry counterfactual This counterfactual (CF1) holds the discounter store network, $\mathcal{S}_t$, at its 2002 configuration, while holding in-store primitives at their observed levels. Household distances to discounter stores are therefore calculated using the 2002 store network. CF1 measures the effect of the subsequent expansion of discounter store networks.

In-store counterfactual This counterfactual (CF2) removes the changes in discounters' in-store primitives since 2002 while holding the store network $\mathcal{S}_t$ at its observed configuration. We modify three in-store primitives: the set of products $\mathcal{J}_t$, their marginal costs, and the mean utilities. We first fix the set of product–retailer options offered at discounters to those available in 2002.[32] We then adjust the evolution of these options' marginal costs $\mathbf{c}_t$ and mean utilities $\boldsymbol{\delta}_t$ so that they

[31] We recover counterfactual manufacturer and retailer markups in each market by jointly solving the retailers' first-order conditions (equation (3.4)) and the Nash bargaining solution (equation (3.7)). Details of the solution algorithm are provided in Appendix M.

[32] Differentiated-product demand models with T1EV errors can overstate welfare gains from product introduction because the inclusive value that drives consumer surplus mechanically increases with the number of products; see Ackerberg and Rysman (2005). Our specification includes consumer heterogeneity and micro moments based on individual-level characteristics and purchase decisions, which reduce the role played by the T1EV errors in determining substitution patterns and counterfactual welfare effects (see Petrin, 2002). Following the diagnostic in Gentzkow (2007), we find that, at the estimated parameters, the variance across household-option pairs of the household-level utility component, μ_{ij}, is 422 times the variance of the T1EV error, $\pi^2/6$. Even after removing heterogeneity in the preference for the inside good, this ratio is 52.

follow the same trend as at traditional retailers. CF2 is informative about changes in discounters' in-store product offering and supply-chain efficiency *conditional on* their network.

Full counterfactual This counterfactual (CF3) combines CF1 and CF2 to measure the overall effect of the rise of the discounters. Comparing CF3 with CF1 and CF2 helps illustrate the quantitative importance of store-network expansion and changes in in-store offerings in shaping market power and economic surplus.

After presenting the main counterfactuals, we also report a policy-motivated variant of the store-entry counterfactual aimed at capturing the role of the Controlled Land Order (2010), which facilitated the rise in new discounter stores.

Discussion The full counterfactual captures the combined effect of changes in discounter entry conditions, including planning rules and the prohibition of restrictive land practices, and changes in their in-store business model on equilibrium pricing and surplus. It allows retailers and manufacturers to adjust prices, but holds traditional retailers' product assortments and store networks at their observed configurations. It therefore does not incorporate potential strategic responses by traditional retailers along these margins. We expect the scope for such responses to be limited in this setting. Traditional retailers already offer broad assortments of breakfast cereals, including branded products and an extensive range of private-label options, which span the product characteristic space and leave little scope for further assortment expansion in response to discounter entry. In addition, traditional retailers faced substantial planning constraints on new stores (see Section 2). Moreover, discounter store-network expansion accelerated after the 2010 prohibition of restrictive land practices previously used by traditional retailers. By releasing mid-sized sites that had previously been inaccessible to discounters, the reform enabled them to expand into locations near incumbent traditional retailers. Given the planning constraints faced by traditional and the fact that discounters primarily entered areas already served by them, large offsetting changes in traditional retailers' geographic footprints are unlikely.

Thus, the household-level preference heterogeneity captured by the model is quantitatively much larger than the T1EV component of utility.

6.2 Results

Table 6.1 summarizes counterfactual results by comparing observed and counterfactual equilibrium outcomes in the final year of our sample, 2021.

Primitives Panel (A) reports changes in the primitives underlying the counterfactuals. The store-entry counterfactual increases the distance of the average household from an Aldi or Lidl store from 4 km to 11 km. The in-store counterfactual reduces the average number of discounter cereal options from 80 to 26 and lowers the mean utility of these options. It reduces average marginal costs when all products are included, reflecting composition changes, but increases them for overlapping options, consistent with discounter efficiency gains over time for a given set of products.

Market equilibrium Panel (B) summarizes the impact of discounter expansion on concentration, prices and markups. In 2021, the retail HHI for breakfast cereals was 1824. Had discounters remained at their 2002 market positions, the retail HHI would have been 242 points higher at 2066. The rise of discounters therefore accounts for most of the decline in breakfast cereal retail concentration from 2013 onward (see Figure 2.2(b)).[33] The rise of discounters similarly reduced manufacturer concentration by 231 HHI points. Both store entry and improvements in discounters' in-store offerings contributed to these declines.

The changes are reflected in prices. In 2021, the average cereal price was £3.93/kg; without the rise of discounters, it would have been £4.10/kg. This decline reflects changes in discounters' marginal costs and product portfolios, as well as the exercise of market power by discounters, competing retailers, and manufacturers. The aggregate vertical additive markup is higher in the counterfactuals that hold discounters at their 2002 positions. In 2021 it was £2.13/kg, compared with £2.25/kg in the full counterfactual. This 12 pence difference accounts for the majority of the 17 pence difference in average prices, with the remaining gap reflecting offsetting changes in marginal costs and product composition.

The total vertical additive markup for products sold by discounters was £1.21/kg in 2021, compared with £0.98/kg had they remained at their 2002 market position. This increase reflects both compositional shifts in discounters' product portfolios and higher markups on products available in both observed and counterfactual

[33]The red lines in Figure 2.2(b) and (d) show the path of breakfast cereal HHIs in the data. The model-based HHIs follow similar paths, albeit with slightly lower levels because our analysis groups together smaller retailers and omits some niche manufacturers.

equilibria.[34] To isolate the latter effect, we also report markups for "overlapping" discounter options—products available in 2021 and in all three counterfactuals. For these products, discounter expansion raises markups for both branded and private-label products. For branded products, both retail and manufacturer markups increase, with the manufacturer component rising more, consistent with the discounters' expanded branded product ranges strengthening manufacturers' bargaining positions. Overall, the increase in markups more than offsets the reduction in marginal costs, implying higher prices for cereals sold by discounters.

Panel (B) also reports markups for products sold by traditional retailers. Competitive pressure from discounters lowers markups for both branded and private-label products. For branded products, the decline is almost entirely driven by the retail component, with manufacturer markups changing little. The intermediate counterfactuals show that both store-network expansion and improvements in discounters' in-store offerings contributed to the erosion of traditional retailers' market power. As traditional-retailer marginal costs are held fixed across the counterfactuals, these markups also summarize the equilibrium price response. For private-label products, price changes correspond to changes in the vertically integrated markup, while for branded products they equal the combined changes in retailer and manufacturer markups. Since manufacturer markups change little, branded-product price responses are well approximated by changes in retailer markups.

Market surplus Panel (C) reports changes in consumer, producer, and total surplus under each counterfactual relative to the 2021 observed equilibrium. Discounter expansion increased consumer surplus in the breakfast cereal market by £73.4 million, equivalent to 5.1% of total breakfast cereal spending in 2021. The intermediate counterfactuals show that both store-network expansion and discounters' in-store improvements are quantitatively important for these gains.

Discounters' profits were £56.7 million higher than they would have been had their market position remained at 2002 levels, equal to 63.4% of their 2021 profits. These gains were more than offset by declines in profits for traditional retailers and manufacturers, amounting to 14.1% and 8.0% of their respective 2021 profits. Overall, consumer gains and discounter profit gains exceed these losses by £43.0 million, yielding a net total-surplus gain equal to 3.0% of total cereal spending. Both store-network expansion and in-store improvements are quantitatively important for this gain.

[34]The set of discounter products differs across counterfactuals: the observed equilibrium and CF1 include a different set of products than CF2 and CF3. Comparisons across these scenarios therefore involve different product sets.

Table 6.1: *Counterfactual analysis*

	Observed equilibrium	Counterfactual equilibrium: Store (CF1)	In-store (CF2)	Full (CF3)
A) Discounter primitives				
Distance (km)	3.99	10.99	-	10.99
Portfolio size	80	-	26	26
Mean utility (δ)				
All options	-3.10	-	-3.42	-3.42
Overlapping options	-2.40	-	-3.16	-3.16
Marginal cost (£/kg)				
All options	1.23	-	1.02	1.02
Overlapping options	0.64	-	0.70	0.70
B) Market equilibrium				
Concentration (HHI)				
Retail	1824	1979	1944	2066
Manufacturer	2123	2269	2257	2354
Average market price (£/kg)				
Unweighted	3.93	3.95	4.08	4.10
Sales-weighted	3.24	3.34	3.20	3.35
Markup (£/kg)	2.13	2.14	2.23	2.25
Discounter markups (£/kg)				
All options	1.21	1.13	1.02	0.98
Overlapping options				
Branded				
Retail component	0.54	0.51	0.47	0.44
Manufacturer component	1.08	1.09	0.78	0.75
Private-label	1.00	0.94	0.96	0.95
Traditional retailer markups (£/kg)				
Branded				
Retail component	1.36	1.39	1.38	1.40
Manufacturer component	1.21	1.21	1.21	1.21
Private-label	1.22	1.26	1.25	1.28
C) Δ annual market surplus (£m)				
Consumer surplus	-	-47.8	-36.9	-73.4
% of spending		*(-3.31%)*	*(-2.55%)*	*(-5.08%)*
Producer surplus				
Traditional retailers	-	41.2	30.7	63.4
% change		*(9.16%)*	*(6.82%)*	*(14.08%)*
Discounters	-	-40.0	-26.9	-56.7
% change		*(-44.72%)*	*(-30.04%)*	*(-63.40%)*
Manufacturers	-	16.1	10.3	23.7
% change		*(5.41%)*	*(3.47%)*	*(7.97%)*
Total surplus	-	-30.5	-22.7	-43.0
% of spending		*(-2.11%)*	*(-1.57%)*	*(-2.98%)*

Notes: The table compares average outcomes in the 2021 observed and counterfactual equilibrium. Unless otherwise stated the averages are unweighted. Panel (A) summarizes the change to market primitives in each counterfactual scenario. Panels (B) and (C) summarize the change in endogenous market outcomes. Marginal costs, prices, and markups are expressed in 2021 £ per kg; annual surplus is expressed in 2021 £ millions.

Planning-policy counterfactual We also consider a policy-motivated variant of CF1 aimed at capturing the role of the post 2010 changes in land use law. Rather than returning discounter store locations to their 2002 configuration, we allow store networks in each region (ONS, 2024a,b) to evolve along their observed path through 2012 and then impose a counterfactual path in which the number of discounter stores continues to grow at its pre-2012 linear trend (see Appendix N).[35] This exercise provides an illustrative benchmark for the extent to which the welfare gains from discounter expansion are associated with the acceleration in store growth after 2012. A broader comparison of alternative planning or land-use policies would require modeling entry decisions by both discounters and traditional retailers.

Relative to the observed 2021 equilibrium, this counterfactual attributes £24.8 million of consumer-surplus gains and £14.7 million of total-surplus gains to the post-2012 acceleration in store growth (see Appendix Q). These gains amount to around half of those from post-2002 store openings (CF1), indicating that the post-2012 acceleration accounts for a substantial share of the gains from discounter store-network expansion.

Distributional effects Figure 6.1 shows how the consumer-surplus gains are distributed across households. We report the difference between consumer surplus in the observed and full counterfactual (CF3) equilibria in 2021, expressed as a fraction of household-level breakfast cereal spending. Panel (a) shows substantial heterogeneity: the interquartile range is 2%–6%, and gains at the 95th percentile are around 11%. Panel (b) shows little systematic relationship with equivalized household income, indicating that the benefits were broadly shared across the income distribution.

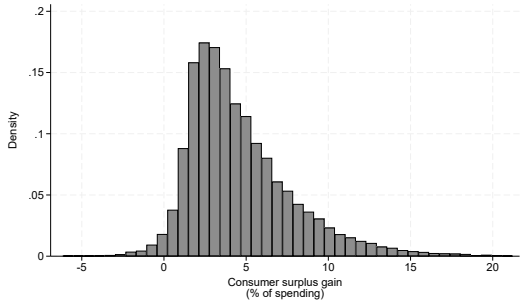
Panel (c) relates consumer gains to changes in proximity to discounter stores. Households whose distance to the nearest discounter does not fall gain around 2.9%, reflecting improvements in discounters' in-store offerings and indirect gains from greater competitive pressure on traditional retailers. Consumer gains increase with the size of the reduction, highlighting the importance of the geography of store expansion.

Cross-category effects Consumers often purchase multiple categories during a store visit (see Smith and Thomassen, 2012), and improvements in discounters' non-cereal offerings may affect cereal demand. We allow the retailer-time component of mean utility to include both cereal-specific factors and non-cereal shopping effects.

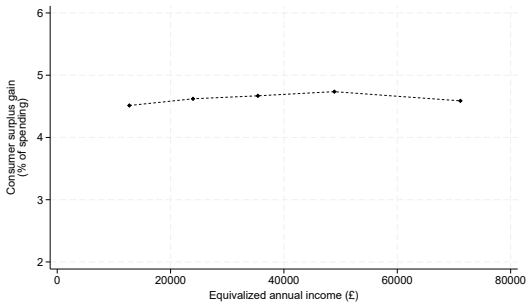
[35] Although the Controlled Land Order (2010) came into force in August 2010, store openings did not begin to increase until 2012, reflecting the time required to plan and build new stores.

Figure 6.1: *Distributional impact*

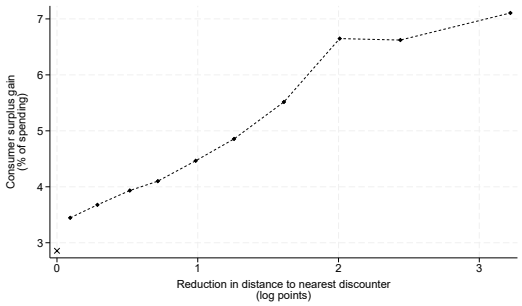
(a) Distribution across households



(b) Impact by income quintiles



(c) Impact by travel distance reduction



Notes: Figures show the difference in consumer surplus between the observed and full counterfactual equilibria for 2021, expressed as a fraction of household breakfast cereal expenditure. Panel (a) reports the distribution of consumer-surplus gains across households. Panel (b) shows the average change for each household income quintile. Panel (c) shows average gains by the reduction in travel distance to the nearest discounter store. The × at zero denotes households whose distance to the nearest discounter does not fall.

Using an auxiliary moment based on continuing cereal products, we isolate the non-cereal component and exclude it when scaling the cereal estimates to aggregate consumer gains; details are in Appendix O.

Retailers may also account for cross-category effects when setting cereal prices. In that case, recovered markups should be interpreted relative to an effective marginal cost, equal to technological marginal cost net of the marginal benefit from additional demand in other categories. Our cost specification includes retailer-year effects, which accommodate such retailer-time variation; Appendix F gives the formal derivation.

Aggregate implications The descriptive analysis in Section 2 shows that concentration trends in breakfast cereals mirror broader patterns across fast-moving consumer goods. To gauge the aggregate magnitude of the consumer gains, we scale our breakfast-cereal estimates to the set of categories sold by discounters.

We decompose the estimated consumer-surplus gain into gains from reduced travel costs and gains arising from cereal-specific factors, including product avail-

ability, prices, and mean utilities, after removing gains attributable to non-cereal retailer utility effects, which we do not scale to other categories (see Appendix P). Of the gain remaining after removing non-cereal retail utility effects, 32% comes from reduced travel costs and 68% from cereal-specific factors. We scale the cereal-specific component by the reciprocal of breakfast cereals' share of total fast-moving consumer goods spending (1.3%), and the travel-cost component by the reciprocal of the share of weekly shopping trips that include breakfast cereals (31.3%). This back-of-the-envelope calculation implies a 2021 consumer-surplus gain of approximately £2.5 billion. While the precise magnitude is uncertain, the exercise suggests that discounter expansion delivered substantial consumer benefits beyond the breakfast cereal market.

Retail conduct and the division of market power The counterfactual results show that the rise of discounters reduced concentration, lowered prices, and raised consumer surplus. It also reduced producer surplus and shifted its distribution across retailers and manufacturers. A feature of our model is that it explicitly models both retailers and manufacturers, which allows us to identify how market power is divided in the vertical chain and how discounter expansion changes that division. The model accounts for differences in manufacturers' bargaining positions across private-label and branded goods, a distinction that shapes the effects of discounter expansion.

Our estimates indicate that retailers account for a substantial share of total vertical markups on branded products: on average, downstream retailer markups account for nearly half of the total markup between retail price and total vertical marginal cost. This is consistent with direct evidence on the division of vertical markups in UK consumer-goods markets (Alvarez-Blaser et al., 2025). Retailers are therefore not simply passive intermediaries. Moreover, the rise of discounters substantially reshaped this downstream component of market power. Markups on products sold by discounters rose as their store networks and assortments expanded, while markups at traditional retailers fell as they faced greater competitive pressure. Almost all of the reduction in traditional-retailer prices induced by discounter expansion is accounted for by lower downstream retailer markups, rather than lower manufacturer markups. At the same time, discounter expansion shifted sales toward discounters, whose assortments contain a disproportionate share of private-label products. This shift raised discounter profits while reducing profits of traditional retailers and branded manufacturers. Thus, the model attributes the competitive response of traditional-retailer prices primarily to changes in downstream competi-

tion, while also showing how discounter expansion redistributed profits across the vertical chain.

To assess which conclusions depend on explicitly modeling both sides of the vertical relationship, we compare our baseline model with the two alternative supply models introduced in Section 3.3. Under retailer pricing, manufacturers earn no wholesale markup, so all vertical market power is attributed to retailers. Under manufacturer pricing, manufacturers set retail prices directly and the downstream retailer markup is absent. Neither benchmark allows both markups to be positive and therefore neither can recover the division of market power between the two sides of the vertical relationship or how that division changes with discounter expansion. Nevertheless, the effects of discounter expansion on concentration, prices, and consumer surplus are qualitatively similar across the three models (Appendix R). Thus, explicitly modeling both retailers and manufacturers is not essential for the conclusion that discounter expansion benefited consumers. It is important for understanding the mechanism behind those gains—in particular, the role of downstream competition—and for determining how changes in market power and profits are divided between retailers and manufacturers.

7 Conclusion

The rise of the discounter format has been a common feature of grocery retailing across many countries in recent decades, with potentially significant implications for market structure and performance. Using rich microdata from the UK, we document that discounter expansion coincided with substantial declines in retail and manufacturing concentration across most narrowly defined product categories. Focusing on the breakfast cereal market, we estimate a model of consumer demand and vertical relationships and find that discounter growth led to lower prices, reduced concentration, and increases in both consumer and total surplus. Our counterfactuals show that these gains reflect store-network expansion, efficiency gains, and changes in product offerings, as well as increased competitive pressure on traditional retailers and branded manufacturers. While discounters increased their own market power over time, the net effect of their expansion was pro-competitive. Our findings highlight the importance of retail format innovation in shaping market outcomes and underscore the value of analyzing narrowly defined product markets to understand broader trends in concentration and market power.

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Appendix: For online publication

The Rise of Discounters and its Impact on Concentration, Market Power and Welfare

Martin O'Connell, Howard Smith and Øyvind Thomassen

October, 2026

A Store and Household Locations

No single data source covers store locations over the full sample period. We therefore combine three sources to construct retailer-year store-location data. The Institute of Grocery Distribution (IGD, 2006) store dataset provides store locations for each retailer over 2002–2006. Geolytix Retail Points provides the same information for 2014–2021 (Geolytix, 2023). For the intervening years, 2007–2013, we use data from Glenigan on new building projects, which records new supermarket store completion dates, retailer identities, and postcodes (Glenigan, 2018).

We construct store locations for 2007–2013 by starting from the set of 2014 Geolytix stores and working backward, removing stores completed in each preceding year according to Glenigan. This approach uses store openings but provides an accurate count of open stores because store closures are rare. As an external validation, we compare the resulting store counts with information from company annual reports and find that they closely match the total number of stores operated by each retailer in each year.

We drop stores of convenience store size (under 280 square meters) operated by the traditional retailers.

To calculate the household-retailer distances, we randomly draw a residential postcode, and its exact grid reference (ONS, 2021b), from each household's postal district and compute the distance to the nearest store of each retailer.

To define whether the household is in an urban area, we match the postal district of the household with the Office for National Statistics Rural Urban Classification, as recorded in the ONS Postcode Directory (ONS, 2021b), and define the binary variable 'urban' as indicating that the postal district is in either a major or minor urban conurbation.

B Cereal Input Prices: Data Description

We use three data sources for crop prices:

1. UK Agricultural Price Index (API) (Defra, 2023). Price index of agricultural outputs in GBP, 2015=100.

2. EU official import prices (European Commission, 2023). Average unit value of EU imports from outside the EU, in euro per 100kg.

3. UN Food and Agriculture Organization (FAO) market data (FAO, 2023). International export prices in USD per tonne.

We use these data sources to construct the eight cost variables in Table 5.1(c). The first seven of these consist of a crop input price interacted with a cereal base dummy, for each of the seven cereal bases. The eighth is a sugar price index interacted with a dummy which takes the value 1 for some of the cereal bases. There are multiple candidate crop input prices to interact with each cereal base, e.g., for the cereal base corn there are different types of maize product, etc. To select a series for each cereal base we run a regression of retail prices on alternative input prices for each of seven cereal bases and select the input price with the highest statistical significance. This results in the following input price for each cereal base:

1. Corn: UN FAO market data, USA (Gulf), maize (US no. 2, yellow).

2. Wheat: EU import prices, common wheat.

3. Rice: UN FAO market data, Pakistan, rice.

4. Oats: EU import prices, oats.

5. Bran: as wheat.

6. Multi: EU import prices, maize.

7. Granola: UK API, oats.

The eighth cost variable is a sugar price index. We use the following sugar price series: UN FAO market data, ICE futures USA sugar. We find this is significant only for cereal bases 1, 3, 5, 6 and 7 so we interact it with a single dummy which indicates whether the cereal is from one of these bases.

We convert the EU and FAO series to GBP using ONS average monthly sterling exchange rates against the euro and the US dollar (ONS, 2021a, 2023a). We normalize each monthly series by its value in September 2001, the first month of our input-price data, average to the quarterly level, and deflate by the all-items CPI (ONS, 2023b).

C Planning Policy and Controlled Land

Competition Commission's store and retailer classifications Two market inquiries by the Competition Commission (CC)—CC, 2000, 2008—considered planning and controlled land from a competition perspective. To define terms, CC (2008) classified *mid-sized* stores as those with a sales area of 280–1,400 square meters, and *larger* stores as those with a sales areas greater than 1,400 square meters respectively (paragraph 4.63). It also classified grocery retailers into *discounters* and *large grocery retailers* (paragraphs 3.3-3.7). In this paper, we refer to the latter

as traditional retailers. The CC found that nearly all of the larger grocery stores were operated by traditional retailers, while discounters exclusively used mid-sized stores.

Restrictive planning policy in the 2000s Before the mid-1990s, planning policy had been relatively liberal and had resulted in traditional retailers opening many larger stores (CC 2000, paragraph 2.165). In the mid-1990s, however, the government changed planning policy by adding two tests that made it harder to open larger stores, primarily affecting traditional retailers. The first was the *sequential* test, which required authorities to favor town-center and retail-park developments, which tend to be mid-sized, over out-of-town developments, which tend to involve larger stores. The second was the *need* test, which required evidence of local need for an additional store. As larger stores were the preferred format of the traditional retailers, these changes greatly inhibited their expansion. This was less true for discounters, because they used mid-sized stores that were less adversely affected by these tests.

These changes constituted a barrier to entry that "had a major impact on store development plans" (CC 2008, paragraph 2.168), "locked in" (CC 2008, paragraph 2.175) the market structure for traditional retailers, and "made entry into and expansion within multiple grocery retailing more difficult for parties wanting to acquire large sites in out of town locations" (CC 2008, paragraph 2.202). Discounters were less seriously affected by these policies (CC 2008, paragraph 2.205). For example, Aldi claimed that because its stores were smaller and not out of town, and could be accommodated in town centers, retail parks, and edge-of-town locations, it was not inhibited by them (CC 2008, paragraph 5.150). The CC concluded that the planning system after these changes "constrains new entry by larger grocery stores" but that "these constraints are less significant for mid-sized grocery stores" given that "for these stores suitable locations that are not subject to planning restrictions are more easily found" (CC 2008, paragraph 7.44).

Proposed changes to planning policy The CC proposed reforms to the operation of the sequential and need tests in the planning system. These proposals were not adopted by the government. As a result, the planning regime did not change, and traditional retailers continued to be constrained throughout 2002–2021.

Controlled land The CC also investigated land-control practices that it viewed as capable of restricting competition by preventing land from being used by rival grocery firms. These included restrictive covenants on land sales, exclusivity arrangements in shopping centers and retail parks, and the leasing or sub-leasing of land sites to third parties not involved in grocery retailing. These practices were viewed particularly problematic when combined with a restrictive planning regime. The combination of a restrictive planning system and controlled landsides frustrated new entry that would otherwise strengthen competition (see paragraph 7.121). Controlled land sites restricted the opening both of mid-sized and larger stores.

The Controlled Land Order CC (2008) proposed remedies to address controlled land, which resulted in the Controlled Land Order 2010. The aim of the Order was to limit large grocery retailers' ability to use land-site control to prevent land from being used by competing grocers. It designated a list of retailers to be restricted by the Order; this list included the traditional retailers but not the discounters. The Order banned restrictive covenants and exclusivity clauses in land deals that prohibited rivals from opening stores. Traditional retailers were required to release existing restrictive covenants and were prohibited from entering new ones. After the Order, discounters could enter some sites that landlords previously could not have leased to them. Because discounters were not restricted by the Order, they retained the ability to use land controls to prevent competitors from opening stores nearby.

D Breakfast Cereal Manufacturers

This appendix provides additional evidence on the manufacturers that supply private-label breakfast cereals to UK supermarkets. We do not observe the supplier of each retailer's private-label cereal product in our transaction data. Public information nevertheless indicates two relevant facts. First, there are multiple specialist manufacturers that supply private-label breakfast cereals but do not sell major branded cereals in our data. Second, some branded manufacturers also appear to have supplied private-label cereals, although the extent of this activity is not observed.

Specialist private-label manufacturers Several firms manufacture private-label breakfast cereals for UK supermarkets without supplying branded cereals, either from UK plants or by exporting from other European countries. The following list is not exhaustive, but provides examples, together with quotes from their websites.

- Cereal Products (production mainly in Portugal) "From breakfast cereals to snacks, we manufacture a wide range of products with various packaging options to meet your specific needs."
- Deeside Cereals (production in the UK) "Deeside Cereals manufactures breakfast cereals for the major supermarket chains."
- Mulder Natural Foods (production in Belgium) "Looking for an expert in breakfast cereals for your private label? Crunchy/granola, muesli, extrusion/co-extrusion or flakes... your wish is our command."
- MPRUK (production in the UK) "we provide an almost full range of Breakfast Cereal under private label."
- LHF (production in the UK) "Welcome to Life Health Foods! We produce breakfast cereals and healthy fruit snacks. We have four fantastic brands, and provide own-label and branded manufacture and packaging services for many of the leading retailers and brands in the UK."
- H J Brueggen KG (production in Germany, Poland and France) "We produce private labels for the leading European and international retail chains.

Whether premium, discount or entry-level price quality, whether conventional or organic: We master all segments thanks to an excellent, graduated price-quality ratio."

Discounter brands Aldi and Lidl sell breakfast cereals under their own brands, including 'Harvest Morn' and 'Crownfield', respectively. These are retailer-owned brands: the trademarks are registered in the UK to Aldi and Lidl, while production is outsourced to private-label manufacturers rather than carried out by the retailers themselves.

Branded manufacturers and private-label production There is also some public information on whether branded breakfast cereal manufacturers produce private-label goods. In some cases they do, although we do not observe details of which products.

- Kelloggs, the largest branded cereal manufacturer, states that it does not make private-label goods for UK supermarkets. For example, during our sample period, Kelloggs says: "We don't make cereal for anyone else so Special K has always had a unique, premium quality but many own brands have attempted to copy both the look, taste and packaging within IP limits," (Louise Thompson-Davies, brand communications manager at Kellogg UK, in Food and Drink Business Europe, May 24, 2013).

- A small number of the products under the Sugar Puffs and later Monster Puffs brand were made by Halo and later Brecks Foods, which acquired the rights in 2016. We found no evidence that these companies made private-label products in the breakfast cereal category.

- Jordans and Dorset came under common ownership in 2014 when a Competition Commission investigation approved their merger. The CMA materials describing these firms' activities do not indicate that Jordans or Dorset produced private-label breakfast cereals.

- Whitworths (initially known as Weetabix Limited) is the firm that makes Weetabix. In 2004 a competition report by the UK's Office of Fair Trading suggested that Weetabix produced private labels, according to the Office of Fair Trading's report on the Latimer Acquisitions Ltd / Weetabix Ltd acquisition, which states that "Weetabix is active in the manufacture and sale of cereal products, mainly breakfast cereals and cereal bars. Its products are sold under the Weetabix, Alpen, Ready Brek and Weetos brands, as well as retailer brands."

- Nestlé's UK breakfast cereal business is operated through Cereal Partners UK (CPUK&I). CPUK&I do make private labels according to the following quote from the news story "Cereal Partners set to close site as sales drop" in *The Grocer* magazine. "Under the proposals, production of Cheerios will transfer to Staverton in Wiltshire, and CPUK&I will cease its supermarket branded cereal business at the end of its current contracts." (Mar 29, 2025).

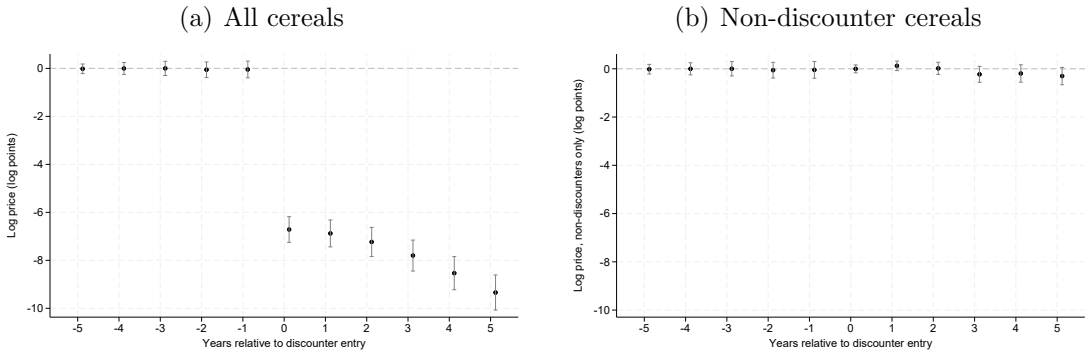
E Additional Evidence on Discounter Entry

E.1 Local Price Responses to Discounter Entry

The decline in the local cereal price index following discounter entry partly reflects the addition of lower-priced discounter products to the local choice set. Figure E.1 compares this baseline price index with an analogous index constructed using only cereals available at non-discounter retailers. Panel (a) repeats the main-text result: the price of cereals locally available to households falls sharply following discounter entry. In contrast, panel (b) shows essentially no relative change in the prices of non-discounter cereals.

This pattern is consistent with the national pricing policies of the major traditional retailers. Unlike settings in which incumbent stores set prices locally, local discounter entry therefore provides little scope for an incumbent price response within the treated area. This distinguishes our setting from Arcidiacono et al. (2020), who study the response of incumbent supermarket prices to local Walmart entry. Our equilibrium demand and supply model captures the equilibrium response of nationally set retailer prices to the broader expansion of the discounter store network.

Figure E.1: *Local cereal prices around discounter entry*



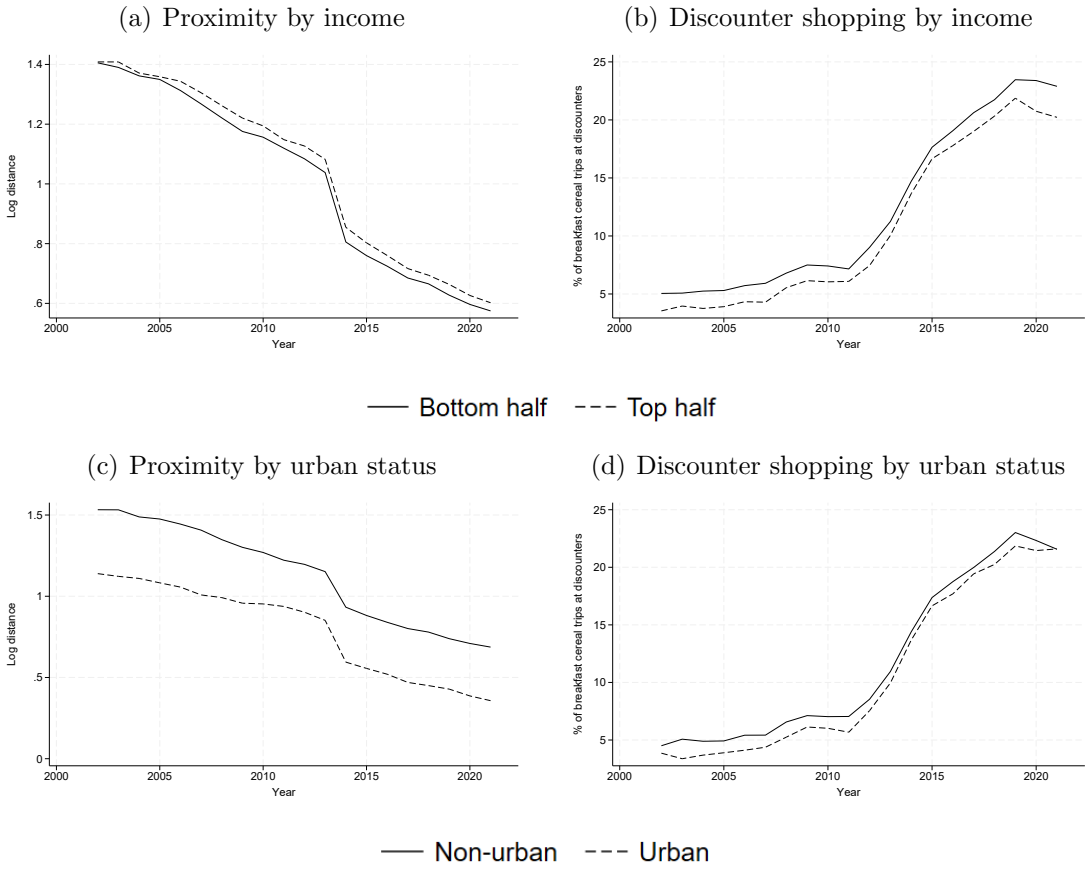
Notes: The figure plots event-study coefficients around the first recorded year in which Aldi or Lidl operates within 5 km of a postal-district centroid; event time 0 denotes this year. Coefficients are estimated using the imputation estimator of Borusyak et al. (2024), with postal-district and calendar-year fixed effects. The estimation sample contains 632 treated postal districts and 321 never-treated postal districts, where never-treated districts have no Aldi or Lidl within 5 km at any point during the sample period. Both price indices are weighted averages of breakfast cereal products available within 5 km, using national market shares as weights. Panel (a) includes all cereal products, while panel (b) excludes products sold by Aldi and Lidl. Vertical lines are 95% confidence intervals.

E.2 Discounter Proximity and Shopping

Figure E.2 documents the evolution of household proximity to discounters and breakfast cereal purchases at discounters by household income and urban residence. Access to discounters increased substantially across all groups as the Aldi and Lidl store networks expanded. Lower-income households live modestly closer to discounters and are somewhat more likely to purchase cereal there than higher-income households. Differences in proximity by urban status are larger: urban households

live substantially closer to discounters throughout the sample, although their share of cereal purchases at discounters is similar to, and for much of the period slightly below, that of non-urban households. Overall, the expansion in both access to and shopping at discounters was broad-based across these observable household groups.

Figure E.2: *Discounter proximity and shopping patterns*



Notes: Panels (a) and (c) show average proximity to the nearest Aldi or Lidl store. Panels (b) and (d) show the share of breakfast cereal shopping trips at discounters. Panels (a) and (b) report results separately for households in the top and bottom halves of the annual equivalized household-income distribution. Panels (c) and (d) report results separately for households living in major urban areas and in other areas.

F Cross-Category Effects in Pricing

Let q_{ort} be the aggregate quantity of other (i.e. non-cereal) categories sold at retailer r and let $\Gamma_{ort} = (p_{ort} - c_{ort})$ be the retailer's markup. We discuss cross-category pricing effects using the baseline bargaining model, which nests the retailer-pricing model. In the manufacturer-pricing model, we assume the manufacturer does not sell products in other categories, so that cross-category effects are not relevant.

The objective function of retailer r in period t in the bargaining model is

$$\Pi_{rt} = \sum_{j' \in \mathcal{J}_{rt}} q_{j't}(p_{j't} - \Gamma^F_{j't} - c_{j't}) + \chi q_{ort} \Gamma_{ort},$$

where $\chi \in \{0, 1\}$ indicates whether the retailer internalizes cross-category effects. The first-order condition with respect to price p_{jt} is

$$\frac{\partial \Pi_{rt}}{\partial p_{jt}} = q_{jt} + \sum_{j' \in \mathcal{J}_{rt}} \frac{\partial q_{j't}}{\partial p_{jt}} (p_{j't} - \Gamma^F_{j't} - c_{j't}) + \chi \frac{\partial q_{ort}}{\partial p_{jt}} \Gamma_{ort} = 0.$$

Dividing by the derivative of demand for option j with respect to its own price, this can be rewritten as

$$\underbrace{p_{jt} + \frac{q_{jt}}{\frac{\partial q_{jt}}{\partial p_{jt}}}}_{\text{marginal revenue, } j} + \underbrace{\sum_{j' \in \mathcal{J}_{rt} \setminus \{j\}} \frac{\frac{\partial q_{j't}}{\partial p_{jt}}}{\frac{\partial q_{jt}}{\partial p_{jt}}} (p_{j't} - \Gamma^F_{j't} - c_{j't})}_{\text{marginal profit, other cereals at } r} = \underbrace{c_{jt} + \Gamma^F_{jt}}_{\substack{\text{perceived} \\ \text{marginal cost, } j}} - \underbrace{\chi \frac{\frac{\partial q_{ort}}{\partial p_{jt}}}{\frac{\partial q_{jt}}{\partial p_{jt}}} \Gamma_{ort}}_{\substack{\text{marginal} \\ \text{externality}}}.$$

The first term on the right-hand side is the retailer's perceived marginal cost: it includes the retailer component of marginal cost and the wholesale price. The second term on the right-hand side captures the effect of changing the cereal price on profits from other categories. Its sign is theoretically ambiguous. For example, if consumers who stop purchasing cereal j at retailer r also switch retailers for other categories, cereal and other-category purchases are complementary and a higher cereal price reduces other-category profits (see Thomassen et al. 2017). The size of this effect may vary over time as the quantity and markups of other goods purchased jointly with cereals change.

For a multiproduct retailer, the cross-category adjustment must account jointly for substitution among all cereal options sold by the retailer. Let $\boldsymbol{m}_{rt} = \boldsymbol{p}_{rt} - \boldsymbol{\Gamma}^F_{rt} - \boldsymbol{c}_{rt}$ denote the vector of retailer markups before accounting for cross-category effects, let $\boldsymbol{\Delta}_{rt}$ be the within-retailer demand-derivative matrix, with element $\Delta_{jj',t} = -\frac{\partial q_{j't}}{\partial p_{jt}}$, $j, j' \in \mathcal{J}_{rt}$, and $\boldsymbol{g}_{rt}$ be the vector with element $g_{jt} = \frac{\partial q_{ort}}{\partial p_{jt}}$. The retailer's pricing first-order conditions can then be written

$$\boldsymbol{q}_{rt} - \boldsymbol{\Delta}_{rt} \boldsymbol{m}_{rt} + \chi \boldsymbol{g}_{rt} \Gamma_{ort} = \boldsymbol{0}.$$

Define the vector of cross-category adjustments as $\boldsymbol{e}_{rt} = -\boldsymbol{\Delta}^{-1}_{rt} \boldsymbol{g}_{rt} \Gamma_{ort}$, and define *effective marginal costs* by

$$\bar{\boldsymbol{c}}_{rt} = \boldsymbol{c}_{rt} - \chi \boldsymbol{e}_{rt}.$$

The corresponding adjusted retailer markups are

$$\tilde{\boldsymbol{\Gamma}}^R_{rt} = \boldsymbol{p}_{rt} - \boldsymbol{\Gamma}^F_{rt} - \bar{\boldsymbol{c}}_{rt}.$$

By construction, the pricing first-order conditions therefore take the same form as in the model without cross-category effects:

$$\tilde{\boldsymbol{\Gamma}}^R_{rt} = \boldsymbol{\Delta}^{-1}_{rt} \boldsymbol{q}_{rt},$$

or, stacking across retailers,

$$\bar{\boldsymbol{c}}_t + \boldsymbol{\Gamma}^F_t = \boldsymbol{p}_t - \boldsymbol{\Delta}_t(\boldsymbol{p}_t)^{-1} \boldsymbol{q}_t.$$

Thus, $\boldsymbol{e}_{rt}$ captures the marginal opportunity-cost adjustment associated with the effect of cereal prices on profits in other categories. If there is no substitution between cereal options sold by the same retailer, this reduces to the scalar adjustment

$$e_{jt} = \frac{\frac{\partial q_{ort}}{\partial p_{jt}}}{\frac{\partial q_{jt}}{\partial p_{jt}}} \Gamma_{ort}.$$

To incorporate cross-category effects in bargaining, we use the same effective-markup adjustment when calculating the retailer's gains from trade, thereby approximating the cross-category profit effect of a negotiation breakdown by its local pricing counterpart. Under the bargaining model, leverage is therefore defined as in equation (4.2), replacing retailer markups with $\tilde{\boldsymbol{\Gamma}}_t^R$. For branded negotiations, $\Gamma_{nt}^F = \rho\tilde{\ell}_{nt}$, while manufacturer markups are zero for private-label products. With estimates of $\boldsymbol{\Delta}_t(\boldsymbol{p}_t)$ from demand estimation, the second-stage regression is

$$[p_{jt} - \tilde{\Gamma}_{jt}^R] = \rho\tilde{\ell}_{n(j)t} + \underbrace{\boldsymbol{\gamma}\boldsymbol{x}_{jt}^s + \omega_{jt}}_{\bar{c}_{jt}},$$

For private-label products the leverage term is zero.

This specification identifies the effective marginal cost $\bar{c}_{jt}$. Separate identification of technological marginal cost c_{jt} and the cross-category adjustment requires additional information on other-category demand and markups and is beyond the scope of the paper.

Counterfactuals In the in-store and full counterfactuals, we adjust discounters' estimated effective marginal costs so that, after 2002, they follow the corresponding trajectory observed for traditional retailers (see Appendix N). Because effective marginal costs combine technological marginal costs with any cross-category pricing adjustment, this also removes differential changes in discounters' cross-category pricing incentives to the extent that they are reflected in estimated effective marginal costs. This treatment is approximate: cross-category effects depend on equilibrium outcomes that may change in the counterfactual, while prices and markups in other categories are held fixed. A full calculation would require modeling demand and prices across categories.

G Efficient Bargaining Case

Our supply model nests, as a special case, the retail-pricing outcome of a two-stage model of negotiated two-part tariffs with bilateral efficiency. This retail-price outcome corresponds to $b_n = 0$ for all negotiations in our baseline model.

We follow the setup in Rey and Verge (2019). In the first stage, each manufacturer–retailer pair bilaterally and simultaneously negotiates a wholesale price and transfer (w_j, T_j) for each product–retailer option j. Firms' payoffs are profits net of transfers, $\Pi_r = \pi_r - \sum_{j \in J_r} T_j$ and $\Pi_f = \pi_f + \sum_{j \in J_f} T_j$. The terms are private to the negotiating pair. In the second stage, wholesale prices remain private and retailers simultaneously set prices, optimizing against their own wholesale prices.

Let $\pi_f(\boldsymbol{w}, J)$ and $\pi_r(\boldsymbol{w}, J)$ denote the payoffs to manufacturer f and retailer r at the Nash equilibrium in retail prices, before transfers. Unlike our main model, this two-part tariff framework assumes sequential price setting: manufacturers anticipate how changes in wholesale prices will affect downstream retail prices, yielding a subgame-perfect equilibrium.

Negotiations are bilateral and take place between individual manufacturer–retailer pairs. Let $g = (f, r)$ denote a negotiating pair, let $\boldsymbol{w}_a$ denote the vector of wholesale prices negotiated by this pair, and let $\boldsymbol{w}_{-g}$ denote the vector of wholesale prices for all other pairs. The vector $\boldsymbol{w}_a$ is *bilaterally efficient* (or *pairwise stable*) if it maximizes the joint surplus of the negotiating pair, taking all other wholesale prices as given:[36]

$$\boldsymbol{w}_a = \arg\max_{\boldsymbol{w}'_a} \pi_f(\boldsymbol{w}'_a, \boldsymbol{w}_{-g}, J) + \pi_r(\boldsymbol{w}'_a, \boldsymbol{w}_{-g}, J). \tag{G.1}$$

Under the conditions in Rey and Verge (2019), bilateral efficiency is achieved when wholesale prices are set equal to the manufacturer's marginal costs, with transfers allocating the resulting joint surplus between the parties. They show that this outcome arises in a unique Nash-in-Nash equilibrium with private contracts and can also be supported as the equilibrium of a non-cooperative game under the delegated-agent assumption, in which there is no information flow across a firm's negotiating pairs. This eliminates vertical externalities within each negotiating pair. Externalities across different negotiating pairs may remain, so the outcome need not maximize total vertical surplus.

The model features transfers and assumes sequential rather than simultaneous price setting, but it yields the same wholesale and retail prices as our model when $b_n = 0$. In that case, total markups coincide with those from a Bertrand–Nash equilibrium in which retailers optimize retail prices against total marginal cost, see conditions (3.4). A key assumption behind this equivalence is that each negotiating pair takes as given the contracts agreed by all other pairs.[37] A related paper, de Fontenay and Gans (2014), obtains the same outcome in terms of retail prices and quantities, but with quantity-forcing in the negotiations.

Retailer pricing The retailer-pricing model is nested in the bargaining model when $\rho_n = 0$ for all n, which implies $\boldsymbol{\Gamma}_t^F = 0$. The retail-price outcome derived above is therefore obtained from the baseline bargaining model by setting $\rho_n = 0$ throughout.

[36] Rey and Verge (2019) show that wholesale prices satisfying pairwise stability may not survive *multilateral* deviations—i.e., coordinated renegotiation across multiple retailer–manufacturer pairs—under certain patterns of own- and cross-price elasticities. To address this concern, one can either verify that profitable multilateral deviations do not exist under the estimated model, or adopt a delegated-agent framework in which each manufacturer negotiates independently with different retailers through agents who do not coordinate, preventing such multilateral deviations from arising.

[37] This assumption, often referred to as "passive beliefs," implies that if an agent receives a non-equilibrium offer, the terms of other contracts remain unchanged—i.e., deviations are unilateral.

H Moments

In this appendix, we detail the market-level BLP instruments and the three sets of micro moments used in demand estimation.

H.1 BLP Instruments

We construct BLP instruments using the following observed product characteristics: indicators for private-label budget, private-label not budget, wheat, rice, oats, bran, multigrain, granola, contains chocolate, nuts, fruit and honey, as well as pack size, and pack size squared. The instruments are based on two alternative ways of summing the observed characteristics; for the lth characteristic these sums are

$$|\mathcal{J}_{rt}|^{-1}\left(\sum_{j'\in\mathcal{J}_{rt}} x^l_{j't}\right) - x^l_{jt} \qquad \text{and} \qquad |\mathcal{J}_t \setminus \mathcal{J}_{rt}|^{-1}\left(\sum_{j'\in\mathcal{J}_t\setminus\mathcal{J}_{rt}} x^l_{j't}\right) - x^l_{jt},$$

for observation (j,t), where r is the retailer for option j. Intuitively, these instruments measure how close option j is to the average of other options sold, (i) in the same retailer and (ii) in competitor retailers, across each dimension of characteristic space, and therefore capture the intensity of competition for the option and are likely to drive to what extent the retailer can set price above marginal cost (see Berry et al. (1995) and Gandhi and Houde (2019) for a discussion).

H.2 Micro Moments

For each moment m, the estimator matches an observed household-level moment Y^m_h, computed using the full set of households N_H, to a predicted counterpart $y^m_h(\boldsymbol{\theta})$, computed using the subsample N^*_H as described in equation (4.1). We use three sets of micro moments.

H.2.1 Cross-Period Persistence Moments

The first set of moments use persistence in household choices across choice occasions. Let I_h denote the set of household-weeks in the full sample for household h, and let I^*_h denote the corresponding set in the subsample used to compute predicted moments. For each characteristic l, let $m(l)$ denote the corresponding moment, let $\mathcal{J}^l_t$ denote the set of options for which characteristic l is defined, and let $\mathcal{P}_2(I'_h)$ denote the set of unordered pairs of household-weeks i from $I'_h \in \{I_h, I^*_h\}$ in which h chooses an option in $\mathcal{J}^l_{t(i)}$. For household-week i, define

$$x^l_i(\boldsymbol{\theta}) = \frac{\sum_{j\in J^l_{t(i)}} s_{ij}(\boldsymbol{\theta}) x^l_{jt(i)}}{\sum_{j\in J^l_{t(i)}} s_{ij}(\boldsymbol{\theta})}, \qquad x^l_i = \frac{\sum_{j\in J^l_{t(i)}} 1[i \text{ chooses } j] x^l_{jt(i)}}{\sum_{j\in J^l_{t(i)}} 1[i \text{ chooses } j]}.$$

Let $\bar{x}^l$ and $\bar{x}^l(\boldsymbol{\theta})$ denote the corresponding sample means, computed with equal weighting across households.

The observed and predicted household-level moments are

$$Y_h^{m(l)} = \frac{1}{|\mathcal{P}_2(I_h)|} \sum_{(i,i') \in \mathcal{P}_2(I_h)} (x_i^l - \bar{x}^l)(x_{i'}^l - \bar{x}^l),$$

and

$$y_h^{m(l)}(\boldsymbol{\theta}) = \frac{1}{|\mathcal{P}_2(I_h^*)|} \sum_{(i,i') \in \mathcal{P}_2(I_h^*)} (x_i^l(\boldsymbol{\theta}) - \bar{x}(\boldsymbol{\theta})^l)(x_{i'}^l(\boldsymbol{\theta}) - \bar{x}^l(\boldsymbol{\theta})).$$

These are centered moments. Centering removes changes in the average level of the characteristic and focuses the moment on whether households persistently choose options with above- or below-average values.

There are seven cross-period persistence moments.

1. *Price*: $x_{jt}^l = \tilde{p}_{jt}$ and $\mathcal{J}_t^l = \mathcal{J}_t$.
2. *Inside good*: $x_{jt}^l = 1[j > 0]$ and $\mathcal{J}_t^l = \mathcal{J}_t \cup \{0\}$.
3. *Retailer*: $x_{jt}^l = 1[r(j) = r]$ and $\mathcal{J}_t^l = \mathcal{J}_t$. This gives a retailer-specific moment for each retailer r, which we aggregate to a single retailer-persistence moment by summing over retailers.
4. *Brand*: $x_{jt}^l = 1[\text{brand}(j) = a]$ and $\mathcal{J}_t^l = \mathcal{J}_t$. This gives a brand-specific moment for each brand a, which we aggregate to a single brand-persistence moment by summing over brands.
5. *Cereal base*: $x_{jt}^l = 1[c(j) = c]$ and $\mathcal{J}_t^l = \mathcal{J}_t$. This gives a cereal-base-specific moment for each base c. We aggregate these into three cereal-base-persistence moments, corresponding to the groups wheat; rice and corn; and oats, bran, multigrain, and granola.

H.2.2 Household-Option Interaction Moments

The second set of moments uses variables that vary across household-option pairs. Let d_{ij}^l denote such a variable and let $m(l)$ denote the corresponding moment. The observed and predicted moment are

$$Y_h^{m(l)} = \frac{1}{|I_h^l|} \sum_{i \in I_h^l} \frac{\sum_{j \in \mathcal{J}_{t(i)}^l} 1[i \text{ chooses } j] d_{ij}^l}{\sum_{j \in \mathcal{J}_{t(i)}^l} 1[i \text{ chooses } j]},$$

and

$$y_h^{m(l)}(\boldsymbol{\theta}) = \frac{1}{|I_h^{*l}|} \sum_{i \in I_h^{*l}} \frac{\sum_{j \in \mathcal{J}_{t(i)}^l} s_{ij}(\boldsymbol{\theta}) d_{ij}^l}{\sum_{j \in \mathcal{J}_{t(i)}^l} s_{ij}(\boldsymbol{\theta})},$$

where $I_h^l = \{i \in I_h : \sum_{j \in \mathcal{J}_{t(i)}^l} 1[i \text{ chooses } j] > 0\}$ and $I_h^{*l} = \{i \in I_h^* : \sum_{j \in \mathcal{J}_{t(i)}^l} s_{ij}(\boldsymbol{\theta}) > 0\}$.

We use five household-option interaction moments:

1. *Price-income interaction*: $d_{ij}^l = \tilde{p}_{jt} \times y_{h(i)t(i)}$ and $\mathcal{J}_t^l = \mathcal{J}_t$.
2. *Distance, major urban areas*: $d_{ij}^l = 1[u_h = 1] \log(\text{dist}_{ir(j)})$ and $\mathcal{J}_t^l = \{j \in \mathcal{J}_t : r(j) \neq \text{Other}\}$.
3. *Distance, other areas*: $d_{ij}^l = 1[u_h = 0] \log(\text{dist}_{ir(j)})$ and $\mathcal{J}_t^l = \{j \in \mathcal{J}_t : r(j) \neq \text{Other}\}$.
4. *Inside good and urban residence*: $d_{ij}^l = 1[u_h = 1]1[j > 0]$ and $\mathcal{J}_t^l = \{0\} \cup \{j \in \mathcal{J}_t : r(j) \neq \text{Other}\}$.
5. *Children and chocolate cereals*: $d_{ij}^l = 1[\text{children}_{h(i)} = 1]1[\text{chocolate}_j = 1]$ and $\mathcal{J}_t^l = \mathcal{J}_t$.

H.2.3 Choice-Set and Inside-Good Moments

The third set of moments helps identify the variance parameter σ^ϕ governing the household-week-specific shock to the utility of buying breakfast cereal. These moments relate household-specific choice-set measures to whether the household chooses any inside good.

Let z_i^l denote a household-week-level measure of the local choice set. The observed and predicted household-level moments are

$$Y_h^{m(l)} = \frac{1}{|I_h|} \sum_{i \in I_h} z_i^l \, 1[i \text{ chooses } j > 0],$$

and

$$y_h^{m(l)}(\boldsymbol{\theta}) = \frac{1}{|I_h^*|} \sum_{i \in I_h^*} z_i^l \, s_{i,j>0}(\boldsymbol{\theta}),$$

where $s_{i,j>0}(\boldsymbol{\theta}) = \sum_{j \in \mathcal{J}_{t(i)}} s_{ij}(\boldsymbol{\theta})$ is the predicted probability that household-week i chooses any breakfast cereal option.

We use four choice-set measures:

1. the number of available options within 2 kilometers;
2. an indicator for whether this number is positive;
3. the distance to the 50th nearest option; and
4. the distance to the 200th nearest option.

I Details of Estimator

The estimator for the demand model is

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \begin{bmatrix} \mathbf{g}_A(\boldsymbol{\theta}) \\ \mathbf{g}_M(\boldsymbol{\theta}) \end{bmatrix}' \begin{bmatrix} \boldsymbol{W}_A & \mathbf{0} \\ \mathbf{0} & \boldsymbol{W}_M \end{bmatrix} \begin{bmatrix} \mathbf{g}_A(\boldsymbol{\theta}) \\ \mathbf{g}_M(\boldsymbol{\theta}) \end{bmatrix}$$

The micro moments $\mathbf{g}_M(\boldsymbol{\theta})$ is a 16×1 vector that stacks the moments given by (4.1), and described in Appendix H.2. The BLP moments $\mathbf{g}_A(\boldsymbol{\theta})$ are

$$\mathbf{g}_A(\boldsymbol{\theta}) \quad = \quad |N_A|^{-1} \boldsymbol{Z}' \boldsymbol{\Delta\xi}(\boldsymbol{\theta})$$

where N_A is the set of jt observations and $|N_A|$ the number of observations, $\boldsymbol{Z}$ is a matrix of instruments, and

$$\boldsymbol{\Delta\xi}(\boldsymbol{\theta}) \quad = \quad \boldsymbol{\delta}(\boldsymbol{\theta}_1) - \boldsymbol{X}\boldsymbol{\theta}_2$$

where the vector $\boldsymbol{\delta}(\boldsymbol{\theta}_1)$ is the value of $\boldsymbol{\delta}$ that, given $\boldsymbol{\theta}_1$, matches observed and predicted market shares, and is found by iterating on the BLP contraction. Here $\boldsymbol{\theta}_1$ are the nonlinear parameters and $\boldsymbol{\theta}_2$ are the linear parameters.

The matrix $\boldsymbol{X}$'s columns are a constant, dummies for all except one market, dummies for all except one retailer, dummies for all except one product. The matrix $\boldsymbol{Z}$ consists of the columns of $\boldsymbol{X}$, plus a set of instruments for price (cost-shifter and BLP instruments).

The vectors $\boldsymbol{\theta}_1$ and $\boldsymbol{\theta}_2$ represent a decomposition of the parameter vector $\boldsymbol{\theta}$ into its nonlinear ($\boldsymbol{\theta}_1$) and linear ($\boldsymbol{\theta}_2$) components. For each trial value $\boldsymbol{\theta}_1$ during numerical minimization, we concentrate out the linear parameters as

$$\hat{\boldsymbol{\theta}}_2(\boldsymbol{\theta}_1) = (\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}_A\boldsymbol{Z}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}_A\boldsymbol{Z}'\boldsymbol{\delta}(\boldsymbol{\theta}_1). \tag{I.1}$$

Rewriting the parameters as $\boldsymbol{\theta} = (\boldsymbol{\theta}_1, \hat{\boldsymbol{\theta}}_2(\boldsymbol{\theta}_1))$, where $\hat{\boldsymbol{\theta}}_2(\boldsymbol{\theta}_1)$ is given by equation (I.1), the estimator is

$$\hat{\boldsymbol{\theta}}_1 = \arg\min_{\theta_1} \begin{bmatrix} \mathbf{g}_A(\boldsymbol{\theta}_1, \hat{\boldsymbol{\theta}}_2(\boldsymbol{\theta}_1)) \\ \mathbf{g}_M(\boldsymbol{\theta}_1, \hat{\boldsymbol{\theta}}_2(\boldsymbol{\theta}_1)) \end{bmatrix}' \begin{bmatrix} \boldsymbol{W}_A & \mathbf{0} \\ \mathbf{0} & \boldsymbol{W}_M \end{bmatrix} \begin{bmatrix} \mathbf{g}_A(\boldsymbol{\theta}_1, \hat{\boldsymbol{\theta}}_2(\boldsymbol{\theta}_1)) \\ \mathbf{g}_M(\boldsymbol{\theta}_1, \hat{\boldsymbol{\theta}}_2(\boldsymbol{\theta}_1)) \end{bmatrix}$$

$$\hat{\boldsymbol{\theta}}_2 \quad = \quad \hat{\boldsymbol{\theta}}_2(\hat{\boldsymbol{\theta}}_1).$$

The matrix $\boldsymbol{W}_A$, corresponding to the BLP moments, is given by $(\mathbf{Z}'\mathbf{Z}/|N_A|)^{-1}$, where $|N_A|$ is the total number of market-option observations, and $\mathbf{Z}$ is the matrix of instruments. The matrix $\mathbf{W}_M$, corresponding to the micro moments, is diagonal, with the entry for moment m equal to the reciprocal of the square of the observed component of that moment: $1/(\bar{Y}^m)^2$, where $\bar{Y}^m$ is defined in equation (4.1). This normalization puts the micro moments—which are not measured in the same units—on a common (units-free) scale as percentage deviation between predicted and observed contributions (see Low and Meghir 2017).

I.1 Simulation of Market Shares

For given $(\boldsymbol{\delta}_t, \boldsymbol{\theta}_1)$, the model-implied market share of option j in market t is

$$S_{jt}(\boldsymbol{\delta}_t, \boldsymbol{\theta}_1) = \int s_{ij}(\boldsymbol{\delta}_t, \boldsymbol{\mu}_i(\boldsymbol{\theta}_1))\, dF_t(\boldsymbol{\mu}|\boldsymbol{\theta}_1),$$

where $s_{ij}(\cdot)$ is the choice probability for household–week i and $F_t(\cdot)$ is the distribution of household attributes (observed and unobserved) in market t (see Section 3.2).

We approximate this integral by simulation using the household panel as the empirical distribution of observables. As described in Section 4.1, we draw a random subset N_H^* of 2,000 households from the full panel and, for each selected household $h \in N_H^*$, we select three household–week observations from different year–quarter markets. This yields a set $I^* \subset I$ of 6,000 household–weeks, which provides the empirical distribution of observed household characteristics used in simulations.

For each simulation draw s, we draw the persistent household-level taste shocks $([\nu_{hs}^l]_{l\in\mathcal{L}}, \nu_{hs}^\alpha)$ independently for each of the 2,000 households and hold them fixed across the three household–weeks selected for that household. We draw the transitory shock ν_{is}^ϕ independently for each of the 6,000 household–weeks. These draws are held fixed across all iterations of the GMM algorithm. Thus, with $S = 1$, the simulation sample contains 2,000 independent realizations of the household-level taste shocks and 6,000 independent realizations of the household–week shock.

For each household–week i and draw s, let

$$\boldsymbol{\nu}_{is} = \big([\nu_{h(i)s}^l]_{l\in\mathcal{L}}, \nu_{is}^\phi, \nu_{h(i)s}^\alpha\big), \qquad s = 1, \ldots, S.$$

Given $(\boldsymbol{\nu}_{is}, y_{h(i)t}, \text{dist}_{ir})$ and the parameter vector $\boldsymbol{\theta}_1$, we compute the individual choice probabilities

$$s_{ij}\big(\boldsymbol{\delta}_t, \boldsymbol{\mu}_{is}(\boldsymbol{\theta}_1)\big)$$

for all options $j \in \mathcal{J}_t$ in each market t where i is observed.

The simulated market share for option j in market t is then

$$\hat{s}_{jt}(\boldsymbol{\delta}_t, \boldsymbol{\theta}_1) = \frac{1}{S|I_t^*|} \sum_{i\in I_t^*} \sum_{s=1}^{S} s_{ij}\big(\boldsymbol{\delta}_t, \boldsymbol{\mu}_{is}(\boldsymbol{\theta}_1)\big),$$

where $I_t^* \subset I^*$ is the set of simulated household–weeks belonging to market t. These simulated shares $\hat{s}_{jt}(\boldsymbol{\delta}_t, \boldsymbol{\theta}_1)$ enter the BLP contraction used to recover $\boldsymbol{\delta}_t(\boldsymbol{\theta}_1)$ and hence the product-level moments $\boldsymbol{g}_A(\boldsymbol{\theta})$.

For post-estimation objects, including elasticities, markups, and counterfactual outcomes, we use the same household–week sampling procedure but increase the number of simulated taste-shock draws to $S = 20$ per household–week.

J Standard Errors

There are three relevant sample sizes: $|N_A|$ for the BLP-style aggregate moments, $|N_H^*|$ for the small household sample, and $|N_H|$ for the large household sample. For the asymptotic analysis, we assume that the ratios $k_0 = |N_A|/|N_H^*|$ and $k = |N_H^*|/|N_H|$ remain fixed as $|N_A| \to \infty$. Equivalently, since we hold k_0 and k constant, we could take $|N_H^*| \to \infty$ as the starting point.

The asymptotic covariance matrix of the estimator depends on the asymptotic distribution of the two sets of moments. By the central limit theorem,

$$\sqrt{|N_A|}g_A(\boldsymbol{\theta}) \to N(\mathbf{0}, \boldsymbol{\Omega}_A) \tag{J.1}$$

$$\sqrt{|N_H^*|}g_M(\boldsymbol{\theta}) \to N(\mathbf{0}, \boldsymbol{\Omega}_M). \tag{J.2}$$

where estimators for the asymptotic covariances of the moments, $\boldsymbol{\Omega}_A$ and $\boldsymbol{\Omega}_M$, are given below. Multiplying by $\sqrt{k_0}$, (J.2) yields

$$\sqrt{|N_A|}g_M(\boldsymbol{\theta}) = \sqrt{|N_A|/|N_H^*|}\sqrt{|N_H^*|}g_M(\boldsymbol{\theta}) \to N(\mathbf{0}, k_0\boldsymbol{\Omega}_M). \tag{J.3}$$

We treat the product-level moments and household-level moments as independent for purposes of inference, and briefly discuss this assumption here. Each predicted micro moment $y_h(\boldsymbol{\theta})$ is constructed as a function of the aggregate data, i.e., the observed shares and the implied mean utilities δ. The household-level micro-moment contribution $Y_h - y_h(\boldsymbol{\theta})$ is sampling noise orthogonal to the aggregate shocks, and therefore has expectation zero conditional on that aggregate data. The product-level moments are themselves functions of the aggregate data, so this conditional-mean-zero property yields zero asymptotic cross-covariance between the two sets of moments. Treating observed shares as population shares would make the cross-covariance exactly zero even though the same panel underlies both sets of moments. In reality, observed shares are not strictly speaking population shares, but the sample is so large (15,000–30,000 households, and more than 72,000 cereal purchases in the median quarter) that each household has a negligible impact on these shares. Moreover, the market shares S_{jt} that go into the BLP contraction are computed using the full dataset, so that the relevant sample size is $|N_H|$, which is assumed to grow with $|N_A|$ since k_0 and k are fixed. These shares therefore converge to the population shares as $|N_A| \to \infty$.

Proceeding with the assumption of independence and stacking (J.1) and (J.3) we get

$$\sqrt{|N_A|}\begin{bmatrix} g_A(\boldsymbol{\theta}) \\ g_M(\boldsymbol{\theta}) \end{bmatrix} \to N\left(\mathbf{0}, \begin{bmatrix} \boldsymbol{\Omega}_A & \mathbf{0} \\ \mathbf{0} & k_0\boldsymbol{\Omega}_M \end{bmatrix}\right)$$

Standard errors are then obtained in the usual way, as the square root of the diagonal of the matrix

$$(\mathbf{G}'\mathbf{W}\mathbf{G})^{-1}\mathbf{G}'\mathbf{W}\boldsymbol{\Omega}\mathbf{W}\mathbf{G}(\mathbf{G}'\mathbf{W}\mathbf{G})^{-1}/|N_A|$$

where

$$\begin{aligned} \mathbf{W} &= \begin{bmatrix} \mathbf{W}_A & \mathbf{0} \\ \mathbf{0} & \mathbf{W}_M \end{bmatrix} \\ \mathbf{G} &= \begin{bmatrix} \nabla \mathbf{g}_A(\boldsymbol{\theta}) \\ \nabla \mathbf{g}_M(\boldsymbol{\theta}) \end{bmatrix} \\ \boldsymbol{\Omega} &= \begin{bmatrix} \boldsymbol{\Omega}_A & \mathbf{0} \\ \mathbf{0} & k_0\boldsymbol{\Omega}_M \end{bmatrix}. \end{aligned}$$

The covariance of the market level moments is

$$\boldsymbol{\Omega}_A = \frac{1}{|N_A|} \sum_{(j,t)\in N_A} \left[\boldsymbol{Z}'_{jt}\Delta\boldsymbol{\xi}_{jt}(\hat{\boldsymbol{\theta}})\right]\left[\boldsymbol{Z}'_{jt}\Delta\boldsymbol{\xi}_{jt}(\hat{\boldsymbol{\theta}})\right]'.$$

The covariance of the micro moments (see O'Connell et al. (2025) for a derivation), is given by

$$\begin{aligned}
\boldsymbol{\Omega}_M &= k\hat{\boldsymbol{\Sigma}}_Y + \hat{\boldsymbol{\Sigma}}_y - k(\hat{\boldsymbol{\Sigma}}_{Yy} + \hat{\boldsymbol{\Sigma}}'_{Yy}) \\
\hat{\boldsymbol{\Sigma}}_Y &= \frac{1}{|N_H|} \sum_{h\in N_H} [\mathbf{Y}_h - \bar{\mathbf{Y}}][\mathbf{Y}_h - \bar{\mathbf{Y}}]' \\
\hat{\boldsymbol{\Sigma}}_y &= \frac{1}{|N^*_H|} \sum_{h\in N^*_H} [\mathbf{y}_h(\hat{\boldsymbol{\theta}}) - \bar{\mathbf{y}}(\hat{\boldsymbol{\theta}})][\mathbf{y}_h(\hat{\boldsymbol{\theta}}) - \bar{\mathbf{y}}(\hat{\boldsymbol{\theta}})]' \\
\hat{\boldsymbol{\Sigma}}_{Yy} &= \frac{1}{|N^*_H|} \sum_{h\in N^*_H} [\mathbf{Y}_h - \bar{\mathbf{Y}}][\mathbf{y}_h(\hat{\boldsymbol{\theta}}) - \bar{\mathbf{y}}(\hat{\boldsymbol{\theta}})]'.
\end{aligned}$$

The vectors $\mathbf{Y}_h$ and $\mathbf{y}_h(\boldsymbol{\theta})$ represent the observed and predicted components of the moment for household h. The vector $\bar{\mathbf{Y}}$ contains the sample means of $\mathbf{Y}_h$ across households in the large sample, while $\bar{\mathbf{y}}(\hat{\boldsymbol{\theta}})$ is the vector of means of $\mathbf{y}_h(\boldsymbol{\theta})$ across households in the small sample. This covariance structure allows for arbitrary within-household correlation across household-weeks i, i.e., standard errors are clustered at the household level h.

J.1 Sampling Variation in Market Shares

The empirical inside shares are constructed from the consumer panel as follows. First, we compute the total share of grocery-shopping occasions in market t that include a breakfast-cereal purchase, which determines the aggregate inside share and hence the outside-good share. Second, conditional on a breakfast-cereal purchase, we compute the fraction of purchases accounted for by each product–retailer option j. The model share S_{jt} is the product of these two objects: the aggregate inside share times option j's conditional share among cereal purchases.

The purchase counts discussed below refer to the conditional distribution of cereal purchases across inside options. The panel contains between 15,000 and 30,000 households, and the median quarter has more than 72,000 cereal purchases. The average market has 637 options, so on average we observe well over 100 purchases per option–market cell. For less popular products we observe fewer purchases: across option–quarter cells, the distribution of purchase counts has a 5–95 percentile range of 17–241 and a median of 49. These low-count cells account for a small fraction of total quantity and surplus, and options in this tail are disproportionately associated with the residual "other retailers" category rather than the main discounters.

In the estimation we treat the resulting empirical shares, S_{jt}, including the outside-good share, as population shares when forming the product-level moments $\boldsymbol{g}_A(\boldsymbol{\theta})$. We do not implement the small-sample bias corrections for share measurement error proposed by Berry et al. (2004) or Freyberger (2015). Given the purchase

counts described above, and the fact that our main objects of interest (elasticities, markups, and counterfactuals) are driven by higher-share options and the major retailers, we do not expect remaining sampling error in the smallest cells to materially affect the results.

K Demand-Side Diagnostics

K.1 Demand-Side Instrument Relevance

The BLP product-level moments $\boldsymbol{g}_A(\boldsymbol{\theta})$ treat option prices p_{jt} as endogenous and use cost-shifter and BLP-style instruments in $\boldsymbol{Z}$ (see Section 4.1). To assess instrument relevance, we run a reduced-form first-stage regression of prices on the full set of exogenous characteristics $\boldsymbol{x}_{jt}$ and excluded instruments $\boldsymbol{z}_{jt}$,

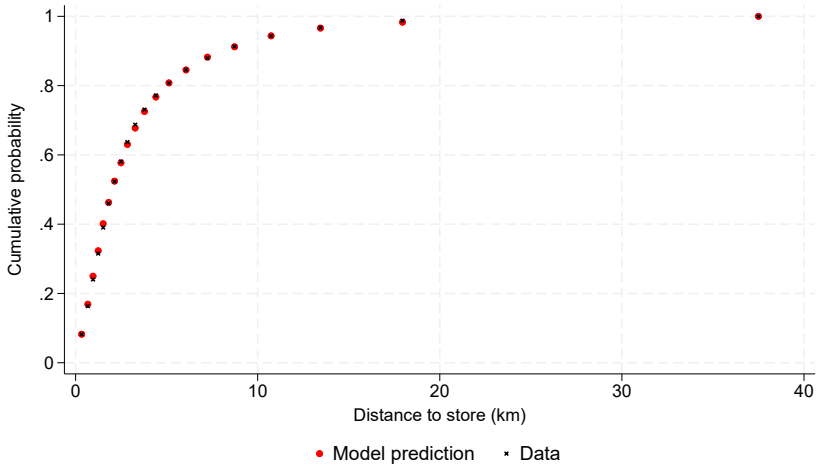
$$p_{jt} = \boldsymbol{x}_{jt}'\pi_x + \boldsymbol{z}_{jt}'\pi_z + \eta_{jt},$$

using the same sample and fixed effects as in the structural estimation. We report the Montiel Olea–Pflueger effective F statistic, which is robust to heteroskedasticity. With our preferred instrument set, the effective F statistic is 16.79. At the 5% significance level, this exceeds the critical value for a 10% worst-case bias threshold (13.09), though not the more stringent 5% threshold (23.77). We therefore view the excluded demand instruments as sufficiently relevant by the conventional 10% benchmark.

K.2 Model Fit

In Section 4.1, we report the fit of the micro moments. In Figure K.1 we show that model-predicted relationship between the average cumulative probability of a household choosing an option and the travel distance to the nearest store selling that option closely matches the corresponding pattern in the data. In other words, the model successfully recovers the how purchase probabilities vary spatial with travel distances.

Figure K.1: *Impact of distance on choice probabilities*



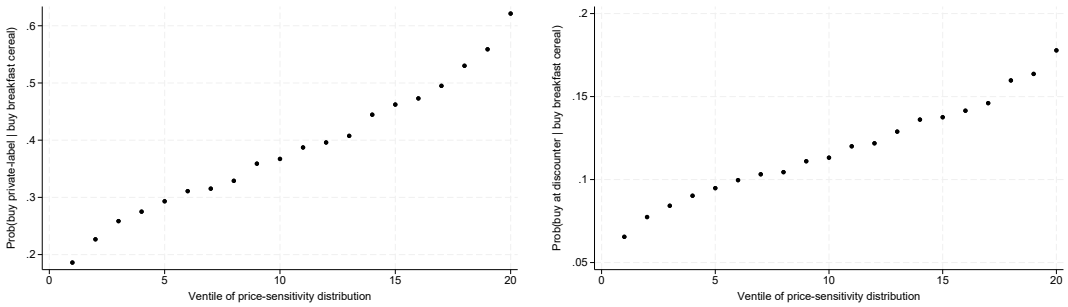
Notes: Each marker shows the cumulative probability of a household choosing an option within the distance indicated on the horizontal axis from their home. Probabilities are conditional on choosing an inside option. Red markers show predictions from our model and black markers are probabilities computed with the data.

L Price Sensitivity and Household Sorting

To illustrate how household sorting varies with estimated price sensitivity, Figure L.1 plots the predicted probability of choosing a private-label option and the predicted probability of choosing a discounter option across ventiles of the estimated price-sensitivity distribution. In each case, we condition on purchasing breakfast cereal. The figure shows that households with higher estimated price sensitivity are systematically more likely to choose both private-label and discounter options.

Figure L.1: *Purchase incidence by price sensitivity*

(a) Private-label options (b) Discounter options



Notes: The figure plots how the average predicted probability of choosing private-label options (panel (a)) and discounter options (panel (b)) varies across ventiles of the estimated price-sensitivity distribution. In each case, we condition on purchasing breakfast cereal. We pool observations across all markets.

M Counterfactual Algorithm

We perform counterfactuals market-by-market. Here, we suppress market subscripts t. Morrow and Skerlos (2011) propose reformulating the standard BLP supply side (equation (3.4)) as

$$\mathbf{p} - \tilde{\mathbf{c}} = \boldsymbol{\zeta}(\mathbf{p}),$$

where $\tilde{\mathbf{c}}$ is the marginal cost that the retailer optimizes against,

$$\boldsymbol{\zeta}(\mathbf{p}) = \boldsymbol{\Lambda}(\mathbf{p})^{-1}\boldsymbol{\Gamma}(\mathbf{p})'(\mathbf{p} - \tilde{\mathbf{c}}) - \boldsymbol{\Lambda}(\mathbf{p})^{-1}\mathbf{s}(\mathbf{p}),$$

$\boldsymbol{\Lambda}$ is a $J \times J$ diagonal matrix with entries

$$\Lambda_{jj} = \textstyle\sum_{i \in I^*} s_{ij} \frac{\partial U_{ij}}{\partial p_j}$$

and $\boldsymbol{\Gamma}$ is a $J \times J$ matrix with entries

$$\boldsymbol{\Gamma}_{jj'} = \begin{cases} \sum_{i \in I^*} s_{ij} s_{ij'} \frac{\partial U_{ij'}}{\partial p_{j'}} & \text{if } j \text{ and } j' \text{ are co-owned} \\ 0 & \text{otherwise.} \end{cases}$$

Morrow and Skerlos (2011) show that the mapping $p \leftarrow \tilde{\mathbf{c}} + \boldsymbol{\zeta}(\mathbf{p})$ is a contraction.

Let $\boldsymbol{\gamma}^R$ be the per-pack (i.e., not per-kilogram) retailer markup and $\tilde{\mathbf{c}}$ the (wholesale price inclusive) retail marginal costs (also per pack), so that $\mathbf{p} \equiv \boldsymbol{\gamma}^R + \tilde{\mathbf{c}}$. We can write the contraction with $\boldsymbol{\gamma}^R$ as an argument instead:

$$\boldsymbol{\gamma}^R \leftarrow \boldsymbol{\zeta}(\boldsymbol{\gamma}^R + \tilde{\boldsymbol{c}}). \tag{M.1}$$

Let kg_j be the pack size in kilograms for option j. Then the per-pack retailer markup $\boldsymbol{\gamma}^R$ relates to the per-kilo retailer markup $\boldsymbol{\Gamma}^R$ as $\Gamma^R_j = \gamma^R_j / \text{kg}_j$, and similarly for prices and costs.

In per-kilogram terms, equation (4.2) gives the vector of negotiation-level manufacturer markups as

$$\boldsymbol{\Gamma}^F_{\mathcal{N}} = \boldsymbol{\rho}\boldsymbol{B}(\boldsymbol{p})^{-1}\boldsymbol{A}(\boldsymbol{p})\boldsymbol{\Gamma}^R. \tag{M.2}$$

The corresponding option-level manufacturer markups are

$$\Gamma^F_j = \left[\boldsymbol{\Gamma}^F_{\mathcal{N}}\right]_{n(j)} \equiv \varphi_j(\boldsymbol{p}, \boldsymbol{\Gamma}^R), \qquad j \in \mathcal{J}, \tag{M.3}$$

and we collect these in the vector $\boldsymbol{\varphi}(\boldsymbol{p}, \boldsymbol{\Gamma}^R) = (\varphi_j(\boldsymbol{p}, \boldsymbol{\Gamma}^R))_{j \in \mathcal{J}}$.

To solve for counterfactual prices we take the following steps.

1. (Step 0). Start with guesses of per-pack retail prices $\mathbf{p}_s$ and per-pack manufacturer markups $\boldsymbol{\gamma}^F_s$, for iteration counter $s = 0$, as well as known total vertical per-pack marginal costs $\boldsymbol{c}$ (which remain fixed throughout this exercise).

2. (Step 1). With $\tilde{\boldsymbol{c}}_s = \boldsymbol{\gamma}^F_s + \boldsymbol{c}$, update retailer markup as $\boldsymbol{\gamma}^R_{s+1} = \boldsymbol{\zeta}(\boldsymbol{p}_s)$, using equation (M.1).

3. (Step 2). Transform $\boldsymbol{\gamma}_{s+1}^R$ to per-kilogram terms, $\boldsymbol{\Gamma}_{s+1}^R$. Use the updated matrices $\boldsymbol{A}(\boldsymbol{p}_s)$ and $\boldsymbol{B}(\boldsymbol{p}_s)$ to calculate the option-level manufacturer markups

$$\boldsymbol{\Gamma}_{s+1}^F = \boldsymbol{\varphi}(\boldsymbol{p}_s, \boldsymbol{\Gamma}_{s+1}^R),$$

and transform these to per-pack terms, $\boldsymbol{\gamma}_{s+1}^F$, by multiplying each element j by the corresponding pack size kg_j.

4. (Step 3). Update retail prices (in per-pack terms) as $\boldsymbol{p}_{s+1} = \boldsymbol{c} + \boldsymbol{\gamma}_{s+1}^R + \boldsymbol{\gamma}_{s+1}^F$.

5. Iterate on steps 1-3 until convergence defined as $\|\boldsymbol{p}_{s+1} - \boldsymbol{p}_s\| < 10^{-8}$.

The foregoing discussion is for the baseline bargaining model. Under retailer pricing, the algorithm is identical except that we fix $\boldsymbol{\Gamma}^F = \boldsymbol{\varphi}(\boldsymbol{p}, \boldsymbol{\Gamma}^R) = 0$ in equation (M.3). We can then skip Step 2. In Step 1, $\tilde{\boldsymbol{c}} = \boldsymbol{c}$, and in Step 3, $\boldsymbol{p}_{s+1} = \boldsymbol{c} + \boldsymbol{\gamma}_{s+1}^R$.

N Counterfactual Specification

Throughout this appendix, $\Delta_{CF} Y_t$ denotes the effect of the observed evolution of discounters, defined as the observed equilibrium outcome minus the corresponding counterfactual outcome. In the results tables, counterfactual changes are reported relative to the observed equilibrium and therefore use the opposite sign.

Store-entry counterfactual (CF1) This counterfactual measures the impact of the expansion in the number of discounter stores, holding the in-store offering as observed in each period. Let $\mathcal{S}_t^D \subset \mathcal{S}_t$ denote the set of discounter stores in period t. In the counterfactual, we fix the set of discounter stores at its initial value $\mathcal{S}_t^D = \mathcal{S}_1^D$, so that the counterfactual set of stores in period t is given by $\mathcal{S}_t' = \mathcal{S}_1^D \cup (\mathcal{S}_t \setminus \mathcal{S}_t^D)$. The change in the endogenous outcome Y_t resulting from this counterfactual (CF1) is:

$$\Delta_{CF1} \mathrm{Y}_t = \mathrm{Y}(\mathcal{S}_t, \mathcal{J}_t, \mathbf{c}_t, \boldsymbol{\delta}_t) - \mathrm{Y}(\mathcal{S}_t', \mathcal{J}_t, \mathbf{c}_t, \boldsymbol{\delta}_t).$$

In this scenario, the distances from consumer i to discounter stores in period t are fixed at their initial values, dist_{ir1}, while distances to all other retailers remain unchanged.

In-store counterfactual (CF2) This counterfactual measures the impact of changes to the in-store offering at discounter stores since 2002, keeping the set of stores $\mathcal{S}_t$ as observed in each period. We modify three primitives: the set of breakfast cereal options $\mathcal{J}_t$, marginal costs $\mathbf{c}_t$, and mean utilities $\boldsymbol{\delta}_t$. The change in the outcome Y_t in period t from this counterfactual (CF2) is:

$$\Delta_{CF2} \mathrm{Y}_t = \mathrm{Y}(\mathcal{S}_t, \mathcal{J}_t, \mathbf{c}_t, \boldsymbol{\delta}_t) - \mathrm{Y}(\mathcal{S}_t, \mathcal{J}_t', \mathbf{c}_t', \boldsymbol{\delta}_t')$$

where counterfactual primitives (denoted with primes) are defined as follows.

1. *Product–retailer options.* In the counterfactual, we replace the set $\mathcal{J}_t^D$ of product–retailer options available at discounters in period t with the set $\mathcal{J}_1^D$ available in period $t = 1$.[38] The resulting counterfactual set of options in period t is $\mathcal{J}_t' = \mathcal{J}_1^D \cup (\mathcal{J}_t \setminus \mathcal{J}_t^D)$.

2. *Marginal costs* Let option j correspond to retailer r, product k, and manufacturer or ownership type f, and let market t correspond to year y and quarter q. Marginal cost is specified as

$$c_{jt} = \gamma_w \mathbf{w}_{kt} + \gamma_{\mathrm{q}(t)} + \gamma_k + \gamma_{ry(t)} + \gamma_{fy(t)} + \omega_{jt}, \tag{N.1}$$

   where $\mathbf{w}_{kt}$ is a vector of input prices, and γ_{ry} and γ_{fy} are retailer–year and branded/private-label year effects, respectively.[39] In the counterfactual, we assume that, for discounters, retailer-year effects γ_{ry} follow the corresponding values for traditional retailers. Formally, for $y > 2002$, we replace these terms with the average across traditional retailer options:

$$\gamma'_{r(j)y} = |\mathcal{J}_y^{Tr}|^{-1} \textstyle\sum_{j' \in \mathcal{J}_y^{Tr}} \gamma_{r(j')y} \quad \text{for all } j \in \mathcal{J}_y^D$$

   where $\mathcal{J}_y^{Tr}$ and $\mathcal{J}_y^D$ are the sets of options in traditional retailers and discounters, respectively, in year y.[40] Substituting these values into equation (N.1) yields counterfactual costs c'_{jt}.

3. *Mean utility.* The mean utility of option $j = (k, r)$—i.e., product k at retailer r—in market t is the sum of a fixed product effect, a fixed retailer effect, and a time-varying product–retailer effect: $\delta_{jt} = \theta_k + \theta_r + \xi_{jt}$.[41] We further decompose ξ_{jt} as $\xi_{jt} = \xi_{rt} + \Delta\xi^*_{jt}$, where ξ_{rt} is a retailer–market–specific effect and $\Delta\xi^*_{jt}$ is a mean-zero product–retailer deviation. Substituting this decomposition, mean utility can be expressed as

$$\delta_{jt} = \theta_k + \theta_r + \xi_{rt} + \Delta\xi^*_{jt}. \tag{N.2}$$

   In the counterfactual, we adjust the retailer–market effects for discounters so that they follow the path observed for the traditional retailers, rather than their discounter-specific trajectories. Specifically, for each period t, we replace $\xi_{r(j)t}$ for discounter products with the average across traditional retailer options:

$$\xi'_{r(j)t} = |\mathcal{J}_t^{Tr}|^{-1} \textstyle\sum_{j' \in \mathcal{J}_t^{Tr}} \xi_{r(j')t} \quad \text{for all } j \in \mathcal{J}_t^D$$

[38] More precisely, for each period t, we use the same quarter in the year 2002—represented as $t = 1$ for notational simplicity. In post-2002 markets, this entails removing products from Aldi and Lidl introduced after 2002 and reinstating the smaller number of products available in the same quarter of 2002 but subsequently discontinued.

[39] Equation (N.1) rewrites equation (4.3) by expanding $\mathbf{x}^s_{jt}$ into its components.

[40] Options in different markets t in the same year are treated as distinct in these sets.

[41] In Section 3.2, equation (3.1), we write $\delta_{jt} = \boldsymbol{\theta}_2 \mathbf{x}_{jt} + \Delta\xi_{jt}$. $\mathbf{x}_{jt}$ comprises product, retailer and market fixed effects, meaning we can rewrite $\delta_{jt} = \theta_k + \theta_r + \theta_t + \Delta\xi_{jt}$. Defining $\xi_{jt} \equiv \theta_t + \Delta\xi_{jt}$ leads to the equation in the text.

where $\mathcal{J}_t^{Tr}$ is the set of options in traditional retailers in market t. Substituting these values into equation $(N.2)$ yields counterfactual mean utility δ'_{jt}.[42]

Full counterfactual (CF3) This counterfactual (CF3) combines the changes from both the store-entry and in-store-offering counterfactuals, to measure the overall effect of the rise of the discounters. The change in outcome Y_t in period t is given by:

$$\Delta_{CF3}\mathrm{Y}_t = \mathrm{Y}(\mathcal{S}_t, \mathcal{J}_t, \mathbf{c}_t, \boldsymbol{\delta}_t) - \mathrm{Y}(\mathcal{S}'_t, \mathcal{J}'_t, \mathbf{c}'_t, \boldsymbol{\delta}'_t).$$

By comparing the results of this counterfactual with those of CF1 and CF2, we can assess the relative importance of store entry versus changes to the in-store offering in shaping changes to market power and economic surplus.

Planning-policy counterfactual This counterfactual is a policy-motivated variant of the store-entry counterfactual. Rather than fixing the set of discounter stores at its 2002 level, we allow the Aldi and Lidl store networks to evolve along their observed paths through 2012 and then impose a counterfactual path in which, within each region of Great Britain, store numbers continue to grow at the average annual rate observed between 2002 and 2012.[43]

Formally, let $g \in \mathcal{G}$ index the nine regions of Great Britain, and let N_{cgy} denote the number of stores operated by chain $c \in \{\text{Aldi}, \text{Lidl}\}$ in region g in year y. For each chain-region pair (c, g), we compute the average annual growth in store numbers over the period 2002–2012 as

$$\bar{m}_{cg} = \frac{N_{cg,2012} - N_{cg,2002}}{10}.$$

By 2021, the counterfactual number of stores is then given by

$$N'_{cg,2021} = N_{cg,2012} + 9\bar{m}_{cg},$$

rounded to the nearest integer.

To implement this counterfactual, we retain all discounter stores observed up to 2012. For post-2012 openings, we then randomly keep only the number of stores required for each chain-region pair to match the implied 2021 counterfactual total $N'_{cg,2021}$. Let $\mathcal{S}_{2021}^{D,\leq 2012}$ denote the set of discounter stores opened by year 2012 and active in 2021, and let $\mathcal{S}_{cg,2021}^{D,>2012}$ denote the set of post-2012 stores for chain c in region g that are active in 2021. For each (c, g), we draw without replacement a subset $\tilde{\mathcal{S}}_{cg,2021}^{D,>2012} \subseteq \mathcal{S}_{cg,2021}^{D,>2012}$ such that the total number of active stores matches the

[42] For discounter options in the counterfactual assortment, we also assume that the counterfactual deviations $\Delta\xi^*_{jt}$ and cost shocks ω_{jt} take the values from the corresponding quarter of the first year, denoted t'. That is, we set $\Delta\xi^*_{jt} = \Delta\xi^*_{jt'}$ and $\omega_{jt} = \omega_{jt'}$.

[43] We construct the nine regions from the ONS country and English region boundaries (ONS, 2024a,b), combining the East and West Midlands, and Wales with the South West.

target implied by $N'_{cg,2021}$. The counterfactual store network for 2021 is then

$$\mathcal{S}^{PP}_{2021} = \mathcal{S}^{D,\leq 2012}_{2021} \cup \bigcup_{c\in\{\text{Aldi,Lidl}\}} \bigcup_{g\in\mathcal{G}} \tilde{\mathcal{S}}^{D,>2012}_{cg,2021} \cup (\mathcal{S}_{2021} \setminus \mathcal{S}^{D}_{2021}),$$

where $\mathcal{S}^{D}_{2021} \subset \mathcal{S}_{2021}$ is the set of discounter stores in 2021.

The change in the endogenous outcome under this planning-policy counterfactual is therefore

$$\Delta_{PP}\mathrm{Y}_{2021} = \mathrm{Y}(\mathcal{S}_{2021}, \mathcal{J}_{2021}, \mathbf{c}_{2021}, \boldsymbol{\delta}_{2021}) - \mathrm{Y}(\mathcal{S}^{PP}_{2021}, \mathcal{J}_{2021}, \mathbf{c}_{2021}, \boldsymbol{\delta}_{2021}).$$

This exercise is intended as an illustrative benchmark for the role of the post-2012 acceleration in discounter store growth, rather than as a fully structural estimate of the effect of any single policy change.

O Decomposition of Cereal and Non-cereal Utility Effects

The mean utility of option $j = (k, r)$—i.e., product k at retailer r—in market t is given by:

$$\delta_{jt} = \theta_k + \theta_r + \xi_{jt}.$$

We decompose the time-varying component as $\xi_{jt} = \xi^*_{jt} + \psi_{rt}$, where ξ^*_{jt} captures factors intrinsic to the breakfast cereal product, and ψ_{rt} reflects retailer-level effects not specific to breakfast cereals (e.g., changes in the quality of the shopping experience or improvements in other product categories purchased alongside cereals).

To separately identify ξ^*_{jt} and ψ_{rt}, we exploit continuing options—specific cereals k sold by retailer r in consecutive periods. Let $\mathcal{J}^{\text{cont}}_{rt}$ be the set of such options for retailer r in period t. We assume the change over time in the intrinsic cereal-specific utility component is mean zero:

$$\mathbb{E}(\xi^*_{jt} - \xi^*_{j,t-1}) = \mathbb{E}([\xi_{jt} - \xi_{j,t-1}] - [\psi_{rt} - \psi_{r,t-1}]) = 0, \quad \forall j \in \mathcal{J}^{\text{cont}}_{rt},$$

for each (r, t). This assumption is motivated by the fact that the characteristics of continuing options do not change, so that average shifts in their option-level mean utilities identify changes in retailer-level non-cereal effects.

The sample analog implies, for all (r, t),

$$\psi_{rt} - \psi_{r,t-1} = |\mathcal{J}^{\text{cont}}_{rt}|^{-1} \textstyle\sum_{j\in\mathcal{J}^{\text{cont}}_{rt}} [\xi_{jt} - \xi_{j,t-1}].$$

Normalizing $\psi_{r1} = 0$ yields ψ_{rt} for all $t > 1$.

In equation (N.2), we re-write mean utility:

$$\delta_{jt} = \theta_k + \theta_r + \xi_{rt} + \Delta\xi^*_{jt},$$

where

$$\xi_{rt} \equiv |\mathcal{J}_{rt}|^{-1} \sum_{j \in \mathcal{J}_{rt}} \xi_{jt}, \quad \Delta\xi^*_{jt} \equiv \xi_{jt} - \xi_{rt}$$

Noting that:

$$\xi_{rt} = |\mathcal{J}_{rt}|^{-1} \sum_{j \in \mathcal{J}_{rt}} \xi^*_{jt} + \psi_{rt} \equiv \xi^*_{rt} + \psi_{rt}$$

we obtain:

$$\delta_{jt} = \theta_k + \theta_r + \psi_{rt} + \xi^*_{rt} + \Delta\xi^*_{jt},$$

where ξ^*_{rt} and $\Delta\xi^*_{jt}$ capture retailer–market-level and idiosyncratic intrinsic cereal effects, respectively, and ψ_{rt} captures retailer–market-level non-cereal effects.

P Consumer Surplus

We write the utility of option $j > 0$ for household–week i as $U_{ijt} = \delta_{jt} + \mu_{ij} + \epsilon_{ij}$, which includes the heterogeneous taste term μ_{ij}, as specified in equation (3.1). We write $\mu_{ij} = \mu_j(\tilde{p}_{jt}, \boldsymbol{\nu}_i, \text{dist}_{ir(j)})$ to make explicit its dependence on equilibrium price $\tilde{p}_{jt}$, the random taste shock vector $\boldsymbol{\nu}_i$, and distance $\text{dist}_{ir(j)}$. Note that distances in year–quarter t depend on the set $\mathcal{S}_t$ of store locations. Let $F(\mu_i|\tilde{\mathbf{p}}_t, \mathcal{S}_t)$ be the distribution function of the vector $\mu_i = (\mu_{ij})_{j \in J}$. For a given realization $(\boldsymbol{\epsilon}_i, \mu_i)$, let the optimal choice be denoted

$$j(\boldsymbol{\epsilon}_i, \mu_i, \mathcal{J}_t) \equiv \underset{j \in \mathcal{J}_t \cup \{0\}}{\arg\max} \{\delta_{jt} + \mu_{ij} + \epsilon_{ij}\}$$

where $\delta_{0t} = \mu_{i0} = 0$. The compensating variation of a consumer in market t being offered options $\mathcal{J}_t$ with mean utilities $\boldsymbol{\delta}_t = \{\delta_{jt}\}_{j \in \mathcal{J}_t}$, is

$$\begin{aligned} \text{CS}_t &= \int_{\mu_i} \frac{1}{\alpha_i} \left[\mathbb{E}_{\boldsymbol{\epsilon}_i} \left(\max_{j \in \mathcal{J}_t \cup \{0\}} \{\delta_{jt} + \mu_{ij} + \epsilon_{ij}\} \right) \right] dF(\mu_i|\tilde{\mathbf{p}}_t, \mathcal{S}_t) \\ &= \int_{\mu_i} \frac{1}{\alpha_i} \left[\ln \left(1 + \sum_{j \in \mathcal{J}_t} \exp(\delta_{jt} + \mu_{ij}) \right) \right] dF(\mu_i|\tilde{\mathbf{p}}_t, \mathcal{S}_t). \end{aligned}$$

The consumer's expected travel cost T_t and the expected contribution Z_t from time-varying non-cereal utility effects $\psi_{r(j)t}$ are respectively given by

$$\begin{aligned} T_t &= -\int_{\mu_i} (\tau_0 + \tau_1 u_i) \frac{1}{\alpha_i} \textstyle\sum_{j \in \mathcal{J}_t} \Big\{ \log \text{dist}_{ij} s_{ijt}(\mu_i) \Big\} dF(\mu_i|\tilde{\mathbf{p}}_t, \mathcal{S}_t), \text{ and} \\ Z_t &= \int_{\mu_i} \frac{1}{\alpha_i} \textstyle\sum_{j \in \mathcal{J}_t} \Big\{ \psi_{r(j)t} s_{ijt}(\mu_i) \Big\} dF(\mu_i|\tilde{\mathbf{p}}_t, \mathcal{S}_t). \end{aligned}$$

See below for the steps in this derivation. Since these effects are additive in utility, the utility component attributable to breakfast cereal products, excluding transport costs and non-cereal retailer choice effects, is

$$B_t = \mathrm{CS}_t - T_t - Z_t.$$

Aggregate surplus To compute aggregate consumer surplus we use the full counterfactual consumer surplus change $\Delta_{CF3}\mathrm{CS}$. In addition, we compute corresponding changes for transport costs and other category effects, i.e.,

$$\Delta_{CF3}\mathrm{T}_t = \mathrm{T}(\mathcal{S}_t, \mathcal{J}_t, \mathbf{c}_t, \boldsymbol{\delta}_t) - \mathrm{T}(\mathcal{S}'_t, \mathcal{J}'_t, \mathbf{c}'_t, \boldsymbol{\delta}'_t)$$

and

$$\Delta_{CF3}\mathrm{Z}_t = \mathrm{Z}(\mathcal{S}_t, \mathcal{J}_t, \mathbf{c}_t, \boldsymbol{\delta}_t) - \mathrm{Z}(\mathcal{S}'_t, \mathcal{J}'_t, \mathbf{c}'_t, \boldsymbol{\delta}'_t),$$

giving

$$\Delta_{CF3}\mathrm{B}_t = \Delta_{CF3}\mathrm{CS}_t - \Delta_{CF3}\mathrm{T}_t - \Delta_{CF3}\mathrm{Z}_t.$$

The implied aggregate consumer surplus change is $\Delta_{CF3}\mathrm{B}_t \times SC_B + \Delta_{CF3}\mathrm{T}_t \times SC_T$ where SC_B is the scale-up factor for the breakfast cereal component (1/revenue share of breakfast cereals) and SC_T is the scale-up factor for transport costs (1/trip share of breakfast cereals).

Derivation of expected surplus components.

$$\begin{aligned}
T_t &= -\int_{\mu_i} (\tau_0 + \tau_1 u_i) \frac{1}{\alpha_i} \mathbb{E}_{\boldsymbol{\epsilon_i}}\big[\log \mathrm{dist}_{ij(\boldsymbol{\epsilon_i},\mu_i)}\big] dF(\mu_i|\tilde{\mathbf{p}}_t, \mathcal{S}_t) \\
&= -\int_{\mu_i} (\tau_0 + \tau_1 u_i) \frac{1}{\alpha_i} \int_{\boldsymbol{\epsilon_i}} \log \mathrm{dist}_{ij(\boldsymbol{\epsilon_i},\mu_i)} dF_\epsilon(\boldsymbol{\epsilon_i}) dF(\mu_i|\tilde{\mathbf{p}}_t, \mathcal{S}_t) \\
&= -\int_{\mu_i} (\tau_0 + \tau_1 u_i) \frac{1}{\alpha_i} \int_{\boldsymbol{\epsilon_i}} \textstyle\sum_{j\in J_t} \Big\{ \log \mathrm{dist}_{ij} 1\big[j = j(\boldsymbol{\epsilon_i}, \mu_i)\big] \Big\} dF_\epsilon(\boldsymbol{\epsilon_i}) dF(\mu_i|\tilde{\mathbf{p}}_t, \mathcal{S}_t) \\
&= -\int_{\mu_i} (\tau_0 + \tau_1 u_i) \frac{1}{\alpha_i} \textstyle\sum_{j\in J_t} \Big\{ \log \mathrm{dist}_{ij} \int_{\boldsymbol{\epsilon_i}} 1\big[j = j(\boldsymbol{\epsilon_i}, \mu_i)\big] dF_\epsilon(\boldsymbol{\epsilon_i}) \Big\} dF(\mu_i|\tilde{\mathbf{p}}_t, \mathcal{S}_t) \\
&= -\int_{\mu_i} (\tau_0 + \tau_1 u_i) \frac{1}{\alpha_i} \textstyle\sum_{j\in J_t} \Big\{ \log \mathrm{dist}_{ij} s_{ijt}(\mu_i) \Big\} dF(\mu_i|\tilde{\mathbf{p}}_t, \mathcal{S}_t)
\end{aligned}$$

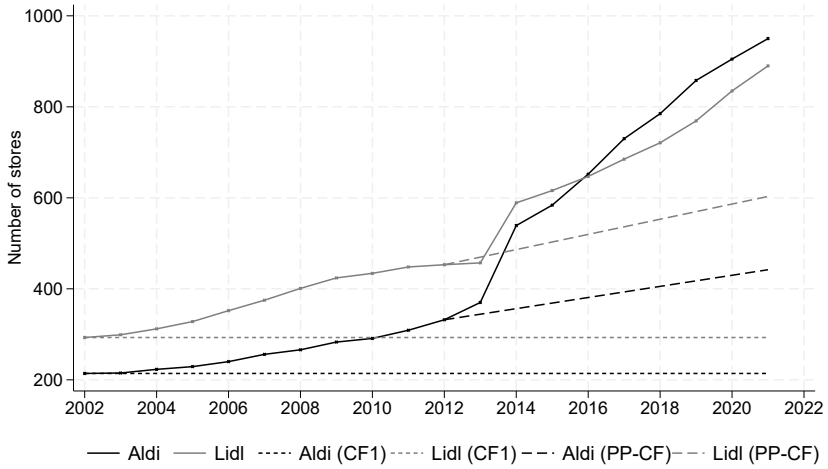
The Z_t term can be derived analogously.

Q Planning-Policy Counterfactual

Figure Q.1 reports the evolution of the number of Aldi and Lidl stores over time, in the data and under both the full store counterfactual (CF1) and the planning-policy counterfactual (see Appendix N). Table Q.1 summarizes the results of CF1 and the planning-policy counterfactual for equilibrium outcomes. The final column

reports the percentage of the full store counterfactual effect (CF1) that is realized under the planning-policy counterfactual. For example, the difference in retail HHI between the observed equilibrium (1824) and the full store counterfactual (1979) is 155, while the corresponding difference under the planning-policy counterfactual (1905) is 81. Thus, the planning-policy counterfactual accounts for 52% of the total CF1 effect.

Figure Q.1: *Discounter store coverage growth over time*



Notes: The figure shows the evolution of the number of Aldi and Lidl stores over time in the data, under the full store counterfactual (CF1), and under the planning-policy counterfactual. The planning-policy counterfactual allows store networks to evolve as observed through 2012 and then imposes a continuation of the pre-2012 linear growth trend.

Table Q.1: *Counterfactual analysis*

	Observed equilibrium	Counterfactual equilibrium:		% of CF1-Δ realized under
		Store (CF1)	Planning-policy	planning-policy CF
A) Discounter primitives				
Distance (km)	3.99	10.99	6.54	*36.4%*
B) Market equilibrium				
Concentration (HHI)				
Retail	1824	1979	1905	*52.4%*
Manufacturer	2123	2269	2214	*62.3%*
Average market price (£/kg)				
Unweighted	3.93	3.95	3.94	*77.0%*
Sales-weighted	3.24	3.34	3.29	*54.0%*
Markup (£/kg)	2.13	2.14	2.14	*77.0%*
Discounter markups (£/kg)				
All options	1.21	1.13	1.20	*9.2%*
Overlapping options				
Branded				
Retail component	0.54	0.51	0.50	*139.1%*
Manufacturer component	1.08	1.09	1.10	*198.2%*
Private-label	1.00	0.94	1.02	*42.5%*
Traditional retailer markups (£/kg)				
Branded				
Retail component	1.36	1.39	1.37	*47.3%*
Manufacturer component	1.21	1.21	1.21	*94.6%*
Private-label	1.22	1.26	1.25	*62.3%*
C) Δ annual market surplus (£m)				
Consumer surplus	-	-47.8	-24.8	*51.8%*
Producer surplus				
Traditional retailers	-	41.2	22.0	*53.2%*
Discounters	-	-40.0	-20.8	*52.1%*
Manufacturers	-	16.1	9.0	*55.9%*
Total surplus	-	-30.5	-14.7	*48.1%*

Notes: The table compares outcomes in the observed 2021 equilibrium with those under the full store counterfactual (CF1) and the planning-policy counterfactual. Unless otherwise stated the averages are unweighted. Panel (A) reports changes in market primitives, while Panels (B) and (C) report the resulting changes in endogenous outcomes. The planning-policy counterfactual allows discounter store networks to evolve as observed through 2012 and then imposes growth at the pre-2012 linear trend. The final column reports the percentage of the total CF1 effect (relative to the observed equilibrium) that is realized under the planning-policy counterfactual. Marginal costs, prices, and markups are expressed in 2021 £ per kg; annual surplus is expressed in 2021 £ millions.

R Results under Alternative Supply Models

R.1 Retailer pricing

Table R.1: *Average elasticities, cost and markups*

		Traditional retailers		Discounters		All options
		Branded	Private-label	Branded	Private-label	
2002	Marginal cost c (£/kg)	3.61	2.14	3.49	1.91	3.22
	Total markup Γ (£/kg)	1.37	1.14	0.60	0.83	1.31
	Lerner index ($\frac{\Gamma}{p}$)	0.28	0.36	0.14	0.31	0.30
2011	Marginal cost c (£/kg)	3.70	1.74	2.76	1.76	3.20
	Total markup Γ (£/kg)	1.58	1.33	0.83	1.01	1.45
	Lerner index ($\frac{\Gamma}{p}$)	0.31	0.44	0.23	0.38	0.33
2021	Marginal cost c (£/kg)	3.32	1.25	2.84	1.12	2.64
	Total markup Γ (£/kg)	1.36	1.22	1.01	1.04	1.29
	Lerner index ($\frac{\Gamma}{p}$)	0.29	0.52	0.26	0.52	0.36

Notes: Table reports average total vertical marginal costs, additive markups and Lerner indexes in 2002, 2011 and 2021. Marginal costs and markups are expressed in 2021 £ per kg.

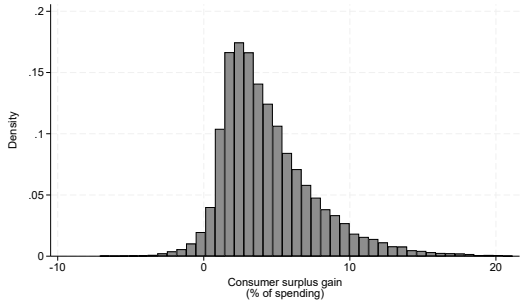
Table R.2: *Counterfactual analysis*

	Observed equilibrium	Counterfactual equilibrium: Store (CF1)	In-store (CF2)	Full (CF3)
A) Discounter primitives				
Marginal cost (£/kg)				
All options	1.40	-	1.06	1.06
Overlapping options	0.79	-	0.78	0.78
B) Market equilibrium				
Concentration (HHI)				
Retail	1824	1979	1949	2068
Manufacturer	2123	2299	2290	2411
Average market price (£/kg)				
Unweighted	3.93	3.95	4.09	4.11
Sales-weighted	3.24	3.34	3.18	3.33
Markup (£/kg)	1.29	1.31	1.33	1.35
Discounter markups (£/kg)				
All options	1.04	0.96	0.95	0.92
Overlapping options				
Branded				
Retail component	0.54	0.51	0.48	0.46
Manufacturer component	-	-	-	-
Private-label	1.00	0.94	0.97	0.96
Traditional retailer markups (£/kg)				
Branded				
Retail component	1.36	1.39	1.38	1.40
Manufacturer component	-	-	-	-
Private-label	1.22	1.26	1.25	1.28
C) Δ annual market surplus (£m)				
Consumer surplus	-	-47.9	-34.1	-70.0
% of spending		*(-3.31%)*	*(-2.36%)*	*(-4.84%)*
Producer surplus				
Traditional retailers	-	41.9	29.2	62.8
% change		*(9.31%)*	*(6.48%)*	*(13.94%)*
Discounters	-	-40.0	-24.7	-55.4
% change		*(-44.69%)*	*(-27.60%)*	*(-62.00%)*
Total surplus	-	-45.9	-29.6	-62.6
% of spending		*(-3.18%)*	*(-2.05%)*	*(-4.33%)*

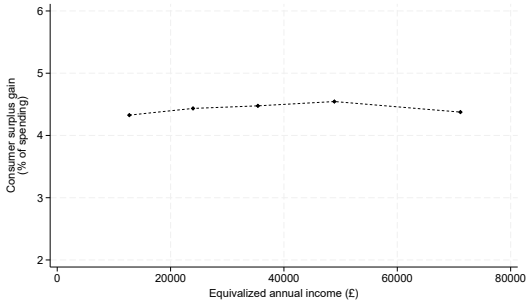
Notes: Table compares average outcomes in the 2021 observed and counterfactual equilibrium. Unless otherwise stated the averages are unweighted. Panel (A) summarizes the change to market primitives in each counterfactual scenario. Panels (B) and (C) summarize the change in endogenous market outcomes. Marginal costs, prices, and markups are expressed in 2021 £ per kg; annual surplus is expressed in 2021 £ millions.

Figure R.1: *Distributional impact*

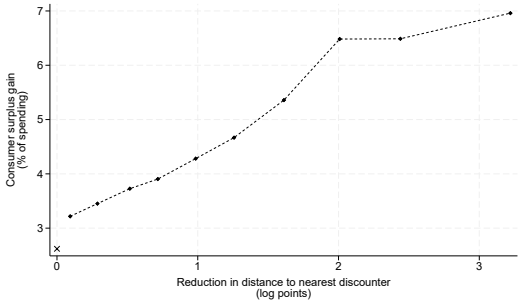
(a) Distribution across households



(b) Impact by income quintiles



(c) Impact by travel distance reduction



Notes: Figures show the difference in consumer surplus between the observed and full counterfactual equilibria for 2021, expressed as a fraction of household breakfast cereal expenditure. Panel (a) reports the distribution of consumer-surplus gains across households. Panel (b) shows the average change for each household income quintile. Panel (c) shows average gains by the reduction in travel distance to the nearest discounter store. The × at zero denotes households whose distance to the nearest discounter does not fall.

R.2 Manufacturer pricing

Table R.3: *Average elasticities, cost and markups*

		Traditional retailers		Discounters		All options
		Branded	Private-label	Branded	Private-label	
2002	Marginal cost c (£/kg)	4.03	2.62	3.53	1.92	3.63
	Total markup Γ (£/kg)	0.96	0.66	0.56	0.82	0.90
	Lerner index ($\frac{\Gamma}{p}$)	0.19	0.21	0.13	0.30	0.20
2011	Marginal cost c (£/kg)	4.27	2.36	2.94	1.78	3.70
	Total markup Γ (£/kg)	1.01	0.71	0.64	0.99	0.95
	Lerner index ($\frac{\Gamma}{p}$)	0.19	0.24	0.18	0.37	0.21
2021	Marginal cost c (£/kg)	3.77	1.68	3.07	1.16	3.01
	Total markup Γ (£/kg)	0.92	0.80	0.78	1.01	0.92
	Lerner index ($\frac{\Gamma}{p}$)	0.20	0.34	0.20	0.50	0.26

Notes: Table reports total vertical marginal costs, additive markups and Lerner indexes in 2002, 2011 and 2021. Marginal costs and markups are expressed in 2021 £ per kg.

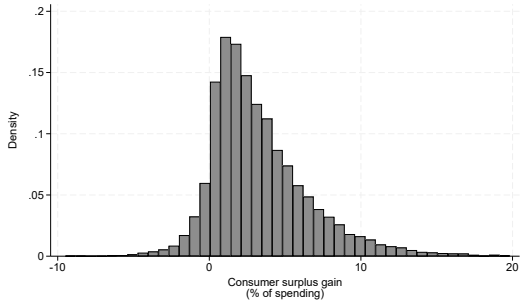
Table R.4: *Counterfactual analysis*

	Observed equilibrium	Counterfactual equilibrium: Store (CF1)	In-store (CF2)	Full (CF3)
A) Discounter primitives				
Marginal cost (£/kg)				
All options	1.47	-	0.97	0.97
Overlapping options	0.81	-	0.70	0.70
B) Market equilibrium				
Concentration (HHI)				
Retail	1824	1985	1951	2073
Manufacturer	2123	2290	2276	2383
Average market price (£/kg)				
Unweighted	3.93	3.93	4.06	4.07
Sales-weighted	3.24	3.32	3.11	3.26
Markup (£/kg)	0.92	0.91	0.92	0.92
Discounter markups (£/kg)				
All options	0.97	0.90	0.95	0.91
Overlapping options				
Branded				
Retail component	-	-	-	-
Manufacturer component	0.55	0.53	0.53	0.54
Private-label	0.97	0.90	0.99	0.96
Traditional retailer markups (£/kg)				
Branded				
Retail component	-	-	-	-
Manufacturer component	0.94	0.95	0.95	0.95
Private-label	0.80	0.81	0.80	0.81
C) Δ annual market surplus (£m)				
Consumer surplus	-	-40.1	-19.2	-51.3
% of spending		*(-2.77%)*	*(-1.33%)*	*(-3.55%)*
Producer surplus				
Trad. retailer private-label	-	10.3	4.2	13.3
% change		*(9.67%)*	*(3.89%)*	*(12.48%)*
Discounter private-label	-	-37.0	-10.5	-44.3
% change		*(-45.62%)*	*(-12.93%)*	*(-54.58%)*
Branded	-	14.8	6.3	19.3
% change		*(6.57%)*	*(2.79%)*	*(8.59%)*
Total surplus	-	-52.0	-19.3	-63.0
% of spending		*(-3.60%)*	*(-1.34%)*	*(-4.36%)*

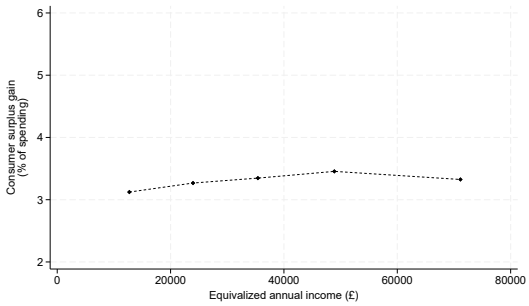
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Figure R.2: *Distributional impact*

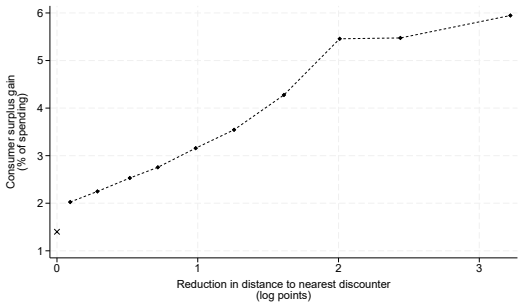
(a) Distribution across households



(b) Impact by income quintiles



(c) Impact by travel distance reduction



Notes: Figures show the difference in consumer surplus between the observed and full counterfactual equilibria for 2021, expressed as a fraction of household breakfast cereal expenditure. Panel (a) reports the distribution of consumer-surplus gains across households. Panel (b) shows the average change for each household income quintile. Panel (c) shows average gains by the reduction in travel distance to the nearest discounter store. The × at zero denotes households whose distance to the nearest discounter does not fall.

S Product categories

Table S.1: *Product categories (1)*

Category	Spending share (%)	Category	Spending share (%)
Ambient Cakes&Pastries	1.61	Pork	0.78
Morning Goods	1.52	Sausages	0.76
Total Bread	1.84	Chilled Cooking Sauces	0.10
Chilled Breads	0.16	Chilled Desserts	0.77
Chilled Cakes	0.36	Chilled Pate&Paste&Spread	0.11
Butter	0.74	Chilled Prepared Salad	0.34
Cheddar	1.55	Chilled Ready Meals	2.83
Eggs	0.78	Chilled Rice	0.03
Fresh Cream	0.34	Chilled Vegetarian	0.09
Margarine And Lard	0.72	Cooked Meats	2.41
Milk	2.98	Fresh Meat&Veg&Pastry	1.21
Non Cheedar Cheese	0.78	Fresh Pasta	0.16
Processed Cheese	0.44	Fresh Soup	0.12
Total Soft White	0.24	Fresh&Chilled Pastry	0.05
Yoghurt	1.82	Other Chilled Convenience	0.28
Yoghurt Drinks And Juices	0.33	Sandwich Fillers	0.14
Apples	0.79	Frozen Bread	0.04
Bananas	0.65	Frozen Meat Products	0.24
Brassicas	0.59	Frozen Pizzas	0.55
Chilled Prepared Frt&Veg	0.96	Frozen Potato Products	0.79
Citrus	0.72	Frozen Processed Poultry	0.48
Legumes	0.22	Frozen Ready Meals	0.84
Nuts - Fruit	0.17	Frozen Savoury Bakery	0.22
Other Vegetables	0.83	Frozen Vegetables	0.57
Pears	0.21	Frozen Vegetarian Prods	0.20
Potatoes	1.16	Other Frozen Foods	0.15
Root Crops	0.81	Ambient Slimming Products	0.03
Salads	1.76	Ambient Soup	0.35
Soft Fruit	1.94	Baked Bean	0.39
Tropical Fruits	0.43	Canned Fish	0.53
Chilled Prepared Fish	0.26	Canned Meats	0.34
Shellfish	0.21	Canned Puddings	0.05
Wet&Smoked Fish	0.85	Canned Rice&Pasta	0.14
Chilled Processed Poultry	0.40	Canned Vegetables	0.16
Cooked Poultry	0.46	Prepared Peas&Beans	0.15
Fresh Poultry	2.20	Tinned Fruit	0.19
Frozen Fish	0.93	Tomato Products	0.24
Frozen Poultry	0.33	Food Drinks	0.19
Bacon	1.39	Herbal Tea	0.08
Beef	2.04	Instant Coffee	0.85
Chilled Burgers&Grills	0.20	Liquid&Grnd Coffee&Beans	0.29
Chld Frnkfurter&Cont Ssgs	0.12	Tea	0.57
Flavoured Meats	0.15	Ambient Pastes&Spreads	0.09
Lamb	0.56	Breakfast Cereals	1.66
Other Meat & Offal	0.13	Honey	0.10

Table S.2: *Product categories (2)*

Category	Spending share (%)	Category	Spending share (%)
Peanut Butter	0.07	Savoury Biscuits	0.24
Porridge Oats	0.20	Cereal&Fruit Bars	0.26
Preserves	0.19	Childrens Biscuits	0.14
Toaster Pastries	0.03	Chocolate Biscuit Bars	0.46
Ambient Condiments	0.09	Confect. & Other Exclusions	0.16
Dips	0.21	Healthier Biscuits	0.36
Olives	0.08	Sweet Biscuits	0.99
Pickles Chutneys&Relish	0.10	Frozen Confectionery	0.36
Salad Accompanimet	0.27	Total Ice Cream	1.00
Sour&Speciality Pickles	0.13	Chocolate Confectionery	2.25
Table Sauces	0.31	Gum Confectionery	0.09
Ambient Rice&Svry Noodles	0.48	Ice Cream Cone	0.01
Cous Cous	0.02	Sugar Confectionery	0.60
Dry Pasta	0.24	Crisps	0.94
Dry Pulses&Cereal	0.06	Nuts - Savoury	0.29
Dry Food	0.03	Popcorn	0.06
Instant Hot Snacks	0.12	Savoury Snacks	0.87
Packet Soup	0.12	Colas	1.01
Ambient Cooking Sauces	0.85	Flavoured Milk	1.34
Cooking Oils	0.33	Lemonade	0.15
Ethnic Ingredients	0.22	Mineral Water	0.40
Flour	0.12	One Shot Drinks	0.43
Herbs&Spices	0.18	Other Soft Drinks	0.65
Meat Extract	0.37	Shandy, Ginger Ale	0.05
Pizza&Bases	0.53	Tonic, Soda Water	0.11
Salt	0.03	Total Fruit Squash	0.61
Stuffing	0.05	Beer&Lager	1.74
Sweet&Savoury Mixes	0.10	Cider	0.43
Vinegar	0.05	Fortifed Wine,Fabs	0.42
Ambient Sponge Puddings	0.02	Sparkling Wine	0.44
Artificial Sweeteners	0.08	Spirits	2.35
Defined Milk&Cream Prd(B)	0.09	Wine	3.37
Home Baking	0.42	Bar Soap	0.05
Instant Milk	0.02	Bath&Shower Products	0.31
Lemon&Lime Juices	0.01	Body Sprays	0.38
Milkshake Mixes	0.03	Liquid Soap	0.13
Mincemeat	0.01	Shaving	0.07
Nuts - Sweet	0.06	Skincare	0.33
Powd Desserts&Custard(B)	0.09	Sun Care	0.05
R.T.S. Custard	0.06	Talcum Powder	0.01
Rts Desserts Long Life	0.08	Hair Colourants	0.10
Ready To Use Icing	0.03	Hair Conditioners	0.15
Sugar	0.31	Hair Styling	0.11
Syrup & Treacle	0.03	Shampoo	0.26
Table&Quick Set Jellies	0.03	Analgesics	0.18

Table S.3: *Product categories (3)*

Category	Spending share (%)	Category	Spending share (%)
Cold Sore Treatment	0.00	Firelighters&Log	0.01
Cold Treatments	0.16	Household Cleaners	0.41
Contact Lens Cleaners	0.01	Household Food Wraps	0.21
Eye Care	0.01	Household Insecticides	0.01
First Aid Dressings	0.02	Kitchen Towels	0.36
Foot Preparations	0.03	Lmscle Rmvrs&Water Softener	0.04
Hayfever Remedies	0.03	Machine Wash Products	1.00
Oral Lesion&Teething Mrkt	0.01	Shoe Care Products	0.01
Sleeping Aids	0.01	Toilet Tissues	1.18
Smoking Cessation	0.04	Wash Additives	0.11
Spray Insecticide	0.00	Washing Up Products	0.49
Stomach Treatments	0.11	Dental Cleaners	0.35
Topical Antiseptics	0.02	Mouthwashes	0.12
Vitamins.Minerals&Splmnts	0.18	Total Toothbrushes	0.15
Air Fresheners	0.33	Cotton Wool	0.04
Batteries	0.18	Feminine Care	0.28
Bin Liners	0.15	Moist Wipes	0.20
Bleaches&Lavatory Clnrs	0.26	Razor Blades	0.17
Carpet Clnrs&Stain Rmvers	0.06	Cat Litter	0.11
Cleaning Accessories	0.13	Cat&Dog Treats	0.36
Electric Light Bulbs	0.06	Dog Food	0.63
Fabric Conditioners	0.37	Fish Foods	0.01
Facial Tissues	0.24	Total Cat Food Inc.Bulk	1.35

Notes: Authors' calculations using Numerator's Take Home Purchase Panel, 2002–2021. Reported spending shares are means across years.

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